Wealth Dynamics and the Persistence of Specialization *

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Abstract

This paper shows how wealth dynamics under financial frictions can make sectoral specialization history-dependent, even when sectors have identical fundamentals. I develop a dynamic multisector general equilibrium model with heterogeneous entrepreneurs facing collateral constraints and persistent productivity shocks. Because borrowing capacity depends on wealth, its allocation across productivity types determines sectoral productivity and accumulation. Persistence depends on the price regime. In a closed economy, relative prices respond to scarcity: initially disadvantaged sectors become more profitable, accelerating wealth accumulation among constrained productive firms and promoting convergence. In a small open economy, sectoral prices are pinned down by exogenous world prices, so this corrective force is absent. Initial differences in wealth allocation therefore generate persistent differences in sectoral specialization. Trade integration changes selection into production and how allocation differences propagate through wealth accumulation. Firm- and sector-level evidence from Peru is consistent with this mechanism. A preliminary quantitative application using input-output and trade data shows that wealth-productivity misalignment generates sizable, persistent differences in specialization, while nontradable prices and sector-specific factors reduce, but do not eliminate, the allocative effects of exogenous world prices.

Keywords: Wealth Distribution, Sectoral Heterogeneity, Persistence, Misallocation
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Preliminary and frequently updated draft. Comments welcome.
Link to the latest version

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1 Introduction

> "Peru needs products from advanced countries, not experiments with Peruvian products."
>
> *President Augusto B. Leguía, to the Peruvian entrepreneur Juan Alberto Grieve.*[1]

While some economies have experienced substantial structural transformation, the productive structure of many economies remains remarkably persistent over long periods of time. A natural interpretation is that these economies are simply specializing according to their comparative advantage. However, many of them also exhibit high levels of labor and capital misallocation, slowly growing firms, and significant financial and operational frictions. If productive sectors coexist with constrained firms and persistent distortions, it is natural to ask whether observed specialization patterns reflect efficient allocation or instead arise from frictions that slow the reallocation of capital across activities. If so, financial frictions may not only slow structural transformation in the short run, but also shape the long-run evolution of sectoral specialization through their interaction with equilibrium prices and wealth accumulation. **This paper** studies this mechanism through a dynamic multisector framework in which financial frictions and the distribution of wealth jointly determine sectoral allocation and specialization dynamics.

Consider an economy in which entrepreneurs differ in productivity but face financial constraints when operating or expanding firms. With frictionless financial markets, capital would flow toward the most productive entrepreneurs within each sector, allowing productive activities to expand regardless of the initial distribution of wealth. However, if access to financing is limited by entrepreneurs' wealth, productive but constrained firms may be unable to scale up, slowing the reallocation of capital toward high-productivity activities.

These distortions can also shape specialization dynamics across sectors. Even when sectors share identical technologies and fundamentals, sectors with a more favorable allocation of wealth toward productive entrepreneurs will initially produce more and accumulate wealth faster. In a closed economy, however, these asymmetries generate relative price adjustments: sectors with lower effective supply become more profitable, accelerating wealth accumulation among constrained but highly-productive firms. As a result, initial differences would tend to dissipate over time. In contrast, in open economies, domestic relative scarcity no longer translates into higher sectoral prices, as relative prices will be tied to global market conditions. In particular, in a small open economy that takes world prices as given, this adjustment mechanism is weakened. Initial allocation

[1] Juan Alberto Grieve (1878–1950) was a Peruvian inventor who designed and built the first internal-combustion engine in South America and one of the first automobiles in the region. His vehicles, adapted to rough local conditions at the time, were reportedly priced around half the cost of importing comparable European models of similar power. Despite this, his project failed to scale, in part due to limited access to financing and weak institutional support. This episode reflects a broader environment in which domestic industrial experimentation faced skepticism and limited support, potentially constraining the reallocation of resources toward more productive local activities. Such patterns are consistent with settings where financial frictions and coordination failures hold back the expansion of efficient firms in constrained sectors. See https://www.britannica.com/biography/Juan-Alberto-Grieve.

advantages are therefore no longer counteracted by relative price adjustment, allowing distortions to translate into persistent specialization patterns.

I explore these mechanisms building on the heterogeneous-agent framework with financial frictions in Moll (2014), extending it to a dynamic multisector general equilibrium environment with endogenous relative prices and sectoral reallocation. Goods produced in different sectors are combined through a constant elasticity of substitution (CES) aggregator and used for consumption and investment. Entrepreneurs operate technologies in a given sector and face collateral constraints and idiosyncratic productivity risk. Since access to capital depends on wealth, wealth accumulation shapes firm expansion and the allocation of resources both within and across sectors over time. I study equilibrium dynamics in two regimes: a closed economy with endogenous relative prices and a small open economy that takes world prices as given under financial autarky, allowing sectoral trade imbalances while maintaining aggregate trade balance. Comparing these environments allows the analysis to isolate the role of price adjustment in shaping the evolution of sectoral allocation.[2]

The model delivers two main theoretical results. First, in a closed economy, endogenous relative prices act as a stabilizing force. Sectors with lower effective supply become relatively more profitable, raising the returns to productive but financially constrained entrepreneurs and accelerating wealth accumulation in initially disadvantaged sectors. As a result, differences in sectoral allocation and productivity tend to dissipate over time.

Second, this adjustment mechanism weakens substantially in open economies. When relative prices are tied to world market conditions, domestic scarcity no longer generates compensating price responses. In a small open economy with exogenous world prices, sectors with an initial allocation advantage accumulate wealth more rapidly and expand persistently, even when sectors share identical technologies and fundamentals. The paper therefore shows a mechanism in which persistent differences in specialization patterns arise solely from initial differences in the distribution of wealth and the allocation of capital.

The contribution is to show that the long run self-correction logic in models with financial frictions changes once production is multisector and prices differ across sectors. In a one-sector economy, wealth dynamics affect aggregate productivity through selection. In a multisector economy, the same wealth dynamics also shape sectoral composition. Whether initial distortions dissipate, persist, or amplify depends on the interaction between price adjustment and sector-level forces. Endogenous domestic prices compress asymmetries; fixed world prices allow them to persist; sector-specific congestion weakens this persistence; and learning or scale externalities can amplify them. Across these cases, the central mechanism is the same: financial frictions make the allocation of wealth across productive firms a state variable for sectoral development.

[2] Appendix K presents a stripped-down analytical benchmark showing how temporary productivity differences can generate persistent specialization patterns in a small open economy with fixed world prices and sector-specific capital accumulation.

The model implies that relatively small initial distortions can generate large and persistent differences in sectoral composition over long horizons. Because wealth accumulation is gradual, the effects of financial frictions propagate dynamically through firms' balance sheets and sectoral profitability. Economies with similar aggregate resources and technologies but different initial allocations of wealth can therefore follow substantially different specialization paths. The model also predicts that the speed of convergence depends critically on the degree of financial frictions and on the extent to which domestic prices adjust to sectoral scarcity. In open economies, persistent relative price differences at the world level amplify these dynamics, slowing reallocation and reinforcing initial allocation advantages.

The baseline model deliberately abstracts from additional sector-level forces in order to isolate this balance-sheet mechanism. In this environment, plants operate constant-returns technologies and the key aggregate state variable is the wealth-productivity distribution within each sector. This benchmark is useful because it delivers a transparent distinction between closed and open economies: when prices are endogenous, sectoral expansion lowers relative profitability; when prices are fixed internationally, this corrective force is absent.

In practice, sectoral expansion may also interact with forces that are not captured by the baseline. Some forces reduce the specialization strength. For example, scarce land, natural resources, infrastructure, managerial capacity, or other sector-specific inputs generate congestion, so that expansion reduces marginal profitability. Other forces amplify specialization. Learning-by-doing, supplier networks, export experience, worker skill accumulation, and agglomeration economies can make expansion raise future productivity and profitability. These forces operate in opposite directions, but both interact with the mechanism emphasized here. Congestion weakens the persistence generated by wealth dynamics, while learning and scale externalities strengthen it. The central point is therefore not that sectors necessarily exhibit constant, decreasing, or increasing returns at the aggregate level. Rather, conditional on these sector-level forces, financial frictions make the distribution of wealth a state variable for the evolution of sectoral productivity and specialization.

Peru provides a useful environment to study these mechanisms. As Figure 1 shows, despite periods of growth, trade integration, and large changes in external demand conditions, production and exports have remained concentrated in a relatively persistent set of sectors over the last few decades. At the same time, firm-level data reveal substantial productivity dispersion and heterogeneous expansion dynamics across firms and sectors. Motivated by these patterns, I use firm-level and sector-level evidence from Peru to evaluate key mechanisms emphasized by the model and provide a quantitative application. Using firm-level panel data *Encuesta Económica Anual* (EEA) (Instituto Nacional de Estadística e Informática (INEI), 2001-2019), the empirical analysis shows that more productive firms tend to expand capital and labor more strongly following favorable external demand shocks, consistent with heterogeneous accumulation dynamics under financial frictions. In addition, sector-level evidence using household survey data (Instituto Nacional de

Estadística e Informática (INEI), 2004-2023) shows that sectors exposed to favorable external demand conditions experience persistent increases in employment shares over subsequent years. Together, these patterns are consistent with the idea that external shocks affect long-run specialization through endogenous accumulation and gradual reallocation dynamics.[3] The quantitative application treats the constant-returns environment as the main theoretical benchmark and introduces sector-specific factors as a conservative discipline on the magnitude of the mechanism. In that extension, plants remain constant returns to scale, but fixed sectoral inputs generate decreasing returns at the sector level and prevent complete specialization. This provides a disciplined way to bring the model to Peruvian data while preserving the core force: wealth dynamics under financial frictions shape how sectors respond to external prices and demand shocks. If learning, supplier networks, or other increasing-returns forces are important in practice, they would amplify rather than overturn the paper's main mechanism.

Related Literature. The paper relates to four strands of literature. First, it relates to work studying the interaction between financial frictions, wealth heterogeneity, and aggregate productivity, including Restuccia and Rogerson (2008), Hsieh and Klenow (2009), Midrigan and Xu (2014), Buera, Kaboski, and Shin (2011), and Moll (2014). This literature emphasizes how distortions to capital allocation affect firm growth and aggregate outcomes. The present paper extends these mechanisms to a multisector environment in which wealth distributions shape sectoral allocation and specialization dynamics over time.

Second, the paper connects to the literature on trade and financial frictions. Antràs and Caballero (2009) study how financial development affects trade patterns, while Manova (2008) shows how credit constraints shape export activity. Relative to this literature, the focus here is not on trade flows themselves, but on how wealth distributions and financial frictions interact with prices to generate persistent differences in sectoral allocation across otherwise similar economies. This extension makes relative price adjustment central for determining whether initial distortions are amplified or reduced over time.

Third, the paper relates to work on reallocation and adjustment dynamics, such as Caliendo, Dvorkin, and Parro (2019) and Dix-Carneiro and Kovak (2017), which study how frictions slow the movement of factors across activities following shocks. In contrast, the mechanism emphasized here operates through within-sector wealth accumulation and endogenous price responses, generating persistence even in the absence of barriers to sectoral mobility.

Fourth, the paper connects to the literature on firm dynamics and selection, including Hopenhayn (1992), Melitz (2003), and Luttmer (2007). In these environments, selection operates through productivity differences. Here, selection is mediated by wealth and financial constraints, so produc-

[3] Appendix A presents complementary cross-country evidence on the persistence of sectoral specialization patterns using international trade data harmonized across countries and sectors. The appendix documents that specialization patterns remain highly persistent over long horizons despite substantial changes in global trade integration and external demand conditions.

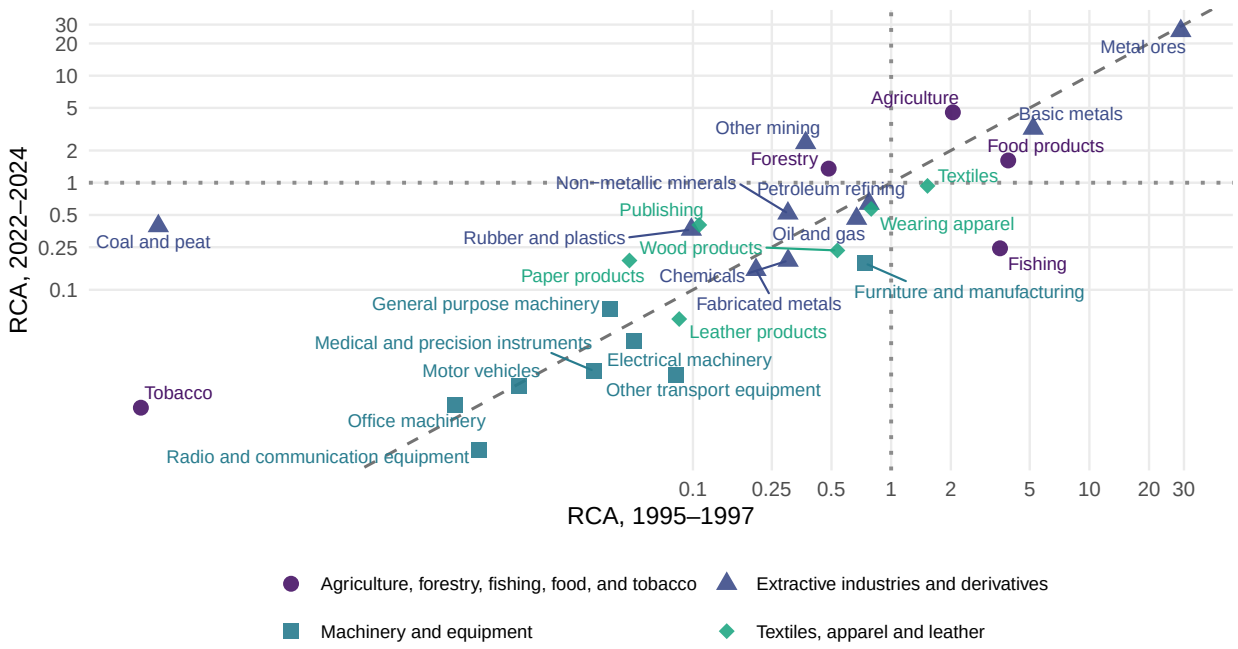


Figure 1: Sectoral Revealed Comparative Advantage in Peru, 1995–1997 vs. 2022–2024

Notes: The figure plots revealed comparative advantage (RCA) for Peruvian exports across ISIC Rev. 3 two-digit sectors in two periods: 1995–1997 (horizontal axis) and 2022–2024 (vertical axis). RCA is defined as $RCA_{i,t} = (X_{Peru,i,t}/X_{Peru,t}) \,/\, (X_{World,i,t}/X_{World,t})$ where $X_{\text{Peru},i,t}$ denotes Peruvian exports in sector i at time t, $X_{\text{Peru},t}$ total Peruvian exports, and $X_{\text{World},i,t}$ and $X_{\text{World},t}$ the corresponding world aggregates. Both axes are reported on a logarithmic scale. The dashed vertical and horizontal lines mark $RCA = 1$, the threshold for comparative advantage. The dashed 45-degree line indicates unchanged specialization between periods; sectors above (below) the line increased (decreased) their revealed comparative advantage over time. Colors and marker shapes identify broad industry groups. Trade data are from the BACI International Trade Database harmonized to ISIC Rev. 3 classifications.

tive but constrained firms may be unable to expand, slowing reallocation toward high-productivity producers.

More generally, the paper contributes to these literatures by showing how wealth distributions and financial frictions interact with price adjustment to shape sectoral allocation and persistence. While Ventura (1997) and Acemoglu and Ventura (2002) emphasize how general equilibrium price adjustments can offset initial differences across economies, the mechanism here highlights how financial frictions can instead propagate initial distortions through endogenous wealth accumulation.

Outline of paper. The rest of the paper proceeds as follows. Section 2 develops a multisector model with heterogeneous entrepreneurs facing financial frictions, in which wealth distributions determine production, prices, and accumulation. Section 3 studies a closed economy, showing how endogenous price adjustment reduces initial asymmetries and promotes convergence through reallocation and selection. Section 4 introduces a small open economy with fixed goods prices, isolating the role of price adjustment and showing how initial differences in wealth and allocation translate

into persistent differences in sectoral composition. Section 5 presents a preliminary quantitative application and firm- and sector-level evidence from Peru. Section 6 concludes.

2 A Multisector Model of Wealth Dynamics and Specialization

This section presents a multisector extension of the heterogeneous-agent model with financial frictions developed by Moll (2014). The key difference is that production takes place in multiple sectors whose outputs are combined through a CES demand system, so that relative prices and specialization patterns emerge endogenously in general equilibrium. To characterize how wealth heterogeneity shapes sectoral allocation, the model introduces sector-specific wealth distribution objects that track both the joint distribution of wealth and productivity within sectors and the allocation of aggregate wealth across sectors. These objects make it possible to characterize how sectoral productivity, prices, and specialization patterns evolve endogenously over time.

The environment described in this section focuses on a closed-economy benchmark that isolates the core mechanism of the paper. Sectors in this setting can be interpreted broadly as industries, regions, or economic activities. More explicit interpretations of sectors will be introduced in the next section, which develops open-economy extensions of the model.

2.1 Setup, Agents, and Final Consumption

I consider an economy with many sectors indexed by $i \in I$. Time is continuous and denoted by t. There are two types of infinitely lived agents: workers (w) and entrepreneurs (e). The total mass of workers is fixed at L. Each worker is endowed with one efficiency unit of labor that they supply inelastically but can allocate across sectors. Workers do not have access to a savings technology, so they live "hand-to-mouth", as in Kaplan and Violante (2014).

Entrepreneurs are attached to a given sector, in which they operate a production technology using capital and labor. In addition, entrepreneurs make forward-looking decisions over consumption and savings based on their stock of financial wealth a. Financial wealth can be transformed into productive capital k, used in the entrepreneur's sector of operation, or consumed.

The difference in mobility between workers and entrepreneurs is similar to the assumption used in Kleinman, Liu, and Redding (2023), where capitalists in a spatial economy are attached to their regions. This restriction helps incorporate forward-looking behavior while keeping the state space tractable. In particular, in the present model the state of the economy at any point in time t can be characterized using $g_i(a, z; t)$, the joint densities over productivity z and wealth a for entrepreneurs operating in each sector i.

I assume that all sectoral output $Y_i(t)$ is transformed into a final composite good through a

constant elasticity of substitution (CES) aggregator:

$$Y(t) = \left(\sum_{i \in I} \upsilon_i^{\frac{1}{\eta}} Y_i(t)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \tag{1}$$

where η denotes the elasticity of substitution across sectors and υ_i are sector-specific weights. This implies the standard CES demand system:

$$D_i(t) = \upsilon_i \left(\frac{p_i(t)}{P(t)} \right)^{-\eta} D(t), \qquad P(t)^{1-\eta} = \sum_{i \in I} \upsilon_i p_i(t)^{1-\eta}. \tag{2}$$

Here, $D_i(t)$ denotes sectoral demand and $D(t)$ denotes total absorption, which includes consumption by both types of agents, the replacement of depreciated capital, and the accumulation of financial wealth.

Workers supply their labor efficiency units to entrepreneurs in a competitive labor market at wage $w(t)$. For each hand-to-mouth worker, consumption expenditure is determined by their labor income, so that $P(t)c_w(t) = w(t)L$. For entrepreneurs, consumption expenditure is determined by an intertemporal optimization problem that depends on prices, productivity, and their stock of financial wealth, and jointly determines their savings behavior.

In equilibrium, sectoral output is absorbed through sectoral demands $\{D_i(t)\}_{i \in I}$, which correspond to the allocation of the final good across sectors. While $D(t)$ summarizes total absorption, it will be useful to distinguish between the portion allocated to current consumption and the portion used for capital replacement and the accumulation of financial wealth. This decomposition will be made explicit in the equilibrium section.

2.2 Production, Technology, and Budgets

Each entrepreneur has access to a technology of production that allows them to use k units of capital and ℓ units of labor to produce

$$y_i = (zk)^{\alpha_i} \ell^{1-\alpha_i} \tag{3}$$

units of output of the good in industry i, where $\alpha_i \in (0, 1)$. Capital depreciates at the same rate δ in every sector. Productivity z follows a continuous-time, sector-specific ergodic Markov process:

$$dz = \mu_i(z)\, dt + \sigma_i(z)\, dW, \tag{4}$$

where $\mu_i(z)$ and $\sigma_i(z)$ denote the drift and diffusion functions, respectively, and W is a standard Brownian motion. I assume that this process admits a stationary distribution in each sector, ensuring that productivity remains bounded in distribution over time. In the quantitative exercises,

I specialize this process to a sector-specific Ornstein–Uhlenbeck process for log productivity, described in Appendix J.

In addition to hiring workers at wages $w(t)$ in a competitive labor market, they can rent capital from other entrepreneurs in a competitive capital rental market at a rental rate $R(t)$, which is equal to the user cost of capital $R(t) = r(t) + \delta$. So, for an entrepreneur in sector i, current productivity z, and wealth $a(t)$, wealth evolves according to:

$$\dot{a} = p_i(zk)^{\alpha_i}\ell^{1-\alpha_i} - w\ell - (r+\delta)k + ra - Pc \tag{5}$$

Entrepreneurs face financial constraints that limit their ability to transform wealth into productive capital. In particular, capital choices are subject to a collateral constraint of the form

$$k \leq \lambda_i a, \tag{6}$$

where $\lambda_i \geq 1$ denotes the tightness of financial constraints in sector i. A higher λ_i corresponds to greater pledgeability of assets and more developed financial intermediation.

This constraint implies that entrepreneurs cannot freely arbitrage between the marginal product of capital and its user cost. In particular, when the constraint binds, capital is proportional to internal wealth, so that more productive but less wealthy entrepreneurs may operate below their efficient scale.

As a result, the allocation of capital depends on the joint distribution of productivity and wealth within each sector. This feature will be central for linking wealth dynamics to sectoral productivity and specialization in general equilibrium.

2.3 Entrepreneur's Problem

Entrepreneurs choose consumption and savings dynamically, taking prices $\{p_i(t), w(t), r(t)\}$ as given. Their decisions determine the evolution of wealth and, through the collateral constraint, their future production capacity.

Let $V_i(a, z; t)$ denote the value function of an entrepreneur in sector i with wealth a and productivity z at time t. The entrepreneur solves

$$V_i(a, z; t) = \max_{\{c(s),k(s),\ell(s)\}_{s\geq t}} \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)}u(c(s))\,ds\right], \tag{7}$$

subject to the law of motion for wealth, Equation 9; the collateral constraint, Equation 6, and the stochastic process for productivity, Equation 4. Throughout, preferences are logarithmic, so that $u(c) = \ln c$.

The entrepreneur's problem can be decomposed into a static and a dynamic component. The static component determines optimal input choices (k, ℓ) at each point in time, given wealth a,

productivity z, and prices, and is summarized by the profit function $\Pi_i(a, z)$ defined in Equation 8. The dynamic component consists of choosing consumption and savings over time, taking as given the profit function and the stochastic evolution of productivity.

This separation is possible because input choices (k, ℓ) affect the entrepreneur's problem only through current-period profits and do not enter the law of motion for wealth except through their impact on current income. Given prices and states (a, z), the static problem determines the maximal flow payoff $\Pi_i(a, z)$ independently of intertemporal considerations. The dynamic problem then consists of allocating this flow of resources over time through consumption and savings decisions. As a result, solving the static and dynamic problems sequentially yields the same solution as the joint problem.

Static problem. It is convenient to define the entrepreneur's static profit function as the solution to the within-period problem:

$$\Pi_i(a, z) = \max_{k,\ell} \left\{ p_i (zk)^{\alpha_i} \ell^{1-\alpha_i} - w\ell - (r + \delta)k \;\text{ s.t. }\; k \leq \lambda_i a \right\}. \tag{8}$$

This formulation summarizes the effect of prices, productivity, and financial constraints on current income. Using this definition, the law of motion for wealth can be written as

$$\dot{a} = \Pi_i(a, z) + ra - Pc. \tag{9}$$

Lemma 1 (Static allocation)**.** *Given prices $\{p_i, w, r\}_{i \in I}$, the solution to the entrepreneurs' static problem implies that capital demand, labor demand, and profits are given by*

$$\begin{aligned}
k_i(a, z) &= \begin{cases} \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise}, \end{cases} \\
\ell_i(a, z) &= \left(\frac{1 - \alpha_i}{w/p_i} \right)^{\frac{1}{\alpha_i}} z \, k_i(a, z), \\
\Pi_i(a, z) &= \lambda_i a \cdot \max \left\{ \pi_i z - (r + \delta),\, 0 \right\},
\end{aligned}$$

where

$$\pi_i = p_i \alpha_i \left(\frac{1 - \alpha_i}{w/p_i} \right)^{\frac{1-\alpha_i}{\alpha_i}},$$

and the productivity cutoff $\underline{z}_i$ satisfies

$$\pi_i \underline{z}_i = r + \delta.$$

Proofs in Appendix B.

A key implication of the static solution is that optimal policies are linear in wealth. Because both production and financial constraints scale proportionally with assets, wealth affects decisions only

through a scaling factor, while productivity determines the intensive margins of behavior. This property will be crucial for aggregation, as it allows the distribution of wealth across productivity levels to summarize the state of the economy and its evolution over time.

Dynamic problem. Using the static allocation results, the entrepreneur's problem reduces to a consumption-savings problem with a stochastic return on wealth. In particular, the law of motion for wealth can be written as

$$\dot{a}(t) = \Gamma_i(z,t)\, a(t) - P(t)c(t), \tag{10}$$

where

$$\Gamma_i(z,t) \equiv \lambda_i \max\left\{\pi_i(t)z - (r(t)+\delta),\, 0\right\} + r(t) \tag{11}$$

denotes the total return on wealth, combining profits from production and interest income.

The entrepreneur's dynamic problem is therefore

$$\max_{\{c(s)\}_{s\geq t}} \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} \ln(c(s))\, ds\right] \quad \text{s.t.} \quad \dot{a}(s) = \Gamma_i(z,s)\, a(s) - P(s)c(s), \tag{12}$$

and can be represented by a Hamilton-Jacobi-Bellman (HJB) equation provided in Appendix C. Under logarithmic preferences, the solution of the problem implies that consumption expenditure is proportional to wealth:

$$P(t)c(t) = \rho a(t), \tag{13}$$

Lemma 2 (Wealth accumulation)**.** *Under logarithmic preferences, individual wealth evolves according to*

$$\dot{a}(t) = s_i(z,t)a(t), \tag{14}$$

where

$$s_i(z,t) \equiv \Gamma_i(z,t) - \rho$$

denotes the net saving rate of entrepreneurs with productivity z in sector i. Proof in Appendix C.

Differences in productivity therefore translate directly into heterogeneous wealth accumulation, governing the evolution of the wealth distribution over time.

2.4 Wealth Distributions and Aggregation

Because optimal policies and wealth accumulation are linear in assets, aggregate outcomes depend only on the distribution of wealth across productivity levels and sectors.

Let $g_i(a,z;t)$ denote the joint density of entrepreneurs with wealth a and productivity z in sector i at time t. Aggregate wealth in the economy and in sector i are given by

$$A(t) = \sum_{i\in I} A_i(t), \qquad A_i(t) = \int a\, g_i(a,z;t)\, da\, dz. \tag{15}$$

Definition 3 (Wealth distributions)**.** Relative to aggregate wealth quantities $A(t)$ and $A_i(t)$, define the following wealth-weighted distributions:

- The share of aggregate wealth held in sector i:

$$\mathcal{O}_i(t) = \frac{A_i(t)}{A(t)}.$$

- The within-sector wealth-weighted productivity density:

$$o_i(z,t) = \frac{1}{A_i(t)} \int a\, g_i(a,z;t)\, da.$$

- The sector-productivity joint wealth density:

$$\omega_i(z,t) = \frac{1}{A(t)} \int a\, g_i(a,z;t)\, da = \mathcal{O}_i(t)\, o_i(z,t).$$

- The economy-wide wealth density

$$\omega(z,t) = \sum_{i \in I} \omega_i(z,t).$$

These objects imply cumulative wealth share distributions within sectors $\Omega_i(z,t) \equiv \int_0^z o_i(x,t)\, dx$, and for the economy as a whole $\Omega(z,t) \equiv \sum_{i \in I} \mathcal{O}_i(t) \cdot \Omega_i(z,t) = \sum_{i \in I} \int_0^z \omega_i(x,t)\, dx$.

The objects $\mathcal{O}_i(t)$ and $o_i(z,t)$ summarize, respectively, the distribution of wealth across sectors and across productivity levels within sectors. Their product, $\omega_i(z,t)$, characterizes the joint distribution of wealth across sectors and productivity levels and is sufficient to determine aggregate allocations and production.

Lemma 4 (Aggregation)**.** *Because optimal policies are linear in wealth, aggregate allocations depend only on the wealth-weighted distribution over productivity. In particular, for any allocation or profit variable $x_i(a,z)$ that is linear in wealth,*

$$\int x_i(a,z)\, g_i(a,z;t)\, da\, dz = A(t) \int x_i(z)\, \omega_i(z,t)\, dz. \tag{16}$$

Hence, the joint wealth density $\omega_i(z,t)$ is sufficient to characterize aggregate allocations and production.

2.5 Closed Economy Equilibrium

Definition 5 (Closed economy equilibrium)**.** Given initial wealth distributions $\{\omega_i(z,0)\}_{i \in I}$, wealth levels $\{A_i(0)\}_{i \in I}$, and labor endowment $\bar{L}$, a competitive closed economy equilibrium is a path of

prices

$$\{w(t), r(t), p_i(t)\}_{i\in I, t\geq 0},$$

allocation rules

$$\{k_i(z,t), \ell_i(z,t), c_i(z,t)\}_{i\in I, t\geq 0},$$

and wealth distributions

$$\{\omega_i(z,t)\}_{i\in I, t\geq 0},$$

such that:

1. Entrepreneurs optimize given prices.

2. Factor markets clear:

$$\bar{L} = A(t)\sum_{i\in I}\int \ell_i(z,t)\,\omega_i(z,t)\,dz,$$
$$A(t) = A(t)\sum_{i\in I}\int k_i(z,t)\,\omega_i(z,t)\,dz.$$

3. Goods markets clear:

$$Y_i(t) = D_i(t), \qquad \forall i \in I,$$

   where sectoral output is

$$Y_i(t) = A(t)\int y_i(z,t)\,\omega_i(z,t)\,dz.$$

4. Sectoral demands satisfy the CES system

$$D_i(t) = \upsilon_i \left(\frac{p_i(t)}{P(t)}\right)^{-\eta} D(t),$$

   with price index

$$P(t)^{1-\eta} = \sum_{i\in I} \upsilon_i p_i(t)^{1-\eta}.$$

5. Aggregate absorption satisfies

$$P(t)D(t) = P(t)C(t) + \dot{A}(t) + \delta A(t),$$

   where aggregate consumption expenditure is

$$P(t)C(t) = w(t)\bar{L} + \rho A(t).$$

6. Wealth distributions evolve consistently with individual wealth accumulation and the pro-

ductivity process.

2.6 Aggregation and Dynamics

The linearity of optimal policies in wealth implies that aggregate dynamics are governed by the distribution of wealth across entrepreneurs. In equilibrium, sectoral productivity reflects endogenous selection, while wealth accumulation depends jointly on profitability, prices, and the allocation of wealth across sectors. The following proposition summarizes these aggregation and dynamic relationships.

Proposition 6 (Wealth distributions, sectoral productivity, and accumulation)**.** *Because individual policies are linear in wealth, sectoral allocations and productivity depend on the wealth-weighted productivity distribution. Sectoral output can be written as*

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

where

$$Z_i(t) = \left(E_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

Sectoral wealth evolves according to

$$\dot{A}_i(t) = \alpha_i p_i(t) Y_i(t) + r(t)\big[A_i(t) - K_i(t)\big] - \delta K_i(t) - \rho A_i(t).$$

Thus, the distribution of wealth across productivity states determines sectoral productivity, profitability, and future accumulation.

Proof and factor-price conditions in Appendix D.

The proposition highlights the channel linking wealth distributions to specialization dynamics. Sectoral productivity depends on how wealth is allocated across productivity types within each sector, while sectoral wealth accumulation depends on profits, prices, and capital use. Relative to the one-sector environment in Moll (2014), sectoral wealth and sectoral capital need not coincide: a sector can use more capital than the wealth held by its entrepreneurs, or it can hold more wealth than the capital used in production. As a result, the common interest rate affects sectoral wealth accumulation through a net asset position channel. An increase in the interest rate raises wealth accumulation in sectors that are net lenders, with $A_i(t) > K_i(t)$, and lowers it in sectors that are net borrowers, with $A_i(t) < K_i(t)$. Equilibrium prices and the interest rate therefore jointly govern the speed at which wealth reallocates across sectors over time, and this reallocation changes future sectoral productivity and specialization.

3 Closed Economy Dynamics

The dynamics of the closed economy are governed by a feedback loop between wealth, production, and prices. Firms with higher productivity accumulate wealth faster, which allows them to operate at larger scale and increases their contribution to sectoral output. As resources are reallocated toward more productive firms, sectoral productivity rises, affecting relative prices and factor returns in general equilibrium. These price adjustments, in turn, influence savings decisions and the allocation of wealth across sectors. In the closed economy, this feedback tends to reduce initial asymmetries, as changes in relative prices partially offset the expansion of sectors that experience faster productivity growth.

3.1 Evolution of Wealth Shares

Given the stochastic evolution of productivity and the optimal saving policy, the individual state (z, a) evolves according to

$$\begin{aligned} dz &= \mu_i(z)\, dt + \sigma_i(z)\, dW, \\ da &= s_i(z,t)\, a\, dt, \end{aligned} \tag{17}$$

where $s_i(z,t) \equiv \Gamma_i(z,t) - \rho$ denotes the net growth rate of wealth implied by optimal behavior.

These dynamics induce a law of motion for the joint distribution $g_i(a, z; t)$ via the associated Kolmogorov Forward equation. As shown in Appendix G, applying this equation to the joint process and exploiting the linearity of policies in wealth implies that the wealth-weighted distribution $\omega_i(z,t)$ satisfies

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] \omega_i(z,t) - \frac{\partial}{\partial z}\left[\mu_i(z)\omega_i(z,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma_i^2(z)\omega_i(z,t)\right]. \tag{18}$$

The first term captures differential wealth accumulation across productivity levels, relative to aggregate wealth growth. The remaining terms reflect the stochastic evolution of productivity.

This law of motion is the point at which the model departs most directly from the one-sector environment in Moll (2014). In Moll's framework, wealth accumulation changes the economy-wide wealth-weighted productivity distribution. Here, the same force operates within each sector, but it also governs the evolution of sectoral wealth shares. In particular, the within-sector distribution evolves according to

$$\frac{\partial o_i(z,t)}{\partial t} = \left[s_i(z,t) - \bar{s}_i(t) \right] o_i(z,t) - \frac{\partial}{\partial z}\left[\mu_i(z) o_i(z,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma_i^2(z) o_i(z,t)\right], \tag{19}$$

where $\bar{s}_i(t) = \int s_i(z,t) o_i(z,t)\, dz$ is the average saving rate in sector i, while sectoral wealth shares

evolve according to

$$\dot{O}_i(t) = \left[\bar{s}_i(t) - \frac{\dot{A}(t)}{A(t)}\right] O_i(t). \tag{20}$$

This distinction is central because sectoral wealth shares determine specialization, and their evolution depends on sector-specific saving rates shaped by relative prices and the common interest rate.

3.2 Aggregated Productivity Dynamics

The model can also be used to decompose aggregate productivity dynamics into within-sector selection and between-sector reallocation components. In the closed economy, this decomposition shows that aggregate productivity gains are driven primarily by within-sector selection, while the between-sector component is quantitatively small and short-lived because relative prices adjust rapidly. Appendix E presents the formal decomposition and the results associated with the quantitative exercise in subsection 3.4.

3.3 Steady State

A steady state is an equilibrium in which sectoral wealth levels and wealth distributions are time-invariant. Formally, for all sectors $i \in I$,

$$\dot{A}_i(t) = 0, \quad g_i(a, z; t) = g_i(a, z), \quad \omega_i(z, t) = \omega_i(z), \quad \text{and} \quad O_i(t) = O_i. \tag{21}$$

In this environment, sectoral productivity and expenditure shares are constant, so specialization is characterized by the stationary allocation of wealth across sectors and productivity states. Appendix F complements the formal steady-state characterization with conditions for sectoral output, factor prices, capital income, and TFP.

3.4 Quantitative Illustration

This section studies how the introduction of sectoral structure alters the dynamics of wealth accumulation, selection, and aggregate productivity in a closed economy. To isolate the role of sectoral reallocation, I construct a one-sector benchmark and compare it to a two-sector economy initialized from the same underlying distributions of wealth and productivity.

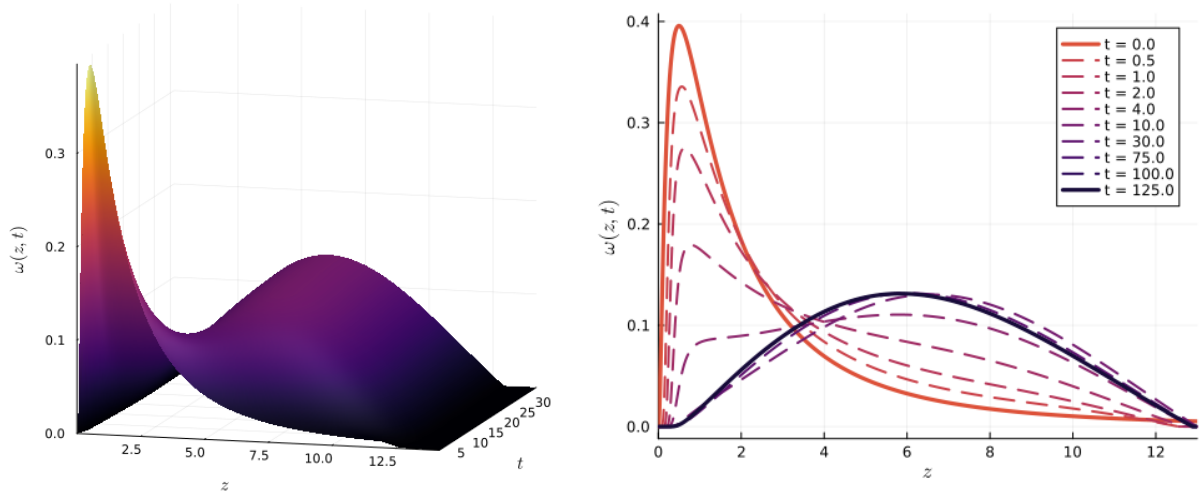
In the one-sector benchmark, I aggregate sectoral distributions into a single economy, so that aggregate initial wealth and the wealth-weighted productivity distribution coincide with those in the multisector environment. In the two-sector economy, I keep the same marginal distributions and total initial wealth, but allow for differences in the allocation of wealth across productivity levels across sectors.

This comparison isolates the role of sectoral structure as an additional margin of adjustment. While the one-sector economy features only within-sector selection and accumulation, the multi-sector economy allows for reallocation of wealth across sectors and endogenous changes in relative prices, which jointly shape the path of aggregate outcomes.

For the following exercises, I use a baseline parameterization summarized in Appendix J. Unless otherwise noted, parameters are common across sectors and chosen to generate a symmetric benchmark economy. Appendix I describes the numerical procedure used, including the fixed-point solution for equilibrium prices and the finite-difference approximation used to update the wealth-share distributions.

3.4.1 Benchmark: One-Sector Economy

I begin with a one-sector version of the model, which serves as a benchmark for the multisector analysis. In this environment, all entrepreneurs operate in the same sector and face the same production technology. Aggregate dynamics therefore reflect only within-sector heterogeneity, financial frictions, and the evolution of the wealth-weighted productivity distribution. There is no sectoral margin, so the model abstracts from changes in specialization.



(a) Initial periods

(b) Selected dates

Figure 2: One-sector wealth distribution dynamics.

The figure shows the transition of the wealth-weighted productivity distribution. The one-sector economy isolates the standard within-sector selection mechanism.

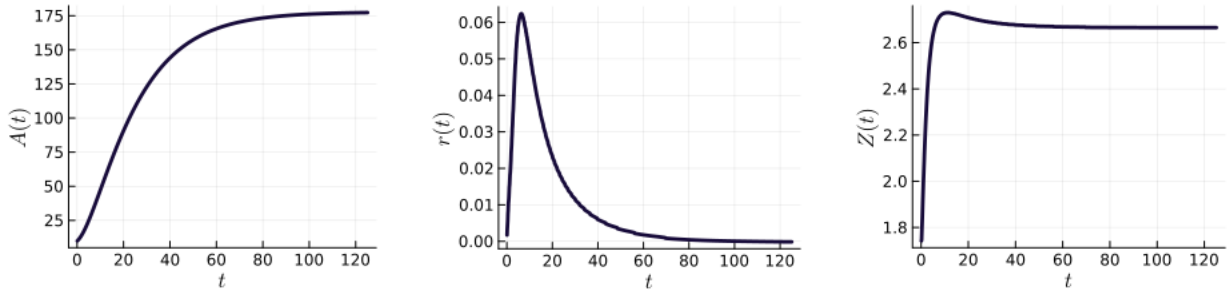
Figure 2 illustrates the central selection mechanism inherited from Moll (2014). Entrepreneurs with higher productivity earn higher returns, save more, and gradually account for a larger share of aggregate wealth. Panel (a) shows this movement over the early transition: the wealth-weighted distribution shifts away from low productivity states and builds mass at higher productivity levels.

Panel (b) shows the same process through selected cross-sections. The mass initially concentrated at low productivity declines, while the right side of the distribution becomes more important over time.

This within-sector reallocation affects aggregate outcomes. As wealth becomes increasingly concentrated among more productive entrepreneurs, aggregate productivity rises. At the same time, aggregate wealth accumulation reduces capital scarcity, which lowers the equilibrium interest rate and weakens the initial force of selection.

Figure 3 summarizes the corresponding aggregate dynamics. Aggregate wealth rises throughout the transition as entrepreneurs save and accumulate assets. The interest rate is high early in the transition, reflecting the high return to scarce capital, but declines as aggregate wealth increases. Aggregate productivity rises sharply at first because the economy reallocates wealth toward more productive entrepreneurs; over time, this process slows as the wealth distribution approaches its stationary shape.

The one-sector benchmark therefore captures the standard self-financing and selection mechanism: productive entrepreneurs gradually accumulate wealth, capital scarcity falls, and aggregate productivity improves. However, because there is only one sector, the benchmark has no notion of sectoral composition or specialization. The multisector economy adds this missing margin by allowing the same wealth-accumulation forces to operate both within sectors and across sectors through relative prices.



(a) Aggregate wealth (b) Interest rate (c) Aggregate productivity

Figure 3: One-sector aggregate dynamics.

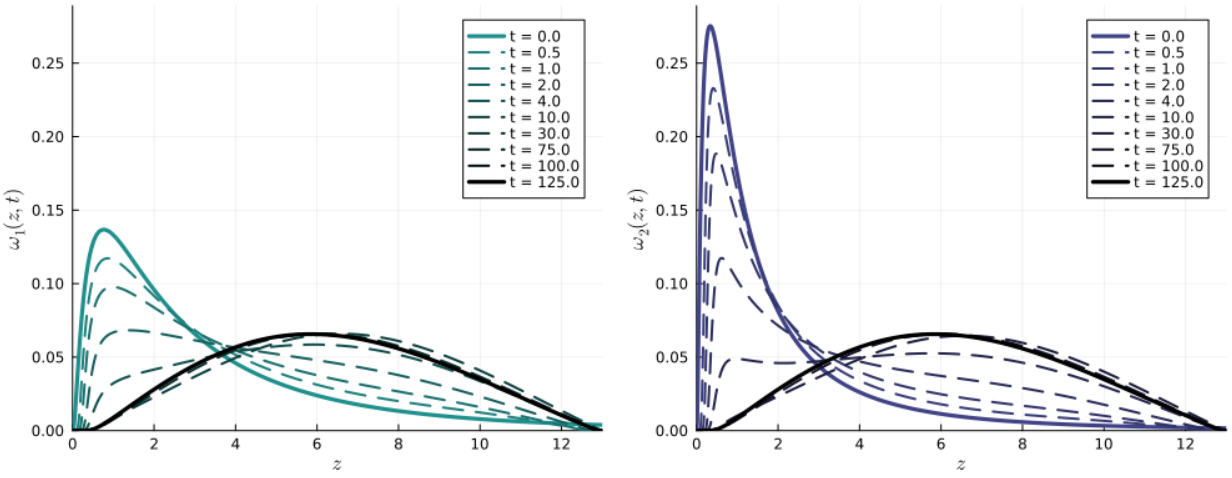
The figure shows the transition of aggregate wealth, the interest rate, and aggregate productivity. Early in the transition, the high return to capital reflects strong selection and reallocation toward productive entrepreneurs. As wealth accumulates, capital scarcity declines, the interest rate falls, and the transition becomes increasingly driven by aggregate accumulation.

3.4.2 Two-Sector Economy (Within- and Across-Sector Dynamics)

I now extend the analysis to a multisector economy with $i \in \{1, 2\}$, allowing for differences in initial wealth shares and productivity distributions across sectors. In particular, the initial correlation of

log-wealth and log-productivity in sector 1 is initialized at $\rho(a,z)_{1,0} = 0.8$ and in sector 2 it is initialized at $\rho(a,z)_{2,0} = 0.2$. This introduces a new margin of adjustment, as resources can be reallocated both within and across sectors, and relative prices adjust endogenously to clear goods markets.

The comparison with the one-sector benchmark illustrates how sectoral structure modifies the dynamics of wealth accumulation, selection, and aggregate productivity.



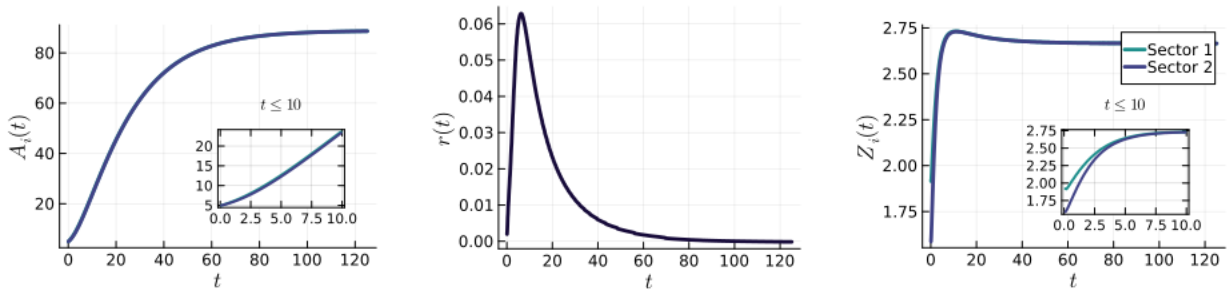
(a) Sector 1 (b) Sector 2

Figure 4: Total wealth sectoral shares distribution dynamics with two sectors.

The figure shows the transition of the sector-specific wealth-weighted productivity distributions in the two-sector closed economy. Differences in initial wealth-productivity alignment generate different short-run selection dynamics across sectors, but closed-economy price adjustment limits the persistence of these differences.

Within-sector selection. Figure 4 shows the evolution of the wealth-weighted productivity distributions $\omega_i(z,t)$ within each sector. As in the one-sector benchmark, more productive firms accumulate wealth and expand, leading to a gradual shift of mass toward higher productivity levels. However, the strength of this selection process differs across sectors, reflecting differences in initial conditions and equilibrium prices.

Sectoral aggregate dynamics The two-sector economy shows that the same selection forces present in the one-sector benchmark now operate alongside sectoral reallocation, which can be seen in Figure 5. At the aggregate level, the interest rate follows a familiar pattern: it rises sharply at the beginning, when productive entrepreneurs are expanding and capital is scarce, and then falls as wealth accumulates. Panel (c) shows that sectoral productivity increases rapidly in both sectors. The early gap between sectoral productivities is small and short-lived, reflecting the fact that closed-economy price adjustment compresses initial asymmetries. Overall, the figure illustrates that initial wealth-



(a) Aggregate wealth (b) Interest rate (c) Aggregate productivity

Figure 5: Two-sector aggregate dynamics.

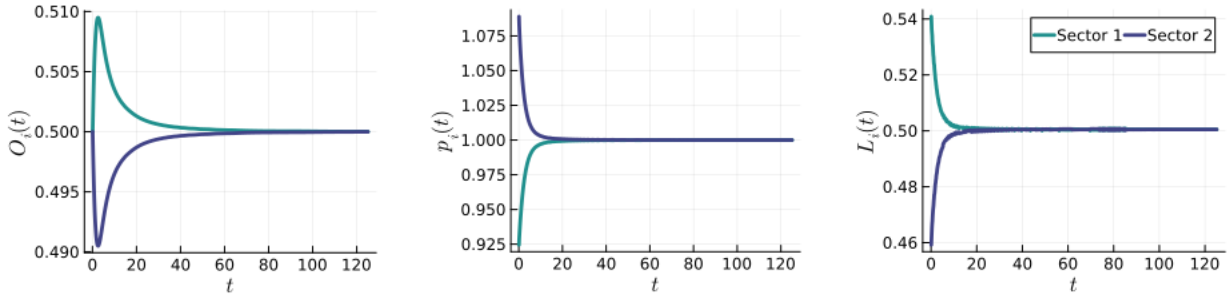
The figure shows transition dynamics in the two-sector closed economy. Panel (a) reports sectoral wealth levels $A_i(t)$, panel (b) reports the common interest rate $r(t)$, and panel (c) reports sectoral productivity $Z_i(t)$. The insets in panels (a) and (c) zooms in on the first ten periods to show the small early differences, especially between sectoral productivity paths, which are hard to see in the full-horizon plot. Colors identify sectors.

productivity differences matter for short-run sectoral dynamics, but general equilibrium forces limit their persistence in the closed economy.

Across-sector reallocation and prices. Figure 6 reports the evolution of sectoral wealth shares $O_i(t)$, output shares $Y_i(t)/Y(t)$, and relative prices $p_i(t)/p_j(t)$. Starting from asymmetric initial conditions, sectors with higher effective productivity attract a larger share of wealth and expand over time.

At the same time, this expansion induces general equilibrium effects through relative prices: as a sector grows, its relative price declines, reducing its profitability and diminishing further accumulation. This price adjustment mechanism partially offsets initial differences across sectors and slows down the reallocation of resources. Appendix E shows that the same force also limits the contribution of between-sector reallocation to aggregate productivity growth in the closed economy.

Taken together, these results show that sectoral structure introduces a reallocation margin that is absent in the one-sector benchmark, but that in a closed economy this margin is self-limiting: endogenous price adjustments act as a correction mechanism, reducing the persistence of initial asymmetries.



(a) Sectoral wealth shares (b) Sectoral prices (c) Labor composition

Figure 6: Across-sector reallocation.

The figure shows the transition dynamics of a two-sector closed economy with asymmetric initial wealth–productivity distributions. Panel (a) reports sectoral wealth shares $O_i(t)$, panel (b) sectoral prices $p_i(t)$, and panel (c) labor allocations $L_i(t)$. The sector with a more favorable initial allocation expands on impact, but general equilibrium price adjustments reduce differences in profitability, leading to convergence in allocations and selection thresholds over time.

4 Small Open Economy

The previous section characterized a closed economy in which relative prices are endogenously determined by domestic conditions. In that environment, sectoral dynamics are shaped jointly by wealth accumulation, financial frictions, and the endogenous adjustment of prices.

This section introduces trade through a small open economy with financial autarky. Goods prices are now taken as given from world markets, while factor prices and allocations remain determined domestically. This isolates the role of trade in shaping the interaction between wealth dynamics and sectoral allocation.

The analysis focuses on how exposure to international prices alters the evolution of sectoral productivity, wealth distributions, and aggregate outcomes. In particular, fixing relative prices eliminates a key adjustment margin present in the closed economy, modifying the feedback between sectoral profitability, savings, and capital allocation. Appendix K develops a simplified analytical benchmark illustrating how fixed world prices can sustain persistent specialization dynamics in a small open economy environment.

4.1 Setup and Equilibrium

The economy is populated by entrepreneurs facing the same production and financial frictions as in the closed economy. The key difference is that sectoral goods prices $\{p_i^*(t)\}_{i\in I}$ are exogenously given by world markets.

Definition 7 (Small open economy equilibrium with financial autarky)**.** Given initial wealth distributions $\{\omega_i(z,0)\}_{i\in I}$, wealth levels $\{A_i(0)\}_{i\in I}$, and a sequence of world prices $\{p_i^*(t)\}_{i\in I, t\geq 0}$, a

competitive small open economy equilibrium with financial autarky is a path of factor prices

$$\{w(t), r(t)\}_{t\geq 0},$$

allocation rules

$$\{k_i(z,t), \ell_i(z,t), c_i(z,t)\}_{i\in I, t\geq 0},$$

net exports

$$\{NX_i(t)\}_{i\in I, t\geq 0},$$

and wealth distributions

$$\{\omega_i(z,t)\}_{i\in I, t\geq 0},$$

such that:

1. Entrepreneurs optimize given world prices and factor prices.

2. Factor markets clear:

$$\bar{L} = A(t)\sum_{i\in I}\int \ell_i(z,t)\,\omega_i(z,t)\,dz,$$
$$A(t) = A(t)\sum_{i\in I}\int k_i(z,t)\,\omega_i(z,t)\,dz.$$

3. Goods markets satisfy

$$Y_i(t) = D_i(t) + NX_i(t), \qquad \forall i \in I,$$

   where domestic absorption satisfies the CES demand system

$$D_i(t) = \upsilon_i \left(\frac{p_i^*(t)}{P^*(t)}\right)^{-\eta} D(t),$$

   with price index

$$P^*(t)^{1-\eta} = \sum_{i\in I} \upsilon_i \left(p_i^*(t)\right)^{1-\eta}.$$

4. External balance holds:

$$\sum_{i\in I} p_i^*(t) NX_i(t) = 0.$$

5. Aggregate absorption satisfies

$$P^*(t)D(t) = P^*(t)C(t) + \dot{A}(t) + \delta A(t),$$

where aggregate consumption expenditure is

$$P^*(t)C(t) = w(t)\bar{L} + \rho A(t).$$

6. Wealth distributions evolve consistently with individual wealth accumulation and the productivity process.

4.2 Persistence Mechanisms

The quantitative exercises in this section should be read as mechanism illustrations rather than a calibration. The baseline parametrization is the same used in subsection 3.4 and summarized in Appendix J. I keep technologies, preferences, and aggregate resources fixed, and vary either the initial wealth-productivity distribution or the world-price regime. This isolates the channel emphasized by the model: when relative prices are fixed, initial differences in effective sectoral capital are not offset by domestic price adjustment and can therefore persist through wealth accumulation.

The results in this section reinterpret standard trade-theory forces in a setting where effective factor supplies are endogenous to wealth distributions. In the standard Rybczynski theorem, changes in factor endowments shift production patterns at fixed goods prices. Here, sectoral wealth and the within-sector allocation of wealth across productivity types play an analogous role: they determine how much effective capital a sector can deploy under collateral constraints. In addition, these effective endowments evolve through saving and wealth accumulation.

Proposition 8. *Consider a small open economy with financial autarky and exogenous goods prices* $\{p_i^*\}_{i\in I}$.

(i) Wealth effect. *Holding fixed the within-sector distribution* $o_i(z,t)$*, an increase in sectoral wealth* $A_i(t)$ *raises capital employed, labor demand, and output in sector* i*:*

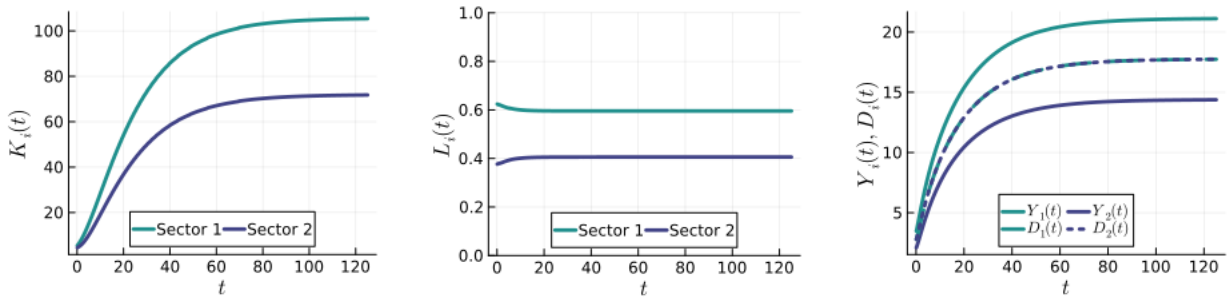
$$\frac{\partial K_i(t)}{\partial A_i(t)} > 0, \qquad \frac{\partial L_i(t)}{\partial A_i(t)} > 0, \qquad \frac{\partial Y_i(t)}{\partial A_i(t)} > 0.$$

(ii) Allocation effect. *Holding fixed sectoral wealth* $A_i(t)$*, a shift in* $o_i(z,t)$ *that raises sectoral productivity* $Z_i(t)$ *increases effective capital use and output in sector* i*:*

$$\frac{\partial K_i(t)}{\partial Z_i(t)} > 0, \qquad \frac{\partial Y_i(t)}{\partial Z_i(t)} > 0.$$

At given world prices, sectoral wealth and within-sector allocation both operate as sources of effective factor endowments. Higher wealth allows entrepreneurs to scale up production under collateral constraints, while a more favorable allocation toward high-productivity entrepreneurs raises the efficiency with which capital is used. As a result, sectors with larger wealth stocks or

better within-sector allocation expand relative to others, reallocating labor and capital across sectors. The model therefore delivers a distributional analogue of the Rybczynski theorem, in which changes in wealth and allocation play the role of factor endowments in shaping production patterns.



(a) Capital (b) Labor (c) Output and Absorption

Figure 7: Sectoral allocation dynamics at given world prices.

The figure shows the transition dynamics of sectoral allocations in a small open economy with fixed world prices. Panel (a) reports capital by sector $K_i(t)$, panel (b) labor allocations $L_i(t)$, and panel (c) output $Y_i(t)$ together with sectoral absorption $D_i(t)$ (dashed lines). Because goods prices are exogenous, differences in initial wealth–productivity distributions translate into persistent differences in capital, labor, and output across sectors. In contrast to the closed economy, the absence of relative price adjustment prevents reallocation forces from equalizing sectoral outcomes over time.

Proposition 9 (Dynamic Rybczynski effect through wealth accumulation)**.** *Consider a small open economy with financial autarky and exogenous goods prices* $\{p_i^*\}_{i\in I}$*, and let sectoral wealth shares be* $O_i(t) = A_i(t)/A(t)$*.*

Suppose that sector i *has either:*

1. *a higher initial wealth level* $A_i(0)$*, holding fixed* $o_i(z, 0)$*; or*
2. *a more favorable initial within-sector allocation* $o_i(z, 0)$ *that raises measured productivity* $Z_i(0)$*.*

Then, at given world prices, sector i *is larger on impact:*

$$K_i(0) \uparrow, \qquad L_i(0) \uparrow, \qquad Y_i(0) \uparrow .$$

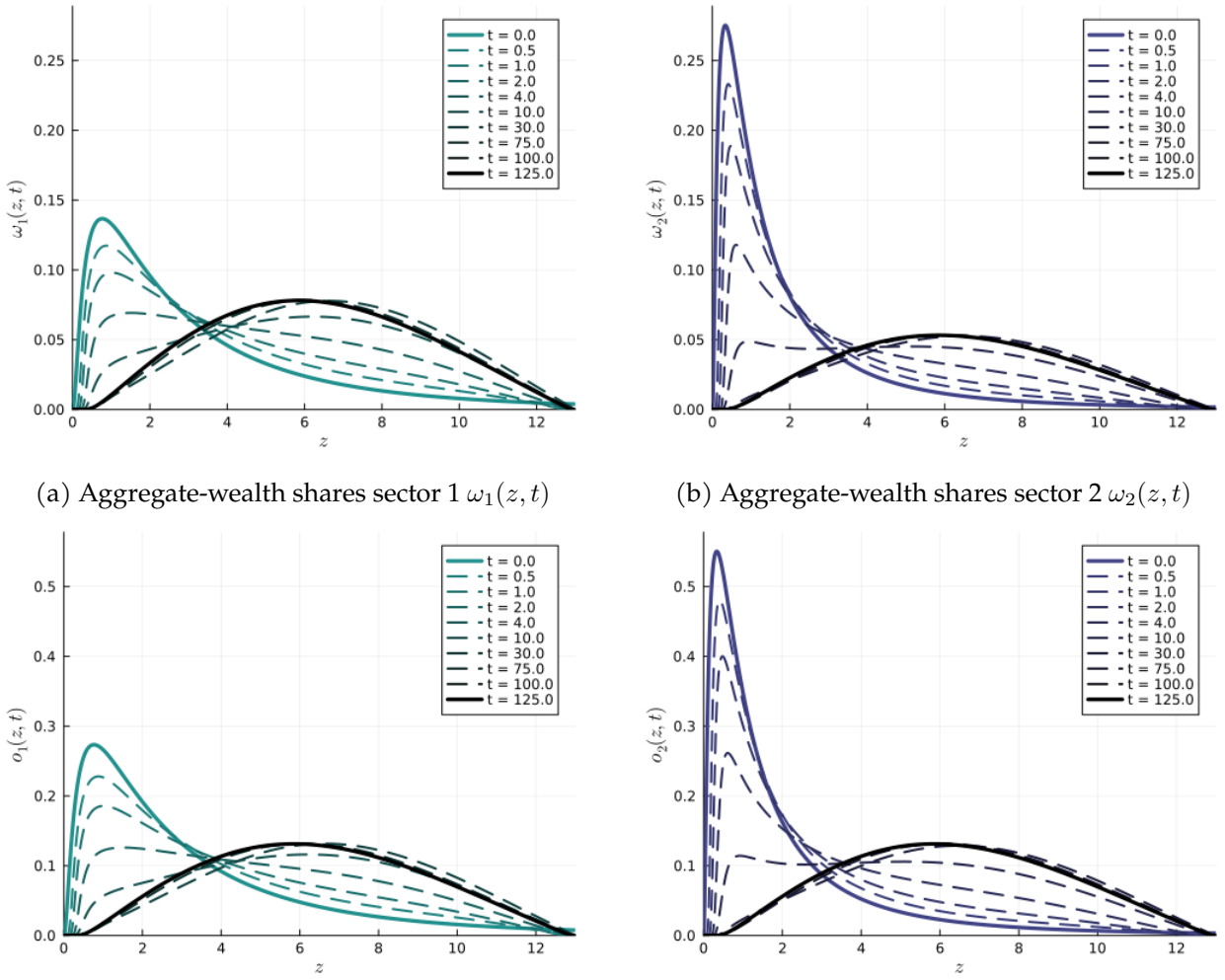
If these initial conditions imply a higher average net saving rate in sector i *relative to the aggregate, then*

$$\frac{d}{dt} \log O_i(t) > 0$$

locally in time.

Consequently, sector i *accumulates wealth faster along the transition dynamics, increasing its relative size over time. In a small open economy, fixed world prices prevent the relative-price adjustment that would otherwise reduce these initial asymmetries, allowing them to persist in the long run.*

Initial advantages in wealth or within-sector allocation translate into higher profitability and higher saving rates when goods prices are fixed. Because entrepreneurs face collateral constraints, higher wealth allows productive agents to expand, reinforcing the initial advantage through accumulation. As a result, sectors that are initially larger or better allocated grow faster and continue to attract resources over time. This generates a dynamic amplification mechanism: differences in initial conditions are not corrected, but instead propagate through wealth accumulation, shaping specialization patterns over the entire transition. Appendix K presents a stripped-down analytical benchmark showing how fixed world prices can generate persistent specialization through sector-specific capital accumulation.



(a) Aggregate-wealth shares sector 1 $\omega_1(z,t)$

(b) Aggregate-wealth shares sector 2 $\omega_2(z,t)$

(c) Within-sector-wealth shares sector 1 $o_1(z,t)$

(d) Within-sector-wealth shares sector 2 $o_2(z,t)$

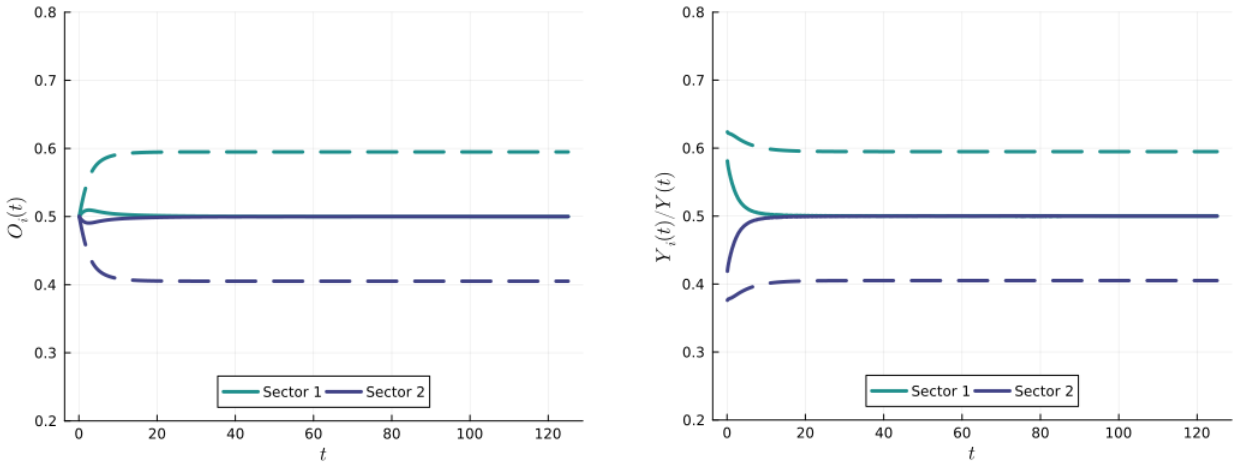
Figure 8: Wealth distribution dynamics at fixed world prices.

The figure shows the evolution of wealth distributions in a two-sector small open economy with fixed world prices. Panels (a) and (b) report the aggregate wealth distributions $\omega_i(z,t)$, which combine sectoral size and within-sector allocation, while panels (c) and (d) report the within-sector distributions $o_i(z,t)$, isolating selection effects.

Wealth distribution dynamics As represented in Figure 8, within each sector, the distribution shifts toward higher productivity types over time, reflecting selection driven by differential returns and savings behavior. However, because goods prices are exogenous, these within-sector forces do not translate into full reallocation across sectors. As a result, differences in initial wealth–productivity distributions generate persistent gaps in sectoral size, visible in the divergence of $\omega_1(z,t)$ and $\omega_2(z,t)$, even as the within-sector distributions $o_i(z,t)$ converge to a common stationary distribution. This illustrates how, in a small open economy, the absence of relative price adjustment limits the equalizing effects of reallocation and sustains long-run differences across sectors.

4.3 Comparison with Closed Economy

To isolate the role of price adjustment, I compare the transition dynamics in the closed economy and the small open economy starting from identical initial wealth–productivity distributions. This comparison holds fixed fundamentals and initial conditions, so differences in outcomes reflect only the behavior of prices across regimes. In the closed economy, relative prices respond to sectoral expansion and reallocation, reducing initial asymmetries. In the small open economy, prices are fixed, removing this adjustment margin. As a result, the two environments generate markedly different dynamics, with convergence in the closed economy and persistent differences in sectoral allocation in the open economy.



(a) Wealth shares: closed vs SOE

(b) Output shares: closed vs SOE

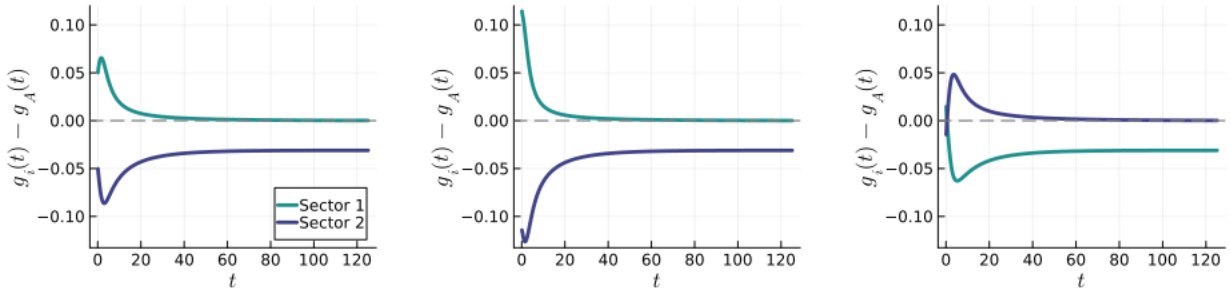
Figure 9: Adjustment and persistence across regimes.

The figure compares sectoral allocation dynamics in a closed economy and a small open economy initialized from the same wealth–productivity distributions. Panel (a) reports sectoral wealth shares $O_i(t)$, and panel (b) reports output shares $Y_i(t)/Y(t)$. Colors identify sectors. Solid lines correspond to the closed economy, while dashed lines correspond to the small open economy.

4.4 Price shocks and endogenous specialization

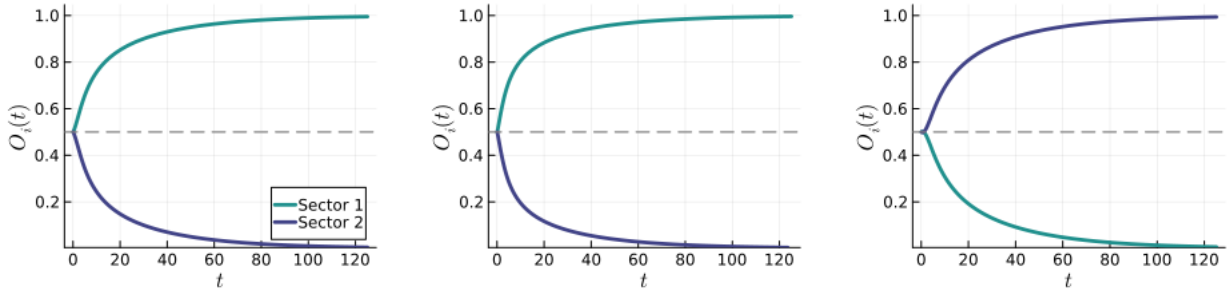
Having shown that fixed prices generate persistence by removing the adjustment margin present in the closed economy, I now study how exogenous price changes affect sectoral dynamics in the open economy. In this environment, world prices act as external drivers of profitability, shifting incentives for accumulation across sectors. Because capital allocation is mediated by wealth and financial frictions, these price changes translate into differential saving rates and, in turn, into changes in sectoral wealth shares over time. The response therefore depends not only on the direction of the price shock, but also on the underlying distribution of wealth within sectors. Corollary 10 formalizes this mechanism, and Figure 10 illustrates how price shocks generate dynamic reallocation and persistent changes in sectoral composition.

Panel A. Saving-rate gap



(a) Symmetric with $p_1^* \uparrow$ (b) Asymmetric with $p_1^* \uparrow$ (c) Asymmetric with $p_2^* \uparrow$

Panel B. Wealth shares



(d) Symmetric with $p_1^* \uparrow$ (e) Asymmetric with $p_1^* \uparrow$ (f) Asymmetric with $p_2^* \uparrow$

Figure 10: Dynamic response to world price shocks.

The figure reports the response of a small open economy to permanent changes in world prices. Panel A plots sectoral saving-rate gaps $g_i(t) - g_A(t)$, and Panel B reports sectoral wealth shares $O_i(t)$. Colors identify sectors. Columns correspond to symmetric and asymmetric environments, with price increases in sector 1 (columns (a)–(b)) and sector 2 (column (c)). In each case, the sector facing the higher price exhibits a positive saving-rate gap, while the other exhibits a negative gap, with corresponding changes in wealth shares.

Corollary 10 (Dynamic price accumulation channel)**.** *Consider a small open economy with financial autarky and exogenous goods prices $\{p_i^*\}_{i\in I}$. Let sectoral wealth shares evolve according to*

$$\frac{d}{dt}\log O_i(t) = g_i(t) - g_A(t),$$

where $g_i(t) = \int s_i(z,t)\, o_i(z,t)\, dz$ is the average net saving rate in sector i, and $g_A(t)$ is the aggregate counterpart.

A permanent increase in the world price of good i, $p_i^* \uparrow$, that raises sectoral profitability and the associated saving rate implies

$$\frac{\partial}{\partial p_i^*}\left(\frac{d}{dt}\log O_i(t)\right) > 0.$$

Moreover, the magnitude of this response depends on the within-sector distribution $o_i(z,t)$: for a given price change, a higher concentration of wealth among high-productivity entrepreneurs increases $g_i(t)$ and strengthens the response of $O_i(t)$.

Interpretation. Changes in goods prices affect sectoral dynamics through their impact on profitability and saving behavior. A higher price for a given sector increases the returns to capital in that sector, raising its average saving rate relative to the aggregate. This generates a positive growth differential in wealth shares, leading the sector to expand over time. Because saving rates depend on the distribution of wealth across entrepreneurs, the response is amplified when wealth is concentrated among more productive agents. As a result, price changes do not only affect contemporaneous allocation, but also the evolution of wealth and the path of specialization through time.

4.5 What does trade do in this model?

Trade changes the role of prices in the adjustment process. In the closed economy, sectoral expansion feeds back into relative prices, which move against the expanding sector and reduce its profitability. This price adjustment limits the persistence of initial differences in wealth and productivity by diminishing the incentives for further accumulation. In contrast, in the small open economy, goods prices are fixed at world levels. As a result, sectoral profitability does not adjust in response to domestic reallocation, and initial differences in wealth and productivity translate directly into differences in growth rates and sectoral expansion.

Because capital allocation is mediated by balance sheets, trade does not eliminate distortions. Instead, it reshapes them. Sectors with initially higher wealth or more favorable within-sector allocation experience higher returns and accumulate wealth faster, leading to persistent differences in specialization. Financial frictions therefore interact with trade by amplifying the effects of initial conditions. While trade integration reallocates production toward sectors with higher world prices, the extent of this reallocation depends on the distribution of wealth, so economies with similar fundamentals but different initial distributions can follow different specialization paths over time.

5 Quantitative Application and Empirical Evidence

The divergence in sectoral specialization documented in the introduction points to the importance of dynamic forces shaping comparative advantage. This section first presents a preliminary quantitative application and then provides firm- and sector-level evidence on the mechanism. The analysis focuses on persistent productivity dispersion, heterogeneous firm expansion, and sectoral responses to external demand shocks.

5.1 Quantitative Application

This section brings the model to a seven-sector representation of the Peruvian economy. The quantitative exercise combines firm-level information on productivity, assets, and wealth–productivity alignment with sectoral moments on absorption, production, employment, and trade. The objective is to assess whether empirically informed initial wealth distributions can generate persistent differences in sectoral composition through the accumulation mechanism emphasized in the theoretical analysis.

Relative to the baseline model, the quantitative environment introduces sector-specific factors, a hybrid price regime, and a resource-export residual.

Plants remain constant returns to scale, but fixed sector-specific inputs generate decreasing returns to sectoral expansion. Tradable-sector prices are determined by exogenous international conditions, while nontradable prices adjust to clear domestic markets. Net exports from the omitted extractive sector finance external imbalances among the modeled sectors.

The current exercise is a preliminary implementation rather than a completed calibration. It reproduces sectoral absorption shares and incorporates empirically informed initial conditions, but sector-specific factor endowments and tradable prices remain provisionally assigned. The results nevertheless show substantial early reallocation followed by persistent differences in sectoral wealth and employment. The following subsections describe the quantitative environment, evaluate the initial fit, and present the resulting transition dynamics.

5.1.1 Environment

The quantitative implementation extends the baseline model along two dimensions. First, it introduces sector-specific factors, which preserve constant returns to scale at the firm level while generating decreasing returns to sectoral expansion. Second, it adopts a hybrid price regime in which tradable-goods prices are determined by international conditions, whereas nontradable-goods prices adjust to clear domestic markets. Details of this model version are presented in Appendix H; extensions with sectoral learning or scale externalities provide a natural way to explore the opposite case in which expansion raises, rather than lowers, future sectoral profitability. Appendix J.3 documents the data sources, parameter mappings, normalizations, initial conditions,

and sectoral moments used in this first implementation, together with the remaining calibration targets and validation moments.

The economy contains seven sectors: agroindustry, commerce, construction, fishing, machinery and transport equipment, other manufacturing, and market services. Firms in sector i produce according to

$$y_{fi} = (z_{fi}k_{fi})^{\alpha_i}\ell_{fi}^{\beta_i}h_{fi}^{\gamma_i}, \qquad \alpha_i + \beta_i + \gamma_i = 1, \tag{22}$$

where h_{fi} denotes the use of a factor specific to sector i. The aggregate supply of this factor is fixed at $\bar{H}_i$. Conditional on prices, optimal labor and specific-factor demands remain proportional to $z_{fi}k_{fi}$. Because active entrepreneurs continue to choose $k_{fi} = \lambda_i a_{fi}$, firm-level input demands and profits remain linear in wealth. The wealth-weighted productivity distribution therefore remains sufficient to characterize sectoral selection, production, and accumulation.

Aggregation yields sectoral production of the form

$$Y_i(t) = Z_i(t)K_i(t)^{\alpha_i}L_i(t)^{\beta_i}\bar{H}_i^{\gamma_i}, \tag{23}$$

where

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}\left[z \mid z \geq \bar{z}_i(t)\right]\right)^{\alpha_i}. \tag{24}$$

Thus, the within-sector allocation of wealth across productivity states continues to determine effective sectoral productivity. At the same time, because $\bar{H}_i$ is fixed, production is decreasing returns to scale in the mobile factors:

$$Y_i(t) = \widetilde{Z}_i(t)K_i(t)^{\alpha_i}L_i(t)^{\beta_i}, \qquad \alpha_i + \beta_i = 1 - \gamma_i < 1.$$

As a sector expands, the rental price of its specific factor rises, reducing profitability and limiting the tendency toward corner specialization. This force provides a conservative discipline on the magnitude of the wealth-accumulation mechanism without changing its central state variables.

Let $\mathcal{I}^T$ and $\mathcal{I}^N$ denote the sets of tradable and nontradable sectors, respectively. Agroindustry, fishing, machinery and transport equipment, and other manufacturing are treated as tradable, while commerce, construction, and market services are nontradable. Sectoral prices satisfy

$$p_i(t) = p_i^*(t), \qquad i \in \mathcal{I}^T, \tag{25}$$

where $p_i^*(t)$ is an exogenous path of international prices. For $i \in \mathcal{I}^N$, $p_i(t)$ adjusts endogenously so that domestic output equals domestic absorption. This structure allows international prices to shape the profitability and accumulation of tradable sectors while preserving domestic relative-price adjustment in nontradable activities.

The seven modeled sectors exclude the extractive resource sector that accounts for an important share of Peruvian exports. I represent this sector through a resource-export residual, $NX_{\text{res},t}$, and

impose aggregate external balance:

$$\sum_{i \in \mathcal{I}^T} p_i(t) NX_i(t) + NX_{\text{res},t} = 0. \tag{26}$$

The residual allows net resource exports to finance trade deficits in the modeled tradable sectors without introducing an additional entrepreneurial sector and wealth distribution.

The current exercise is a preliminary implementation rather than a completed calibration. Sector-specific factor endowments are set proportional to observed employment shares, tradable-sector prices are initially normalized to $p_i^* = 1$, the collateral parameter is set to the common value $\lambda_i = 1.10$, and the elasticity of substitution is provisionally set to $\eta = 0.99$. The comparison below uses this parameterization to evaluate whether the richer environment generates economically meaningful sectoral dynamics and to identify the main remaining calibration margins.

5.1.2 Initial Fit and Calibration

Table 1: Initial target-versus-model comparison

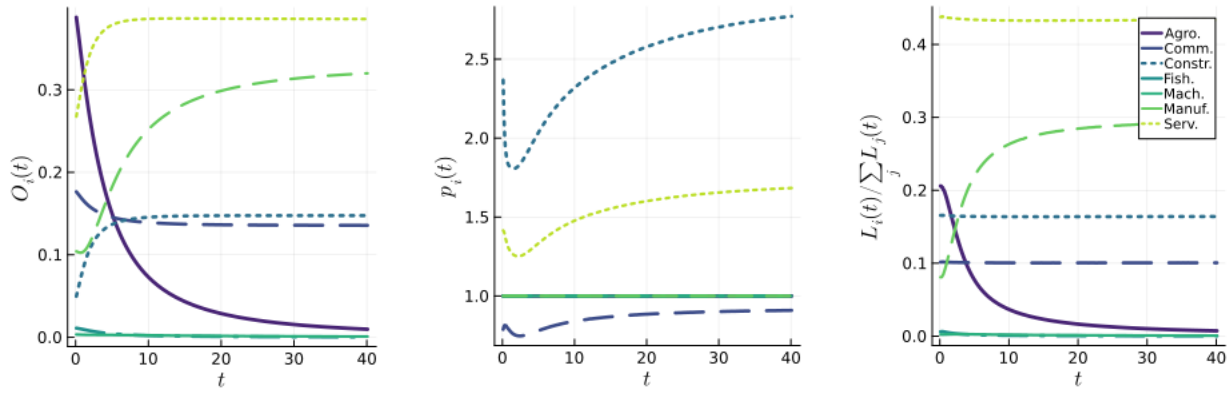
	Labor share		Absorption share		Output share		Trade balance	
Sector	Data	Model	Data	Model	Data	Model	Data	Model
Agroindustry	0.3266	0.2059	0.0303	0.0302	0.0629	0.2148	-0.0034	0.1829
Commerce	0.1844	0.1016	0.1219	0.1213	0.1124	0.1278	–	–
Construction	0.0491	0.1656	0.1592	0.1602	0.0886	0.1687	–	–
Fishing	0.0077	0.0062	0.0027	0.0027	0.0097	0.0078	0.0000	0.0050
Machinery and transport	0.0038	0.0024	0.0894	0.0892	0.0145	0.0023	-0.0503	-0.0917
Other manufacturing	0.1200	0.0808	0.2188	0.2183	0.3334	0.0804	-0.0021	-0.1497
Market services	0.3084	0.4375	0.3777	0.3781	0.3784	0.3982	–	–

Notes: The table compares sectoral moments in the data with those generated by the preliminary quantitative implementation. The CES expenditure weights are chosen to reproduce sectoral absorption shares. Sector-specific factor endowments and tradable-goods prices remain provisionally assigned rather than calibrated to employment shares and sectoral trade balances.

Table 1 reports the initial fit of the quantitative implementation. Sectoral absorption shares are reproduced closely, largely by construction, because the CES expenditure weights are disciplined by the input–output data. The remaining moments reveal the calibration margins that still need to be disciplined. Construction and market services employ too much labor, while agroindustry and manufacturing employ too little. These differences are also reflected in the sectoral composition of output.

The trade-balance discrepancies partly reflect the provisional assumption of equal prices across tradable sectors. Under this parameterization, the model generates excessive net exports in agroindustry and excessive trade deficits in machinery and other manufacturing. The next calibration step therefore chooses sector-specific factor endowments to discipline employment shares and tradable prices to discipline sectoral trade balances.

5.1.3 Transition Dynamics



(a) Sectoral wealth shares (b) Sectoral prices (c) Labor composition

Figure 11: Transition dynamics in the preliminary quantitative implementation

Notes: The figure reports transition paths from the empirically informed initial sectoral wealth allocation under the preliminary parameterization. Panel (a) reports each sector's share of aggregate entrepreneurial wealth. Panel (b) reports sectoral goods prices; tradable prices are determined by exogenous international prices, while nontradable prices adjust to clear domestic goods markets. Panel (c) reports sectoral shares of total employment. Sector-specific factor endowments and tradable prices remain provisionally assigned rather than fully calibrated.

Figure 11 illustrates the transition generated by the preliminary quantitative implementation. Most of the sectoral reallocation occurs early in the simulation. Entrepreneurial wealth moves sharply away from agroindustry and toward manufacturing, construction, and market services. After this initial adjustment, the economy approaches a path with persistent differences in sectoral wealth and employment shares.

The price responses reflect the hybrid structure of the economy. Prices remain externally determined in tradable sectors, whereas prices in nontradable sectors adjust in response to domestic scarcity. In particular, the prices of construction and market services rise as demand for these sectors increases relative to their effective productive capacity. These endogenous price responses limit the extent of sectoral expansion but do not undo the persistent reallocation generated by wealth accumulation.

Sectoral wealth and employment shares do not move one-for-one. Entrepreneurs can lend and borrow through the common capital market, while fixed sector-specific inputs constrain the expansion of individual sectors. Consequently, changes in the location of entrepreneurial wealth affect employment through equilibrium wages, goods prices, capital allocation, and sector-specific-factor rents. Although the exercise is not yet fully calibrated, it shows that initial differences in wealth allocation can generate large early reallocations and persistent differences in sectoral composition in an economy with nontradables and sector-specific factors.

5.2 Data

The empirical analysis combines firm-level survey data from Peru with international trade data from BACI International Trade Database (Gaulier & Zignago, 2010). The analysis uses firm-level data from the Annual Economic Survey (*Encuesta Económica Anual*, EEA), conducted by Instituto Nacional de Estadística e Informática (INEI) (2001-2019). The survey is administered annually for national accounts and sectoral analysis and is mandatory for all firms above a minimum sales threshold, effectively covering the universe of medium and large formal firms. The data include detailed balance-sheet and income-statement information, such as sales, value added, costs, exports, employment, wages, and capital stocks. Coverage spans all major sectors except mining and extractive industries, which report to a different administrative source.

International trade data are constructed using bilateral trade flows from BACI at the HS6-product level for the period 1995–2024. Products are mapped to ISIC Revision 3 sectors using concordance tables, and sector-level measures of external demand, supply, trade exposure, and unit-value price proxies are constructed using trade flows excluding Peru.

The combined dataset allows sectoral trade shocks to be linked to firm-level dynamics in productivity, capital accumulation, and sectoral specialization patterns over time.

Figure 12 shows that firm-level productivity exhibits substantial dispersion within sectors. The distributions are wide and persistent over time, and their dispersion varies systematically across sectors. These patterns indicate that sectors are characterized by stable heterogeneity in firm-level productivity, providing scope for selection and reallocation dynamics.



(a): Empirical distr. of z^{α}

(b): TFPR weighted by capital

Figure 12: Distribution of firm-level productivity by sector.

For both panels, productivity is constructed as a residual $\alpha_i \log z_{fi}^* = \log(p_i y_{fi}) - \alpha_i \log k_{fi} - (1-\alpha_i) \log l_{fi}$. a) Calculated using product-capital elasticities variables by sector: $\alpha_{\text{Agroindustry}} = 0.452$, $\alpha_{\text{Fishing}} = 0.417$, $\alpha_{\text{Manufacturing}} = 0.343$, $\alpha_{\text{Services}} = 0.328$, estimated with IV. b) Trends show capital-weighted sample estimates of $\log(Z_i) = \log\left(\mathbb{E}[z_i(t)]^{\alpha_i}\right)$ (log-TFPR). Bands indicate 95% bootstrapped confidence intervals.

Table 2 shows that more productive firms expand faster, but that this relationship varies substantially across sectors. In the baseline specification, lagged productivity is positively associated with both capital and labor growth, consistent with selection driven by higher returns and wealth accumulation.

Table 2: Productivity driven expansion differs across sectors.

	Capital growth		Labor growth	
	Coef.	SE	Coef.	SE
Baseline (*Ref.: Agroindustry*)				
Lagged productivity	0.081***	(0.004)	0.194***	(0.011)
Sector interactions with lagged productivity				
Commerce	-0.042***	(0.001)	-0.068***	(0.005)
Construction	-0.034***	(0.004)	0.026**	(0.007)
Electricity and services	-0.115***	(0.004)	-0.153***	(0.005)
Hydrocarbons and mining	-0.047***	(0.003)	-0.185***	(0.004)
Manufacturing	-0.030***	(0.002)	0.187***	(0.002)
Fishing	0.083***	(0.004)	0.099***	(0.007)
Services	-0.039***	(0.003)	-0.044***	(0.002)
Observations	22,942		22,942	
R-squared	0.032		0.132	

Firm-level regressions of capital and labor growth on lagged productivity and its interaction with sector fixed effects. All specifications include sector and year fixed effects. Standard errors clustered at the sector level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

However, the strength of this relationship differs across sectors. Relative to agroindustry, most sectors exhibit a weaker response of capital growth to productivity, while a few sectors display a stronger relationship. For labor growth, the heterogeneity is even more pronounced: some sectors show attenuated or even negative responses, while others, such as manufacturing and fishing, exhibit a strong positive relationship between productivity and expansion.

These patterns are consistent with a setting in which sectoral environments shape the extent to which productive firms can scale up. Differences in the responsiveness of growth to productivity suggest that the process of reallocation toward high-productivity firms is uneven across sectors. In the context of the model, this heterogeneity can be interpreted as reflecting differences in the interaction between wealth distributions and financial frictions, which affect the ability of productive firms to expand and accumulate capital over time.

5.3 Empirical Strategy

The model emphasizes a dynamic amplification mechanism through which sectoral shocks affect wealth accumulation, which in turn shapes future specialization patterns. In particular, sectors exposed to favorable price or demand shocks experience higher profitability and saving rates, al-

lowing productive firms to accumulate wealth and expand persistently over time.

This subsection outlines an empirical strategy to discipline this mechanism using sector-level trade shocks and firm-level balance sheet data.

5.3.1 Trade shocks and sectoral exposure indicators

The empirical strategy uses international trade data excluding Peru to construct measures of changes in external conditions across sectors. The main object of interest is a measure of external demand conditions that allows sectoral exposure to evolve over time, consistent with the endogenous specialization dynamics emphasized in the model. I also construct complementary measures based on fixed exposure shares, international unit values, and world export growth.

The baseline measure used throughout subsection 5.4 captures changes in external demand faced by Peruvian sectors while allowing exposure shares to vary over time:

$$DynamicDemandShock_{i,t} = \sum_{d} \omega_{id,t-1} \left(\log M_{d,i,t}^{World \setminus Peru} - \log M_{d,i,t-1}^{World \setminus Peru} \right),$$

where $\omega_{id,t-1}$ denotes lagged export exposure of sector i to destination market d, and $M_{d,i,t}^{World \setminus Peru}$ denotes imports by destination d in sector i from the rest of the world excluding Peru.

This specification captures changes in external demand conditions weighted by the evolving export composition of Peruvian sectors. In the model, sectoral exposure changes endogenously through wealth accumulation and specialization dynamics: sectors experiencing favorable profitability conditions gradually expand and become more exposed to growing destination markets. The dynamic exposure measure therefore incorporates both changes in foreign demand and endogenous amplification through evolving specialization patterns.

For comparison, I also construct a conventional shift-share measure using fixed initial exposure shares:

$$DemandShock_{i,t} = \sum_{d} \omega_{id,0} \left(\log M_{d,i,t}^{World \setminus Peru} - \log M_{d,i,t-1}^{World \setminus Peru} \right),$$

where $\omega_{id,0}$ denotes initial destination exposure shares. This specification isolates variation arising from changes in foreign import demand while abstracting from contemporaneous changes in Peru's export composition.[4]

To proxy for changes in international prices, I additionally construct sector-level unit-value shocks. For each HS6 product h, define the foreign unit value as

$$uv_{h,t}^{World \setminus Peru} = \frac{\sum_{o \neq Peru} \sum_{d} v_{o,d,h,t}}{\sum_{o \neq Peru} \sum_{d} q_{o,d,h,t}},$$

[4] A useful special case is the expansion in Chinese import demand following China's accession to the WTO, which can be interpreted as a large realization of the broader external-demand framework considered here.

where $v_{o,d,h,t}$ and $q_{o,d,h,t}$ denote the value and quantity of exports from origin o to destination d in product h and year t. Sector-level price changes are then constructed as

$$PriceShock_{i,t} = \sum_{h \in i} \omega_{ih,0} \left(\log uv_{h,t}^{World \setminus Peru} - \log uv_{h,t-1}^{World \setminus Peru} \right).$$

This measure captures changes in international unit values faced by Peruvian sectors. Since unit values may also reflect quality upgrading, compositional changes, and measurement error, I interpret this object as a proxy for external price conditions rather than a direct measure of prices.

Finally, I construct a measure of external supply conditions based on export growth in the rest of the world:

$$SupplyShock_{i,t} = \Delta \log X_{i,t}^{World \setminus Peru},$$

where $X_{i,t}^{World \setminus Peru}$ denotes exports by the rest of the world in sector i.

Together, these measures provide complementary indicators of changes in external conditions. Dynamic demand shocks capture evolving exposure to foreign demand growth, fixed-weight demand shocks isolate exogenous variation in destination demand, price shocks proxy for changes in international profitability conditions, and supply shocks capture changes in competitive pressure from the rest of the world.

5.3.2 Empirical Specifications and Mechanism Discipline

This subsection outlines a set of empirical specifications designed to connect the model's dynamic mechanism to observed firm- and sector-level responses in the data. The goal is not to estimate static treatment effects, but rather to evaluate whether external trade shocks generate persistent changes in accumulation, reallocation, and specialization patterns consistent with the mechanism emphasized in the model.

The specifications focus on three related objects: firm-level expansion following sectoral trade shocks, the evolution of sectoral shares over time, and the persistence of specialization patterns through endogenous wealth accumulation. A central prediction of the model is that sectors exposed to favorable external conditions should experience stronger and more persistent expansion when productive firms are able to accumulate wealth and scale up under financial frictions.

Firm expansion and external demand shocks. The model predicts that favorable external demand conditions raise sectoral profitability and affect firm expansion heterogeneously within sectors. Because firms face collateral constraints, more productive firms are better positioned to translate improved external demand conditions into subsequent capital and labor expansion.

To evaluate this mechanism, I combine firm-level panel data from the Peruvian *Encuesta Económica Anual* (EEA) with sector-level measures of external demand shocks constructed from international

trade data. The empirical specification relates one-period firm growth to lagged sectoral shocks and lagged firm productivity:

$$\Delta \log x_{f,i,t} = \beta_\ell Shock_{i,t-\ell} + \gamma_\ell z_{f,i,t-1} + \delta_\ell Shock_{i,t-\ell} \times z_{f,i,t-1} + \mu_i + \lambda_t + \varepsilon_{f,i,t},$$

where $\Delta \log x_{f,i,t}$ denotes one-period firm growth in capital or employment, $Shock_{i,t-\ell}$ is the sector-level external demand shock lagged by ℓ years, and $z_{f,i,t-1}$ is lagged firm productivity. I estimate the specification separately for lags $\ell = 1, \ldots, 5$. All specifications include sector and year fixed effects, and standard errors are clustered at the sector level.

The interaction term captures a central implication of the model: favorable external demand conditions should disproportionately affect productive firms, which are better able to expand under financial constraints. The estimates therefore test whether productive firms in sectors exposed to stronger past external demand shocks experience faster subsequent growth in capital and labor.

Persistence and sectoral specialization. A central prediction of the model is that trade shocks affect not only contemporaneous allocation, but also future specialization patterns through endogenous accumulation dynamics. To evaluate the persistence of sectoral responses over time, I estimate local-projection specifications relating future changes in sectoral shares to current external trade shocks.

Specifically, for each horizon $h \geq 0$, consider regressions of the form

$$\Delta_h \log O_{i,t+h} = \beta_h Shock_{i,t} + \phi_h \Delta \log A_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t+h},$$

where

$$\Delta_h \log O_{i,t+h} = \log O_{i,t+h} - \log O_{i,t},$$

$O_{i,t}$ denotes sectoral output, export, or employment shares, and $A_{i,t}$ measures accumulated sectoral assets or capital.

The sequence of coefficients $\{\beta_h\}_{h \geq 0}$ traces the dynamic response of sectoral specialization to external shocks over time. In the model, sectors receiving favorable trade shocks experience persistent increases in sectoral shares because higher profitability induces wealth accumulation and gradual expansion among productive firms.

More generally, the framework implies that temporary external shocks may generate long-lasting differences in specialization patterns through endogenous accumulation and reallocation dynamics. Comparing the persistence of estimated responses in the data and in simulated model economies therefore provides a direct way to discipline the mechanism emphasized in the model.

Discussion. Taken together, these specifications are designed to discipline the model's dynamic mechanism using both firm-level and sector-level evidence. The key object of interest is the persis-

tence of sectoral responses over time and their relationship with wealth accumulation, productivity, and reallocation.

More broadly, the empirical analysis seeks to evaluate whether external trade shocks generate persistent differences in specialization through balance-sheet dynamics, as predicted by the model.

5.4 Results

Table 3 reports firm-level regressions relating subsequent firm growth to lagged external demand shocks, lagged productivity, and their interaction. Across specifications, lagged productivity is strongly associated with subsequent capital and labor growth, consistent with persistent heterogeneity in firm expansion dynamics. More importantly, the interaction between external demand shocks and lagged productivity is positive in several specifications, indicating that productive firms tend to expand more strongly following favorable external demand conditions. This pattern is consistent with the mechanism emphasized in the model, in which productive firms are better able to translate improved profitability conditions into subsequent expansion under financial frictions.

The estimates also suggest that the effects of external demand conditions are heterogeneous across firms and evolve gradually over time. In several specifications, the direct effect of the shock is small or negative, while the interaction with productivity remains positive. This pattern is consistent with reallocation dynamics within sectors, where favorable external conditions disproportionately benefit productive firms rather than generating uniform expansion across all producers.

Figure 13 reports local-projection estimates of the dynamic response of sectoral employment shares to external demand shocks. The estimates display relatively flat pre-trends and positive post-shock responses over subsequent years, suggesting that favorable external demand conditions are associated with persistent changes in sectoral employment composition. Although the responses are gradual and somewhat noisy across horizons, the overall pattern is consistent with the idea that temporary external shocks can generate persistent specialization dynamics through endogenous accumulation and reallocation mechanisms.

Taken together, these results provide preliminary empirical support for the paper's central mechanism. External demand shocks appear to affect both firm-level expansion and medium-run sectoral reallocation in ways consistent with wealth-dependent accumulation dynamics under financial frictions. Alternative results that use a demand shock measure using fixed initial exposure shares are available in Appendix L.

Table 3: External demand shocks and firm expansion

	Panel A. Capital growth				
	ℓ=1	ℓ=2	ℓ=3	ℓ=4	ℓ=5
Dynamic demand shock	-0.202	-0.153	-0.237	-0.153***	-0.066
	(0.146)	(0.121)	(0.233)	(0.044)	(0.062)
Lagged productivity	0.065***	0.064***	0.063***	0.062***	0.062***
	(0.010)	(0.010)	(0.011)	(0.011)	(0.011)
Shock $\times$ productivity	0.027	0.027*	0.018	0.031**	-0.001
	(0.020)	(0.015)	(0.033)	(0.013)	(0.020)
Observations	4,521	4,521	4,521	4,521	4,521
	Panel B. Labor growth				
	ℓ=1	ℓ=2	ℓ=3	ℓ=4	ℓ=5
Dynamic demand shock	-0.250**	-0.004	-0.278	-0.163	0.108
	(0.094)	(0.189)	(0.414)	(0.131)	(0.084)
Lagged productivity	0.067***	0.065***	0.064***	0.064***	0.065***
	(0.018)	(0.019)	(0.017)	(0.017)	(0.018)
Shock $\times$ productivity	0.033**	0.020	0.063	0.040	-0.026
	(0.013)	(0.025)	(0.056)	(0.025)	(0.024)
Observations	4,532	4,532	4,532	4,532	4,532

Notes: The table reports firm-level regressions of capital and labor growth on lagged sector-level dynamic demand shocks, lagged firm productivity, and their interaction. Panel A uses capital growth as the dependent variable, and Panel B uses labor growth. Columns correspond to shock lags from one to five years. Dynamic demand shocks are constructed from changes in external import demand using lagged Peruvian export shares as exposure weights. Lagged productivity is measured using the firm-level productivity residual described in the text. All regressions include sector and year fixed effects. Standard errors are clustered at the sector level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

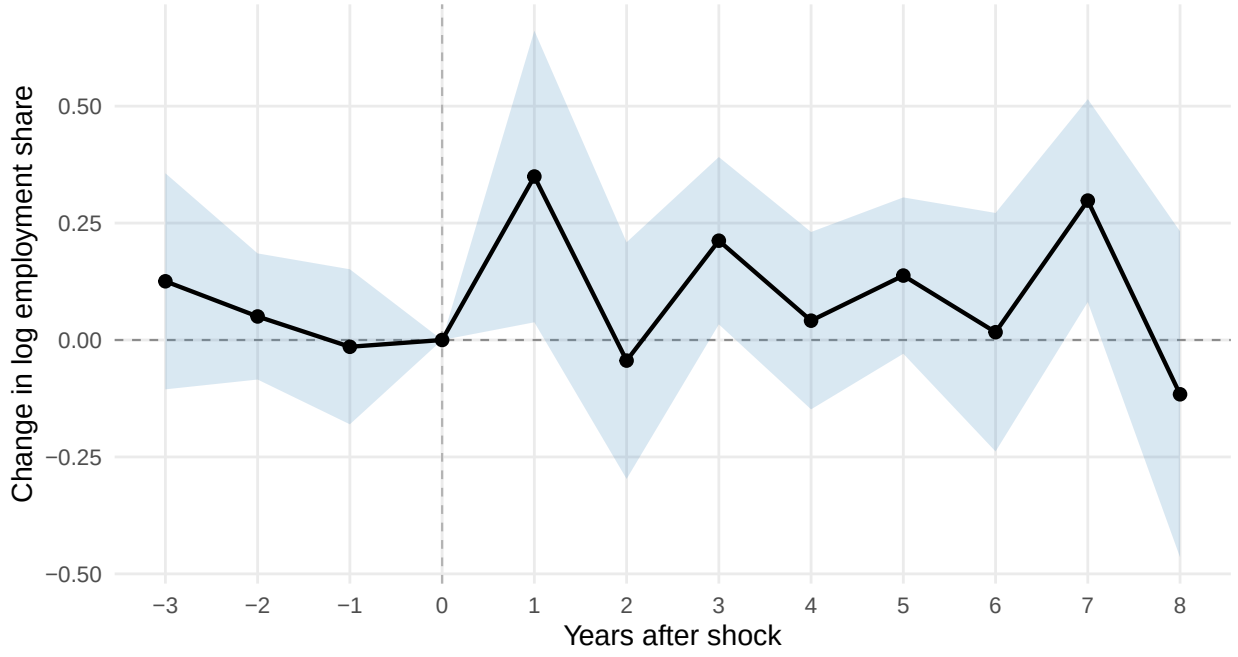


Figure 13: Dynamic Response of Sectoral Employment Shares to External Demand Shocks

Notes: The figure plots local-projection estimates of the dynamic response of sectoral employment shares to external demand shocks in Peru. The horizontal axis reports years relative to the shock, where negative values correspond to pre-trends and positive values to post-shock responses. The vertical axis reports cumulative changes in log sectoral employment shares. The shock measure is constructed using changes in external import demand weighted by lagged Peruvian export exposure across destination markets and sectors. Shaded bands denote 95 percent confidence intervals based on standard errors clustered at the sector level. Sectoral employment shares are computed using household survey data from ENAHO aggregated to ISIC Rev. 3 two-digit sectors. The dashed vertical line marks the shock period ($h = 0$), and the dashed horizontal line marks zero change in sectoral employment shares.

6 Conclusion

This paper studies how wealth distribution and financial frictions shape specialization and aggregate dynamics. The central mechanism is a feedback loop between wealth, production, and prices. Because entrepreneurs face collateral constraints, their ability to expand depends on their balance sheets. Wealth accumulation then governs future production, while sectoral output affects prices, which in turn shape profits and savings. This interaction makes the distribution of wealth a state variable that determines both the path of adjustment and the allocation of resources across sectors.

A key implication is that the role of prices differs sharply across environments. In a closed economy, relative prices respond to sectoral expansion and act as a correction mechanism, reducing initial asymmetries and promoting convergence in allocation and productivity. In a small open economy, prices are fixed at world levels, removing this adjustment margin. As a result, differences in initial wealth and within-sector allocation translate into persistent differences in specialization. Trade does not eliminate distortions; instead, it allows them to propagate over time through wealth accumulation.

These results highlight that specialization is not determined solely by fundamentals such as technology or endowments. The distribution of wealth and the presence of financial frictions play a central role in shaping both transition dynamics and long-run outcomes. More broadly, the findings suggest that the gains from trade depend on initial conditions and domestic financial structure, and that policies affecting access to capital can influence not only aggregate growth but also the direction of specialization.

The empirical analysis provides preliminary evidence consistent with the mechanism using sector-level external trade shocks combined with firm-level and sector-level data from Peru. Rather than focusing on a single historical episode, I construct measures of changes in external demand, supply, and international prices using global trade flows excluding Peru. These measures capture variation in external profitability conditions across sectors and can be interpreted as changes in the environment faced by a small open economy.

Combining these sectoral trade shocks with firm-level balance-sheet data, the results show that firm expansion is heterogeneous in ways consistent with the model. More productive firms tend to expand capital and labor more strongly following favorable external demand shocks, suggesting that external conditions affect reallocation partly through firm-level accumulation dynamics. Sector-level local projections also show that external demand shocks are followed by gradual and persistent changes in employment shares, consistent with the idea that trade shocks affect specialization over time rather than only contemporaneous allocation.

These results should be interpreted as preliminary evidence on the mechanism. The quantitative application complements this evidence by showing that empirically informed wealth–productivity misalignment can generate substantial early reallocation and persistent differences in sectoral composition, even when nontradable prices and sector-specific factors limit sectoral expansion. The

current implementation reproduces sectoral absorption shares but leaves employment allocations and sectoral trade balances as the main remaining calibration margins. Together, the theory, evidence, and preliminary quantification suggest that specialization in open economies may reflect not only fundamentals or static comparative advantage, but also the interaction of external prices with endogenous wealth accumulation and domestic financial frictions.

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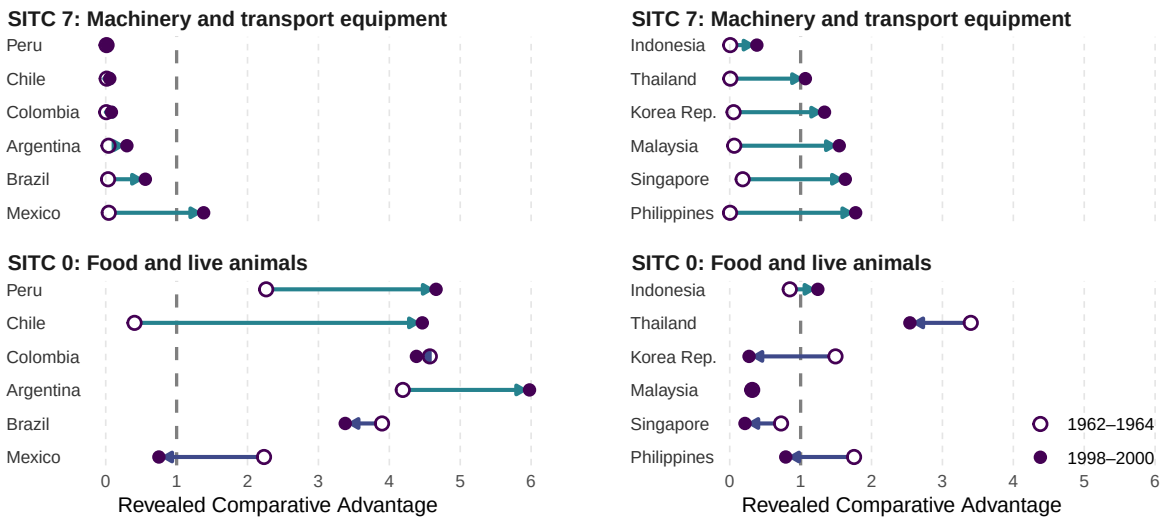
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A Cross-Country Evidence on Divergence in Sectoral Specialization

This appendix presents cross-country evidence on the evolution of sectoral specialization that connects the paper with the divergence between East Asia and Latin America (Evans, 1995; Rodrik, 1995; Ocampo, Rada, & Taylor, 2009). Figure A.1 plots changes in revealed comparative advantage (RCA) for a set of Latin American and Asian economies between the early 1960s and the late 1990s.

Two patterns emerge. First, divergence in specialization is concentrated in capital-intensive sectors, such as machinery and transport equipment. Countries with similar initial RCA levels experience markedly different trajectories, with Asian economies moving toward strong comparative advantage and Latin American economies remaining below the threshold. Second, this divergence is not present in less capital-intensive sectors, such as food products, where changes are more muted and do not display a systematic regional pattern.

These patterns suggest that specialization paths are not determined solely by initial conditions or static comparative advantage. Instead, they point to mechanisms linked to capital accumulation and sectoral dynamics that can generate persistent divergence across otherwise similar economies. While this evidence is not used directly in the main text, it provides background motivation for the model's focus on wealth distribution, financial frictions, and dynamic specialization.



(a) Latin America (b) Asia

Figure A.1: Divergence in Sectoral Specialization between Asian and Latin American Economies

Notes: The figure plots revealed comparative advantage (RCA) following Balassa (1965) for a set of Latin American and Asian economies. RCA is defined as $RCA_{c,i,t} = \frac{X_{c,i,t}/X_{c,t}}{X_{w,i,t}/X_{w,t}}$, where $X_{c,i,t}$ denotes exports of country c in sector i at time t, $X_{c,t}$ total exports of country c, and $X_{w,i,t}$ and $X_{w,t}$ the corresponding world aggregates. Each row corresponds to a country. Hollow circles denote average RCA in 1962–1964, and filled circles denote average RCA in 1998–2000. Arrows connect initial to final values, indicating the direction and magnitude of changes in specialization. The horizontal axis reports RCA levels, and the dashed vertical line marks $RCA = 1$, the threshold for comparative advantage. Trade data are from Feenstra, Lipsey, Deng, Ma, and Mo (2005).

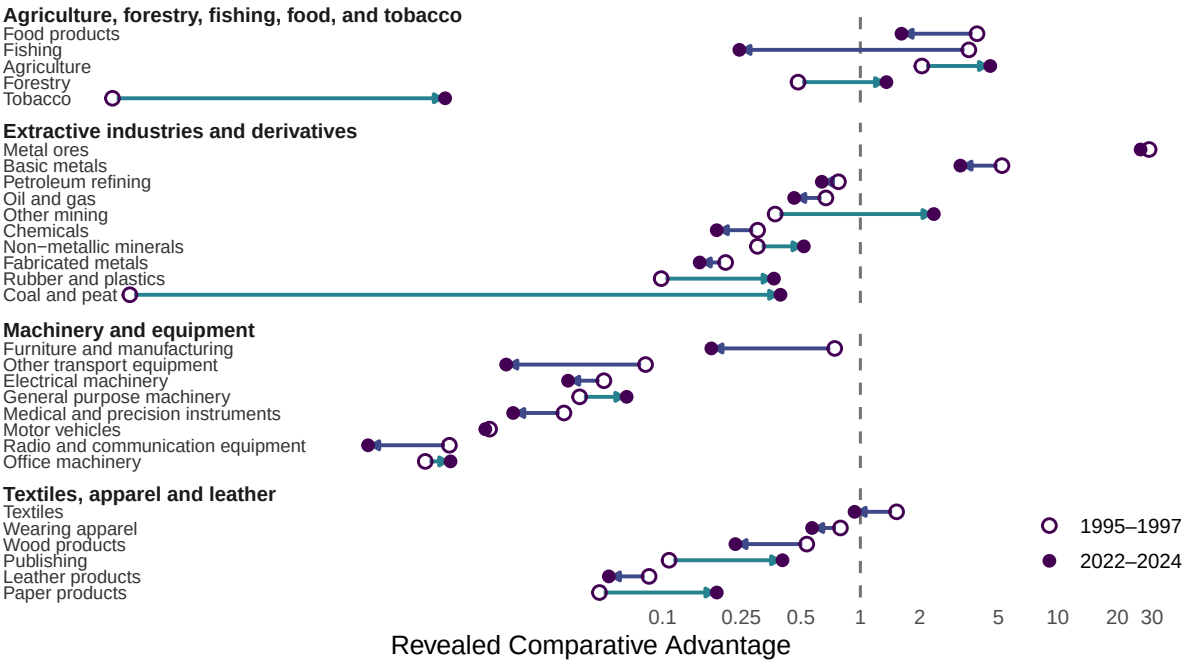


Figure A.2: Direction of Sectoral Specialization in Peru

Notes: The figure plots revealed comparative advantage (RCA) for Peruvian exports across ISIC Rev. 3 two-digit sectors. RCA is defined as $RCA_{i,t} = (X_{Peru,i,t}/X_{Peru,t}) \,/\, (X_{World,i,t}/X_{World,t})$, where $X_{Peru,i,t}$ denotes Peruvian exports in sector i at time t, $X_{Peru,t}$ total Peruvian exports, and $X_{World,i,t}$ and $X_{World,t}$ the corresponding world aggregates. Hollow circles denote average RCA over 1995–1997, and filled circles denote average RCA over 2022–2024. Arrows connect initial to final values, indicating the direction and magnitude of changes in specialization patterns over time. The horizontal axis reports RCA levels on a logarithmic scale, and the dashed vertical line marks $RCA = 1$, the threshold for comparative advantage. Sectors are grouped into broad industry categories to highlight the persistence of specialization patterns across resource-intensive, manufacturing, and machinery-related activities. Trade data are from the BACI International Trade Database harmonized to ISIC Rev. 3 classifications.

B Solution of the Static Problem of the Firm

The entrepreneur's static problem is:

$$\Pi_i(a,z) = \max_{k,\ell} \left\{ p_i (zk)^{\alpha_i} \ell^{1-\alpha_i} - w\ell - (r+\delta)k \;\text{ s.t. }\; k \leq \lambda_i a \right\}. \tag{B.1}$$

Step 1: Optimal labor demand. The first-order condition with respect to ℓ is:

$$\frac{\partial \Pi_i}{\partial \ell} = 0 \iff p_i(1-\alpha_i)(zk)^{\alpha_i}\ell^{-\alpha_i} = w \tag{B.2}$$

$$\iff \ell = \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1}{\alpha_i}} zk. \tag{B.3}$$

Step 2: Reduced-form profit function. Substituting optimal labor into profits yields:

$$\Pi_i(a,z) = \max_{k \leq \lambda_i a} \left\{ p_i (zk)^{\alpha_i} \ell^{1-\alpha_i} - w\ell - (r+\delta)k \right\} \tag{B.4}$$

$$= \max_{k \leq \lambda_i a} \left\{ \left[p_i \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1-\alpha_i}{\alpha_i}} - w \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1}{\alpha_i}} \right] zk - (r+\delta)k \right\}. \tag{B.5}$$

Define

$$\pi_i \equiv p_i \alpha_i \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1-\alpha_i}{\alpha_i}}, \tag{B.6}$$

so that profits can be written as

$$\Pi_i(a,z) = \max_{k \leq \lambda_i a} \left\{ \left[\pi_i z - (r+\delta)\right] k \right\}. \tag{B.7}$$

Step 3: Optimal capital choice. Since profits are linear in k, the optimal choice is at a corner:

$$k(a,z) = \begin{cases} \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{B.8}$$

The productivity cutoff $\underline{z}_i$ is defined by $\pi_i \underline{z}_i = r + \delta$.

Resulting profit function. Substituting optimal $k(a,z)$ yields:

$$\Pi_i(a,z) = \lambda_i \cdot \max\left\{\pi_i z - (r+\delta),\, 0\right\} a. \tag{B.9}$$

C Solution of the Dynamic Problem of the Entrepreneur

The entrepreneur's dynamic problem

$$\max_{\{c(s)\}_{s\geq t}} \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} \ln(c(s))\, ds\right] \quad \text{s.t.} \quad \dot{a}(s) = \Gamma_i(z,s)\, a(s) - P(s)c(s), \tag{C.1}$$

can be represented by the following Hamilton-Jacobi-Bellman (HJB) equation:

$$\rho V_i(a,z,t) = \max_c \left\{\ln(c) + V_{i,a}(a,z,t)\left[\Gamma_i(z,t)a - P(t)c\right] + V_{i,t}(a,z,t) + \mathcal{L}_z V_i(a,z,t)\right\}, \tag{C.2}$$

where $\mathcal{L}_z$ denotes the infinitesimal generator associated with the productivity process in Equation 4:

$$\mathcal{L}_z V_i(a,z,t) \equiv \mu_i(z)\, V_{i,z}(a,z,t) + \frac{1}{2}\sigma_i(z)^2\, V_{i,zz}(a,z,t). \tag{C.3}$$

This section provides details on the solution of the entrepreneur's dynamic problem.

The first-order and envelope conditions associated with the Hamilton-Jacobi-Bellman equation are:

$$u'(c) = P(t)\, V_{i,a}(a,z,t), \tag{C.4}$$

$$\rho V_{i,a} = V_{i,a}\,\Gamma_i(z,t) + V_{i,at} + \mathcal{L}_z V_{i,a}. \tag{C.5}$$

The following subsections present two alternative derivations of the optimal consumption rule and the induced wealth dynamics. The first uses a Hamilton-Jacobi-Bellman approach with guess and verify, while the second uses a perfect foresight optimal control formulation.

C.1 Hamilton-Jacobi-Bellman with Guess and Verify Approach

Following Moll (2014), consider a guess-and-verify solution that exploits the homothetic structure of the problem. Suppressing sector indices for notational simplicity, the Hamilton–Jacobi–Bellman equation becomes

$$\rho V(a,z,t) = \max_c \left\{\ln c + V_a(a,z,t)\left[\Gamma_i(z,t)a - P(t)c\right] + V_t(a,z,t) + \mathcal{L}_z V(a,z,t)\right\}.$$

The first-order condition with respect to consumption is

$$\frac{1}{c} = P(t)V_a(a,z,t),$$

or equivalently,

$$P(t)c(a,z,t) = \frac{1}{V_a(a,z,t)}.$$

Consider the guess

$$V(a,z,t) = D(z,t) + E(z,t)\log a.$$

Then

$$V_a(a,z,t) = \frac{E(z,t)}{a},$$

implying

$$P(t)c(a,z,t) = \frac{a}{E(z,t)}.$$

Substituting the guess and optimal consumption rule into the HJB equation and collecting terms proportional to $\log a$ yields

$$\rho E(z,t) = 1 + E_t(z,t).$$

A time-invariant solution therefore satisfies

$$E(z) = \frac{1}{\rho}.$$

Hence,

$$P(t)c(a,t) = \rho a(t).$$

Substituting into the wealth accumulation equation yields

$$\dot{a}(t) = \left[\Gamma_i(z,t) - \rho\right] a(t).$$

C.2 Perfect Foresight Approach

As an alternative, the problem can be solved using a standard optimal control approach. Consider the Hamiltonian:

$$\mathcal{H} = \log c(t) + \mu(t)\left[\Gamma_i(z,t)a(t) - P(t)c(t)\right].$$

The first-order condition with respect to consumption is:

$$\frac{\partial \mathcal{H}}{\partial c(t)} = 0 \quad \Longleftrightarrow \quad \frac{1}{c(t)} = \mu(t)P(t) \quad \Longleftrightarrow \quad \mu(t) = \frac{1}{P(t)c(t)}.$$

The costate equation is:

$$\dot{\mu}(t) = \rho\mu(t) - \frac{\partial \mathcal{H}}{\partial a(t)} = \rho\mu(t) - \mu(t)\Gamma_i(z,t).$$

Substituting for $\mu(t)$ yields:

$$\frac{(\dot{P(t)c(t)})}{P(t)c(t)} = \Gamma_i(z,t) - \rho.$$

Let $m(t) \equiv P(t)c(t)$. The costate equation can be further rewritten as

$$\dot{m}(t) = [\Gamma_i(z,t) - \rho]\, m(t).$$

Together with the law of motion for wealth,

$$\dot{a}(t) = \Gamma_i(z,t)a(t) - m(t),$$

this implies that the ratio $x(t) \equiv m(t)/a(t)$ evolves according to

$$\dot{x}(t) = x(t)\big(x(t) - \rho\big).$$

The optimal path consistent with the transversality condition is the stationary ratio $x(t) = \rho$, so that

$$P(t)c(t) = \rho a(t).$$

Hence,

$$\dot{a}(t) = [\Gamma_i(z,t) - \rho]\, a(t).$$

D Aggregation and factor prices

This appendix derives the aggregation results used in Proposition 6. The derivation proceeds in four steps. First, I aggregate individual wealth accumulation to the sectoral level. Second, I derive sectoral capital and labor demand. Third, I obtain the sectoral production representation. Fourth, I derive the factor-price conditions.

Useful definitions. Let

$$X_i(t) \equiv \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz, \qquad M_i(t) \equiv 1 - \Omega_i(\underline{z}_i(t)).$$

where $M_i(t)$ is the share of sectoral wealth held by active entrepreneurs. Then sectoral capital demand is

$$K_i(t) = \lambda_i A_i(t) M_i(t),$$

and sectoral productivity is

$$Z_i(t) = \left(\frac{X_i(t)}{M_i(t)}\right)^{\alpha_i} = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

D.1 Sectoral wealth accumulation

Using the individual law of motion $\dot{a} = s_i(z,t)a$, sectoral wealth evolves as

$$\dot{A}_i(t) = \int \dot{a}\, g_i(a,z;t)\, da\, dz = \int s_i(z,t) a\, g_i(a,z;t)\, da\, dz.$$

Using the definition of the within-sector wealth-weighted density,

$$o_i(z,t) = \frac{1}{A_i(t)} \int a g_i(a,z;t)\, da,$$

this becomes

$$\dot{A}_i(t) = A_i(t) \int s_i(z,t) o_i(z,t)\, dz.$$

Aggregating across sectors,

$$\dot{A}(t) = \sum_{i \in I} \dot{A}_i(t) = \sum_{i \in I} A_i(t) \int s_i(z,t) o_i(z,t)\, dz = A(t) \sum_{i \in I} \int s_i(z,t) \omega_i(z,t)\, dz.$$

D.2 Sectoral input demands

From the static problem, active entrepreneurs satisfy

$$k_i(a,z;t) = \lambda_i a, \qquad \ell_i(a,z;t) = \left(\frac{\pi_i(t)}{p_i(t)\alpha_i}\right)^{\frac{1}{1-\alpha_i}} z\lambda_i a \quad \text{for } z \geq \underline{z}_i(t).$$

Therefore,

$$K_i(t) = \int_{\underline{z}_i(t)}^{\infty}\int_0^{\infty} \lambda_i a g_i(a,z;t)\, da\, dz = \lambda_i A_i(t) M_i(t),$$

and

$$L_i(t) = \left(\frac{\pi_i(t)}{p_i(t)\alpha_i}\right)^{\frac{1}{1-\alpha_i}} \lambda_i A_i(t) X_i(t).$$

Solving the last expression for $\pi_i(t)$ gives

$$\pi_i(t) = p_i(t)\alpha_i \left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i - 1} L_i(t)^{1-\alpha_i}.$$

D.3 Sectoral output and measured productivity

For active entrepreneurs, substituting optimal labor demand into production gives

$$y_i(a,z;t) = z\frac{\pi_i(t)}{p_i(t)\alpha_i}\lambda_i a.$$

Aggregating,

$$Y_i(t) = \lambda_i \frac{\pi_i(t)}{p_i(t)\alpha_i} A_i(t) X_i(t).$$

Using the expression for $\pi_i(t)$ above,

$$Y_i(t) = \left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i} L_i(t)^{1-\alpha_i}.$$

Since

$$K_i(t) = \lambda_i A_i(t) M_i(t),$$

this can be rewritten as

$$Y_i(t) = \left(\frac{X_i(t)}{M_i(t)}\right)^{\alpha_i} K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i}.$$

Thus,

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

where

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

D.4 Sectoral wealth law of motion

Using $s_i(z,t) = \Gamma_i(z,t) - \rho$, sectoral wealth growth is

$$\dot{A}_i(t) = A_i(t) \int \left[\lambda_i \max\{\pi_i(t)z - (r(t)+\delta), 0\} + r(t) - \rho\right] o_i(z,t)\, dz.$$

Splitting the integral at the cutoff $\underline{z}_i(t)$,

$$\dot{A}_i(t) = \lambda_i \pi_i(t) A_i(t) X_i(t) - (r(t)+\delta) K_i(t) + (r(t)-\rho) A_i(t).$$

Since

$$\lambda_i \pi_i(t) A_i(t) X_i(t) = \alpha_i p_i(t) Y_i(t),$$

we obtain

$$\dot{A}_i(t) = \alpha_i p_i(t) Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t).$$

D.5 Factor-price conditions

Wage condition. The aggregate labor demand expression derived above is

$$L_i(t) = \left(\frac{\pi_i(t)}{p_i(t)\alpha_i}\right)^{\frac{1}{1-\alpha_i}} \lambda_i A_i(t) X_i(t),$$

where

$$X_i(t) \equiv \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz$$

and

$$\pi_i(t) = p_i(t)\alpha_i \left(\frac{1-\alpha_i}{w(t)/p_i(t)}\right)^{\frac{1-\alpha_i}{\alpha_i}}.$$

Substituting this expression for $\pi_i(t)$ into sectoral labor demand gives

$$\begin{aligned} L_i(t) &= \left[\left(\frac{1-\alpha_i}{w(t)/p_i(t)}\right)^{\frac{1-\alpha_i}{\alpha_i}}\right]^{\frac{1}{1-\alpha_i}} \lambda_i A_i(t) X_i(t) \\ &= \left(\frac{1-\alpha_i}{w(t)/p_i(t)}\right)^{\frac{1}{\alpha_i}} \lambda_i A_i(t) X_i(t). \end{aligned}$$

Raising both sides to the power α_i and rearranging yields

$$L_i(t)^{\alpha_i} = \frac{1-\alpha_i}{w(t)/p_i(t)} \left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i},$$

or equivalently,

$$\frac{w(t)}{p_i(t)} = (1-\alpha_i)\left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i} L_i(t)^{-\alpha_i}.$$

Using the sectoral production representation derived above,

$$Y_i(t) = \left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i} L_i(t)^{1-\alpha_i} = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

we have

$$\left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i} = Z_i(t) K_i(t)^{\alpha_i}.$$

Substituting this expression into the real wage equation gives

$$\frac{w(t)}{p_i(t)} = (1-\alpha_i) Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{-\alpha_i}.$$

Equivalently,

$$w(t) L_i(t) = (1-\alpha_i) p_i(t) Y_i(t).$$

Interest-rate condition. The productivity cutoff satisfies

$$\pi_i(t) \underline{z}_i(t) = r(t) + \delta.$$

Using

$$\pi_i(t) = p_i(t) \alpha_i \left(\frac{1-\alpha_i}{w(t)/p_i(t)}\right)^{\frac{1-\alpha_i}{\alpha_i}},$$

and the real wage condition derived above,

$$\frac{w(t)}{p_i(t)} = (1-\alpha_i) Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{-\alpha_i},$$

we can rewrite the term inside the profit coefficient as

$$\frac{1-\alpha_i}{w(t)/p_i(t)} = Z_i(t)^{-1} K_i(t)^{-\alpha_i} L_i(t)^{\alpha_i}.$$

Therefore,

$$\begin{aligned} r(t) + \delta &= \underline{z}_i(t) p_i(t) \alpha_i \left[Z_i(t)^{-1} K_i(t)^{-\alpha_i} L_i(t)^{\alpha_i}\right]^{\frac{1-\alpha_i}{\alpha_i}} \\ &= p_i(t) \alpha_i \underline{z}_i(t) Z_i(t)^{\frac{\alpha_i - 1}{\alpha_i}} K_i(t)^{\alpha_i - 1} L_i(t)^{1-\alpha_i}. \end{aligned}$$

Since

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}\left[z \mid z \geq \underline{z}_i(t)\right]\right)^{\alpha_i},$$

define the selection wedge

$$\zeta_i(t) \equiv \frac{\underline{z}_i(t)}{Z_i(t)^{1/\alpha_i}} = \frac{\underline{z}_i(t)}{\mathbb{E}_{o_i,t}\left[z \mid z \geq \underline{z}_i(t)\right]} \in [0,1].$$

Then the interest-rate condition becomes

$$r(t) = p_i(t)\alpha_i\zeta_i(t)Z_i(t)K_i(t)^{\alpha_i-1}L_i(t)^{1-\alpha_i} - \delta.$$

Aggregate factor-price conditions. Aggregating across sectors and using labor and capital market clearing gives

$$w(t) = \frac{1}{\bar{L}}\sum_{i\in I}(1-\alpha_i)p_i(t)Y_i(t),$$

and

$$r(t) = \frac{1}{K(t)}\sum_{i\in I} p_i(t)\alpha_i\zeta_i(t)Y_i(t) - \delta,$$

where $K(t) = \sum_{i\in I} K_i(t)$.

D.6 Summary

Collecting the results, sectoral aggregates satisfy

$$Y_i(t) = Z_i(t)K_i(t)^{\alpha_i}L_i(t)^{1-\alpha_i},$$

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i},$$

and

$$\dot{A}_i(t) = \alpha_i p_i(t)Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t).$$

These are the aggregation results stated in Proposition 6.

E Aggregate Productivity and Decomposition

While aggregation at the sectoral level is tractable given the linearity of individual policies, heterogeneity in production elasticities across sectors precludes representing aggregate output with a single Cobb–Douglas production function. As a result, changes in sectoral composition affect aggregate productivity through their impact on the relative importance of sectors with different technologies. Aggregation therefore follows the Domar–Hulten approach to growth accounting, whereby aggregate productivity growth can be expressed as a Domar-weighted average of sectoral productivity growth (Domar, 1961; Hulten, 1978).

The within–between decomposition is related to the firm dynamics literature, which separates productivity changes into within-unit improvements and reallocation across units (Baily, Hulten, & Campbell, 1992; Foster, Haltiwanger, & Krizan, 2001). In this environment, however, both sectoral productivity and Domar weights are endogenous, as they are jointly determined by wealth dynamics, selection, and equilibrium relative prices.

Proposition 11 (Aggregate Productivity and Sectoral Reallocation)**.** *Let nominal gross output be*

$$P(t)Y(t) \equiv \sum_{i \in I} p_i(t) Y_i(t),$$

and define the Domar weight (expenditure share) of sector i as

$$\psi_i(t) \equiv \frac{p_i(t) Y_i(t)}{P(t) Y(t)}.$$

(*i*) *Aggregate productivity growth.* *Can be written as:*

$$\dot{\log} Z^A(t) = \sum_{i \in I} \psi_i(t) \dot{\log} Z_i(t) = \sum_{i \in I} \psi_i(t) \alpha_i \frac{d}{dt} \log \mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)].$$

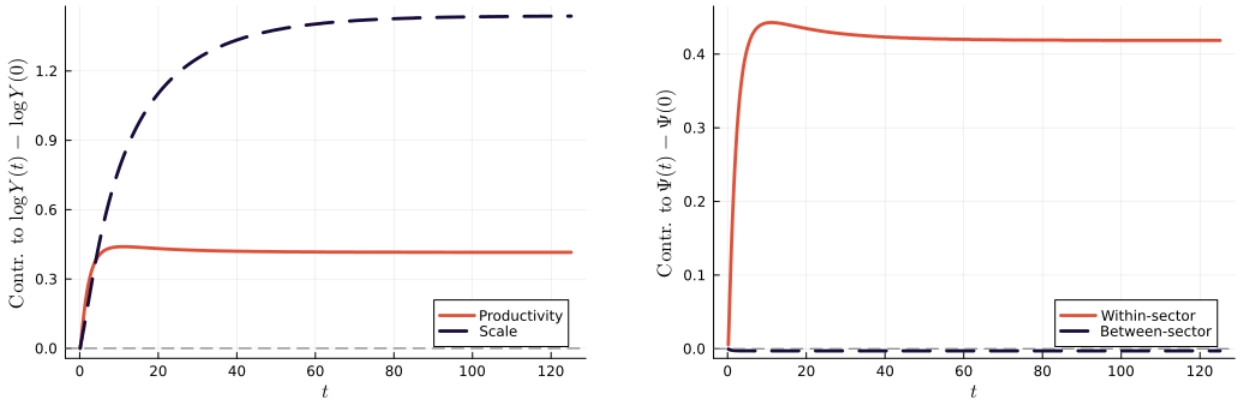
(*ii*) *Within–between decomposition.* *The Domar-weighted log productivity index*

$$\Psi(t) \equiv \sum_{i \in I} \psi_i(t) \log Z_i(t)$$

satisfies

$$\dot{\Psi}(t) = \underbrace{\sum_{i \in I} \psi_i(t) \dot{\log} Z_i(t)}_{\textit{within-sector}} + \underbrace{\sum_{i \in I} \dot{\psi}_i(t) \log Z_i(t)}_{\textit{between-sector}}.$$

This decomposition separates aggregate productivity dynamics into a within-sector component and a between-sector component. The within-sector term reflects changes in productivity driven by selection and wealth dynamics within sectors, while the between-sector term captures reallocation across sectors induced by changes in sectoral profitability and relative prices. Since both $\psi_i(t)$



(a) Growth Decomposition

(b) Cumulative decomposition

Figure E.1: Output, aggregate productivity and sectoral reallocation.

The figure reports aggregate output dynamics and its decomposition in a two-sector closed economy with asymmetric initial wealth–productivity distributions. Panel (a) decomposes output growth, $\log Y(t) - \log Y(0)$, into a productivity component given by the Domar-weighted index $\Psi(t)$ and a residual accumulation/allocation component. Panel (b) decomposes cumulative productivity growth into within-sector and between-sector components, corresponding to $\sum_i \psi_i(t) \log \dot{Z}_i(t)$ and $\sum_i \dot{\psi}_i(t) \log Z_i(t)$, respectively. Aggregate productivity gains are driven primarily by within-sector selection, while the contribution of sectoral reallocation is quantitatively small and short-lived due to the rapid adjustment of relative prices in equilibrium.

and $Z_i(t)$ are endogenous, aggregate productivity dynamics reflect the joint evolution of sectoral composition and within-sector efficiency.

To quantify the role of sectoral reallocation in the closed economy quantitative exercise, Figure E.1 decomposes aggregate productivity dynamics into within-sector and between-sector components using the Domar-based decomposition introduced above.

The within-sector component captures improvements in productivity driven by selection within sectors, while the between-sector component reflects changes in sectoral composition. In the closed economy, the between-sector component tends to decline over time, as relative price adjustments reduce incentives for further reallocation toward initially expanding sectors.

Proof. Proceed in two steps.

Step 1: Aggregate productivity growth. Let nominal gross output be

$$P(t)Y(t) = \sum_{i \in I} p_i(t) Y_i(t),$$

and define Domar weights as

$$\psi_i(t) = \frac{p_i(t) Y_i(t)}{P(t) Y(t)}.$$

Aggregate productivity growth is given by the Divisia index

$$\dot{\log} Z^A(t) = \sum_{i \in I} \psi_i(t) \dot{\log} Z_i(t).$$

Using

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i},$$

it follows that

$$\log Z_i(t) = \alpha_i \log \mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)],$$

and therefore

$$\dot{\log} Z_i(t) = \alpha_i \frac{d}{dt} \log \mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)].$$

Substituting into the Divisia index yields

$$\dot{\log} Z^A(t) = \sum_{i \in I} \psi_i(t) \alpha_i \frac{d}{dt} \log \mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)].$$

Step 2: Within–between decomposition. Define the Domar-weighted log productivity index

$$\Psi(t) \equiv \sum_{i \in I} \psi_i(t) \log Z_i(t).$$

Differentiating and applying the product rule,

$$\dot{\Psi}(t) = \sum_{i \in I} \psi_i(t) \dot{\log} Z_i(t) + \sum_{i \in I} \dot{\psi}_i(t) \log Z_i(t).$$

The first term captures within-sector productivity changes, holding expenditure shares fixed. The second term captures reallocation across sectors through changes in expenditure shares. Since $\psi_i(t)$ depends on both sectoral outputs and prices, this term reflects reallocation driven by endogenous changes in quantities and relative prices.

□

F Steady State

A steady state is an equilibrium in which aggregate wealth and its distribution are time-invariant. Formally, for all sectors $i \in I$,

$$\dot{A}_i(t) = 0, \quad g_i(a, z; t) = g_i(a, z), \quad \omega_i(z, t) = \omega_i(z), \quad \text{and} \quad O_i(t) = O_i. \tag{F.1}$$

In this environment, sectoral output is given by

$$Y_i = Z_i K_i^{\alpha_i} L_i^{1-\alpha_i}, \tag{F.2}$$

where Z_i denotes sectoral productivity, which depends on the stationary distribution of active entrepreneurs.

Using the optimality conditions of firms and the definition of profits, capital income in each sector satisfies

$$\alpha_i p_i Y_i = (r + \delta) K_i - (r - \rho) A_i. \tag{F.3}$$

Aggregating across sectors, factor prices are pinned down by market clearing. The wage satisfies

$$w = \frac{1}{\bar{L}} \left(\sum_{i \in I} p_i Y_i - (\rho + \delta) K \right), \tag{F.4}$$

while the interest rate satisfies

$$r = \frac{1}{K} \sum_{i \in I} \zeta_i \left[(r + \delta) K_i - (r - \rho) A_i \right] - \delta. \tag{F.5}$$

In steady state, the objects underlying aggregate productivity admit a simple characterization. Since sectoral productivities and expenditure shares are constant over time, $\dot{\log} Z_i(t) = 0$ and $\dot{D}_i(t) = 0$ for all i, implying zero aggregate productivity growth, $\dot{\log} Z^A(t) = 0$. The Domar weights D_i become time-invariant and reflect the stationary allocation of expenditure across sectors, while sectoral productivity levels Z_i are pinned down by the stationary distribution of entrepreneurs and the corresponding selection thresholds.

As a result, the within–between decomposition reduces to a static accounting identity: aggregate productivity is fully characterized by the cross-sectional distribution of sectoral productivities and their Domar weights, both of which are endogenously determined in equilibrium through wealth dynamics, selection, and relative prices.

G Wealth Dynamics

This appendix derives the evolution of wealth shares used in the main text. The procedure follows three steps. First, it states the general Fokker–Planck equation that governs the evolution of densities associated with Itô diffusion processes. The derivation and intuition behind this equation are standard and can be found in Risken (1996); Gardiner (2009); Pavliotis (2014); Øksendal (2013).

Second, the Fokker–Planck equation is applied to the joint process for productivity and wealth to derive the law of motion for the wealth-share density. Third, these dynamics are decomposed into within-sector and across-sector components, which are used in the analysis in the main text.

G.1 The General Fokker-Planck Equation

Consider a d-dimensional Itô diffusion process $X_t \in \mathbb{R}^d$ governed by:

$$dX_t = \mu(X_t, t)\, dt + \Sigma(X_t, t)\, dW_t,$$

where:

- $\mu : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^d$ is the drift vector,
- $\Sigma : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^{d \times m}$ is the diffusion matrix,
- W_t is an m-dimensional Wiener process.

Then, the probability density $\phi(x, t)$ of X_t evolves according to the Fokker–Planck equation:

$$\frac{\partial \phi(x,t)}{\partial t} = -\nabla \cdot [\mu(x,t)\phi(x,t)] + \frac{1}{2} \sum_{i,j=1}^{d} \frac{\partial^2}{\partial x_i \partial x_j} \left[D_{ij}(x,t)\phi(x,t) \right],$$

where $D(x,t) = \Sigma(x,t)\Sigma(x,t)^\top$ is the diffusion (covariance) matrix.

G.2 Derivation of the Wealth Share Dynamics

Apply the general Fokker–Planck equation to the joint process $X_t = (z(t), a(t))$, where $\phi(x,t) = g_i(z, a, t)$:

1. Start from the bivariate system implied by: **sectoral productivity dynamics** and the **optimal savings policy** for sector i:

$$\begin{aligned} dz &= \mu_i(z)\, dt + \sigma_i(z)\, dW_t \\ da &= s_i(z,t)\, a\, dt \end{aligned}$$

Note that wealth follows a deterministic drift (conditional on z), so there is no diffusion term in the a-dimension.

2. Then, $g_i(z, a, t)$ evolves according to the bivariate Fokker–Planck equation:

$$\frac{\partial g_i}{\partial t} = -\frac{\partial}{\partial z}[\mu_i(z) g_i] - \frac{\partial}{\partial a}[s_i(z,t) a g_i] + \frac{1}{2}\frac{\partial^2}{\partial z^2}[\sigma_i^2(z) g_i]$$

Since shocks affect only productivity, there are no cross-partial diffusion terms involving a.

3. Define $\omega_i(z, t)$ as the wealth-share density over productivity z in sector i:

$$\omega_i(z,t) := \frac{1}{A(t)} \int_0^\infty a\, g_i(z, a, t)\, da$$

Then, its time derivative is:

$$\frac{\partial \omega_i(z,t)}{\partial t} = \frac{1}{A(t)} \int_0^\infty a \frac{\partial g_i}{\partial t}\, da - \frac{\dot{A}(t)}{A(t)} \omega_i(z,t)$$

4. Substituting from the Fokker–Planck equation into the expression for $\partial_t \omega_i(z, t)$ yields:

$$\frac{\partial \omega_i(z,t)}{\partial t} = \frac{1}{A(t)} \int_0^\infty a \left\{ -\frac{\partial}{\partial z}\big[\mu_i(z) g_i\big] - \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\big[\sigma_i^2(z) g_i\big] \right\} da - \frac{\dot{A}(t)}{A(t)} \omega_i(z,t).$$

Consider the term involving derivatives with respect to a:

$$\int_0^\infty a \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big]\, da.$$

Applying integration by parts,

$$\int_0^\infty u\, dv = [uv]_0^\infty - \int_0^\infty v\, du,$$

with $u = a$ and $dv = \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big] da$, yields

$$\int_0^\infty a \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big]\, da = [a\, s_i(z,t)\, a\, g_i(z,a,t)]_0^\infty - \int_0^\infty s_i(z,t)\, a\, g_i(z,a,t)\, da.$$

Assuming $a^2 g_i(a, z, t) \to 0$ as $a \to 0$ and $a \to \infty$; the boundary term vanishes, so

$$\int_0^\infty a \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big]\, da = -\int_0^\infty s_i(z,t)\, a\, g_i(z,a,t)\, da.$$

Similarly, derivatives with respect to z can be taken outside the integral:

$$\int_0^\infty a \frac{\partial}{\partial z}\big[\mu_i(z) g_i\big]\, da = \frac{\partial}{\partial z}\left[\mu_i(z) \int_0^\infty a\, g_i\, da\right],$$

and analogously for the second derivative term.

Using the definition of $\omega_i(z,t)$, this implies

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] \omega_i(z,t) - \frac{\partial}{\partial z} \left[\mu_i(z) \omega_i(z,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial z^2} \left[\sigma_i^2(z) \omega_i(z,t) \right]. \tag{G.1}$$

G.3 Decomposition of Wealth Dynamics

This appendix derives the decomposition of the evolution of wealth shares into within- and across-sector components.

Step 1: Evolution of wealth-weighted distribution. Recall that the wealth-weighted density $\omega_i(z,t)$ evolves according to

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] \omega_i(z,t) - \frac{\partial}{\partial z} \left[\mu_i(z) \omega_i(z,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial z^2} \left[\sigma_i^2(z) \omega_i(z,t) \right]. \tag{G.2}$$

Step 2: Decomposition. Using the identity

$$\omega_i(z,t) = O_i(t)\, o_i(z,t), \tag{G.3}$$

its time derivative satisfies

$$\frac{\partial \omega_i(z,t)}{\partial t} = \dot{O}_i(t)\, o_i(z,t) + O_i(t)\, \frac{\partial o_i(z,t)}{\partial t}. \tag{G.4}$$

Substituting (G.3) and (G.4) into (G.2) and dividing by $O_i(t)$ yields

$$\begin{aligned} \frac{\dot{O}_i(t)}{O_i(t)}\, o_i(z,t) + \frac{\partial o_i(z,t)}{\partial t} = & \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] o_i(z,t) \\ & - \frac{\partial}{\partial z} \left[\mu_i(z) o_i(z,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial z^2} \left[\sigma_i^2(z) o_i(z,t) \right]. \end{aligned} \tag{G.5}$$

Step 3: Law of motion for sectoral wealth. Integrating (G.2) over z and using that $\int \omega_i(z,t)\, dz = O_i(t)$, we obtain

$$\dot{O}_i(t) = \int s_i(z,t)\, \omega_i(z,t)\, dz - \frac{\dot{A}(t)}{A(t)} O_i(t). \tag{G.6}$$

Using $\omega_i(z,t) = O_i(t) o_i(z,t)$, this becomes

$$\frac{\dot{O}_i(t)}{O_i(t)} = \bar{s}_i(t) - \frac{\dot{A}(t)}{A(t)}, \tag{G.7}$$

where

$$\bar{s}_i(t) \equiv \int s_i(z,t)\, o_i(z,t)\, dz. \tag{G.8}$$

Step 4: Within-sector dynamics. Substituting (G.7) into (G.5) yields

$$\frac{\partial o_i(z,t)}{\partial t} = \left[s_i(z,t) - \bar{s}_i(t)\right] o_i(z,t) - \frac{\partial}{\partial z}\left[\mu_i(z) o_i(z,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma_i^2(z) o_i(z,t)\right]. \tag{G.9}$$

Conclusion. The evolution of wealth shares can therefore be decomposed into: (i) within-sector dynamics governed by deviations from the sectoral average savings rate, and (ii) across-sector dynamics governed by differences in average savings rates across sectors.

H Quantitative Open Economy with Nontradables, Sector-Specific Factors, and Domestic-Origin Prices

This appendix extends the model along three dimensions used in the quantitative application. First, it introduces factors that are fixed in supply at the sector level. Firms remain constant returns to scale in capital, labor, and the sector-specific factor jointly, but the fixed sectoral supply of the latter generates decreasing returns in the scalable inputs at the sector level. Second, it allows for nontradable sectors in the open-economy environment. Third, it distinguishes, within tradable sectors, between the world price at which goods can be imported or exported and the domestic-origin price received by domestic producers. Let $D_i(t)$ denote domestic use of sector i, and let $\varphi_i \in [0,1]$ denote the Armington home-bias parameter governing substitution between domestic-origin and imported varieties. The imported variety is available at the world price, while the domestic-origin variety is supplied by domestic producers. When domestic production exceeds domestic-origin demand, the surplus is exported and the domestic-origin price equals the world price. When domestic-origin demand exhausts domestic production, the domestic-origin price may rise above the world price, generating a scarcity premium for domestic producers. This structure disciplines gross trade flows while allowing domestic scarcity to affect sectoral profitability. I first solve the firm's static problem, then derive the corresponding aggregation and factor-price conditions, and finally define the quantitative open-economy equilibrium.

In the open-economy definition, I explicitly allow the set of modeled entrepreneurial sectors to exclude resource-intensive activities such as oil and mining. Let $NX_{\text{res}}(t)$ denote the exogenous net exports generated by these omitted resource activities. This residual closes the aggregate external-balance condition of the complete economy and finances part of expenditure in the modeled economy. Resource income is rebated lump-sum to a separate hand-to-mouth resource owner and therefore contributes to aggregate domestic absorption without entering entrepreneurs' individual wealth accounts. Consequently, $NX_{\text{res}}(t)$ modifies only the aggregate external-balance and expenditure conditions; the firm problem, aggregation results, and sectoral wealth laws derived below are unchanged. In closed-economy special cases, $NX_{\text{res}}(t) = 0$.

H.1 Static Problem of the Firm with Sector-Specific Factors and Solution

The entrepreneur's static problem is:

$$\Pi_i(a,z) = \max_{k,\ell,h} \left\{ p_i (zk)^{\alpha_i} \ell^{\beta_i} h^{\gamma_i} - w\ell - q_i h - (r+\delta)k \;\text{ s.t. }\; k \le \lambda_i a \right\}, \tag{H.1}$$

where $\alpha_i, \beta_i, \gamma_i > 0$ and

$$\alpha_i + \beta_i + \gamma_i = 1. \tag{H.2}$$

The variable h denotes a sector-specific factor rented by firms at price q_i. Its aggregate supply is fixed at the sector level and will be imposed in equilibrium. The price p_i denotes the price received by domestic producers in sector i. For tradable sectors, this is the domestic-origin price p_i^H, which may differ from the world import/export price p_i^* when domestic-origin goods are scarce. For nontradable sectors, p_i is the domestic market-clearing price. The domestic-use price system introduced below affects final demand and the allocation between domestic-origin and imported varieties, but firms take the domestic producer price as given, so the static problem remains unchanged.

Step 1: Optimal labor and specific-factor demand. For a given level of capital k, the first-order conditions with respect to ℓ and h are:

$$p_i \beta_i (zk)^{\alpha_i} \ell^{\beta_i - 1} h^{\gamma_i} = w, \tag{H.3}$$

$$p_i \gamma_i (zk)^{\alpha_i} \ell^{\beta_i} h^{\gamma_i - 1} = q_i. \tag{H.4}$$

Dividing the first-order condition for labor by the first-order condition for the specific factor gives

$$h = \frac{\gamma_i}{\beta_i} \frac{w}{q_i} \ell. \tag{H.5}$$

Substituting this expression into the labor first-order condition yields

$$p_i \beta_i z^{\alpha_i} k^{\alpha_i} \left(\frac{\gamma_i}{\beta_i} \frac{w}{q_i} \right)^{\gamma_i} \ell^{\beta_i + \gamma_i - 1} = w. \tag{H.6}$$

Since $\alpha_i + \beta_i + \gamma_i = 1$, this becomes

$$p_i \beta_i z^{\alpha_i} k^{\alpha_i} \left(\frac{\gamma_i}{\beta_i} \frac{w}{q_i} \right)^{\gamma_i} \ell^{-\alpha_i} = w. \tag{H.7}$$

Therefore,

$$\ell_i(k, z) = \left[\frac{p_i \beta_i}{w} \left(\frac{\gamma_i}{\beta_i} \frac{w}{q_i} \right)^{\gamma_i} \right]^{1/\alpha_i} zk. \tag{H.8}$$

Using the input ratio again,

$$h_i(k, z) = \frac{\gamma_i}{\beta_i} \frac{w}{q_i} \left[\frac{p_i \beta_i}{w} \left(\frac{\gamma_i}{\beta_i} \frac{w}{q_i} \right)^{\gamma_i} \right]^{1/\alpha_i} zk. \tag{H.9}$$

Equivalently, the optimal input demands can be written as

$$\ell_i(k,z) = zk\left[\frac{p_i\beta_i}{w}\left(\frac{\gamma_i w}{\beta_i q_i}\right)^{\gamma_i}\right]^{\frac{1}{\alpha_i}}, \tag{H.10}$$

$$h_i(k,z) = zk\left[\frac{p_i\gamma_i}{q_i}\left(\frac{\beta_i q_i}{\gamma_i w}\right)^{\beta_i}\right]^{\frac{1}{\alpha_i}}. \tag{H.11}$$

Thus both non-capital input demands are proportional to zk. Since optimal capital is proportional to wealth whenever the collateral constraint binds, labor and specific-factor demand are also linear in wealth.

Step 2: Reduced-form profit function. From Step 1, optimal non-capital input demands can be written as

$$\ell_i(k,z) = m_{\ell i} zk, \qquad h_i(k,z) = m_{hi} zk, \tag{H.12}$$

where

$$m_{\ell i} \equiv \left[\frac{p_i\beta_i}{w}\left(\frac{\gamma_i w}{\beta_i q_i}\right)^{\gamma_i}\right]^{\frac{1}{\alpha_i}}, \tag{H.13}$$

and

$$m_{hi} \equiv \left[\frac{p_i\gamma_i}{q_i}\left(\frac{\beta_i q_i}{\gamma_i w}\right)^{\beta_i}\right]^{\frac{1}{\alpha_i}}. \tag{H.14}$$

Substituting these expressions into output gives

$$y_i(k,z) = (zk)^{\alpha_i}(m_{\ell i} zk)^{\beta_i}(m_{hi} zk)^{\gamma_i} \tag{H.15}$$

$$= m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} z^{\alpha_i+\beta_i+\gamma_i} k^{\alpha_i+\beta_i+\gamma_i}. \tag{H.16}$$

Since $\alpha_i + \beta_i + \gamma_i = 1$, this becomes

$$y_i(k,z) = m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} zk. \tag{H.17}$$

Operating profits after choosing ℓ and h are therefore

$$\max_{\ell,h}\left\{p_i(zk)^{\alpha_i}\ell^{\beta_i}h^{\gamma_i} - w\ell - q_i h\right\} = \left[p_i m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} - w m_{\ell i} - q_i m_{hi}\right] zk. \tag{H.18}$$

Using the Cobb-Douglas first-order conditions, payments to labor and the specific factor satisfy

$$w\ell_i = \beta_i p_i y_i, \qquad q_i h_i = \gamma_i p_i y_i. \tag{H.19}$$

Hence operating surplus equals the capital share of revenue:

$$p_i y_i - w\ell_i - q_i h_i = \alpha_i p_i y_i. \tag{H.20}$$

Therefore,

$$\max_{\ell,h} \left\{ p_i (zk)^{\alpha_i} \ell^{\beta_i} h^{\gamma_i} - w\ell - q_i h \right\} = \pi_i z k, \tag{H.21}$$

where

$$\pi_i \equiv \alpha_i p_i m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i}. \tag{H.22}$$

Substituting the expressions for $m_{\ell i}$ and m_{hi}, this can be written as

$$\pi_i \equiv \alpha_i p_i \left(\frac{\beta_i p_i}{w} \right)^{\frac{\beta_i}{\alpha_i}} \left(\frac{\gamma_i p_i}{q_i} \right)^{\frac{\gamma_i}{\alpha_i}}. \tag{H.23}$$

Equivalently, since $\alpha_i + \beta_i + \gamma_i = 1$,

$$\pi_i = \alpha_i p_i^{1/\alpha_i} \left(\frac{\beta_i}{w} \right)^{\beta_i/\alpha_i} \left(\frac{\gamma_i}{q_i} \right)^{\gamma_i/\alpha_i}. \tag{H.24}$$

Thus, the entrepreneur's static problem becomes

$$\Pi_i(a, z) = \max_{k \leq \lambda_i a} \left\{ \left[\pi_i z - (r + \delta) \right] k \right\}. \tag{H.25}$$

Step 3: Optimal capital choice. Since profits are linear in k, the optimal choice is at a corner:

$$k_i(a, z) = \begin{cases} \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{H.26}$$

The productivity cutoff $\underline{z}_i$ is defined by

$$\pi_i \underline{z}_i = r + \delta. \tag{H.27}$$

Thus,

$$\underline{z}_i = \frac{r + \delta}{\pi_i}. \tag{H.28}$$

Resulting input demands. Substituting the optimal capital choice into the input demands gives:

$$\ell_i(a, z) = \begin{cases} m_{\ell i} z \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise,} \end{cases} \tag{H.29}$$

and

$$h_i(a,z) = \begin{cases} m_{hi} z \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{H.30}$$

Hence capital, labor, and specific-factor demand remain linear in wealth.

Resulting profit function. Substituting optimal capital demand into the reduced-form problem yields

$$\Pi_i(a,z) = \lambda_i \max\{\pi_i z - (r+\delta), 0\}\, a. \tag{H.31}$$

H.2 Specific-Factor Market Clearing and Sector-Level Decreasing Returns to Scale

The sector-specific factor is in fixed supply. In equilibrium, its rental price q_i adjusts so that

$$H_i = \int h_i(a,z) g_i(a,z)\, da\, dz. \tag{H.32}$$

Using the linear policy rules derived above, this condition can be written as

$$H_i = \lambda_i A_i \int_{\underline{z}_i}^{\infty} m_{hi} z o_i(z)\, dz. \tag{H.33}$$

Since m_{hi} depends on the rental price q_i, this condition pins down q_i in equilibrium. As sector i expands, demand for the sector-specific factor rises. Because H_i is fixed, the rental price q_i increases, lowering the operating-surplus coefficient π_i. This creates a congestion force that attenuates complete specialization while preserving the linearity of firm policies in wealth.

Sector-level decreasing returns in scalable inputs. The same aggregation logic used in the baseline model continues to apply. Since individual policies are linear in wealth, sectoral output can be written as

$$Y_i = Z_i K_i^{\alpha_i} L_i^{\beta_i} H_i^{\gamma_i}, \tag{H.34}$$

where

$$\alpha_i + \beta_i + \gamma_i = 1. \tag{H.35}$$

Because H_i is fixed at the sector level, equilibrium output can be interpreted as a decreasing-returns function of the scalable inputs K_i and L_i:

$$Y_i = \widetilde{Z}_i K_i^{\alpha_i} L_i^{\beta_i}, \qquad \widetilde{Z}_i \equiv Z_i H_i^{\gamma_i}. \tag{H.36}$$

Since

$$\alpha_i + \beta_i = 1 - \gamma_i < 1, \tag{H.37}$$

sectoral expansion faces decreasing returns in capital and labor. This provides an additional force limiting complete specialization: as a sector grows, its fixed factor becomes scarcer, the rental price q_i rises, and the sector's profitability advantage declines.

H.3 Aggregation, Prices, and Domestic-Origin Prices

The derivation of the aggregation and factor-price conditions for this extension follows the same steps as in Appendix D. First, I define the relevant wealth-weighted objects. Second, I aggregate input demands. Third, I obtain the sectoral production representation. Fourth, I derive the sectoral wealth law of motion and the factor-price conditions.

Throughout this subsection, let $p_i^P(t)$ denote the price received by domestic producers in sector i. For tradable sectors, this producer price will be the domestic-origin price $p_i^H(t)$; for nontradable sectors, it will be the domestic market-clearing price $p_i^N(t)$. The world trade price $p_i^*(t)$ and the domestic-use price faced by final demand are introduced below. The aggregation results in this subsection use only the producer price because they follow from firms' revenue and input choices.

Useful definitions. To avoid conflict with the notation for gross exports and imports introduced below, I denote the active wealth share by $n_i(t)$ and the productivity-weighted active wealth share by $e_i(t)$ in this appendix.

Let

$$n_i(t) \equiv 1 - \Omega_i(\underline{z}_i(t)), \qquad e_i(t) \equiv \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz.$$

Here, $n_i(t)$ is the share of sectoral wealth held by active entrepreneurs, while $e_i(t)$ is the corresponding productivity-weighted active wealth share. Since active entrepreneurs choose $k_i(a,z;t) = \lambda_i a$, sectoral capital demand is

$$K_i(t) = \lambda_i A_i(t) n_i(t).$$

Define sectoral productivity as

$$Z_i(t) \equiv \left(\frac{e_i(t)}{n_i(t)}\right)^{\alpha_i} = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

H.3.1 Sectoral input demands

From the static problem with sector-specific factors, active entrepreneurs satisfy

$$k_i(a,z;t) = \lambda_i a,$$

$$\ell_i(a,z;t) = m_{\ell i}(t) z \lambda_i a, \qquad h_i(a,z;t) = m_{hi}(t) z \lambda_i a,$$

for $z \geq \underline{z}_i(t)$, where

$$m_{\ell i}(t) \equiv \left[\frac{p_i^P(t)\beta_i}{w(t)} \left(\frac{\gamma_i w(t)}{\beta_i q_i(t)} \right)^{\gamma_i} \right]^{1/\alpha_i},$$

and

$$m_{hi}(t) \equiv \left[\frac{p_i^P(t)\gamma_i}{q_i(t)} \left(\frac{\beta_i q_i(t)}{\gamma_i w(t)} \right)^{\beta_i} \right]^{1/\alpha_i}.$$

Aggregating capital demand gives

$$K_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} \lambda_i a g_i(a, z; t)\, da\, dz = \lambda_i A_i(t) n_i(t).$$

Aggregating labor demand gives

$$L_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} m_{\ell i}(t) z \lambda_i a g_i(a, z; t)\, da\, dz = m_{\ell i}(t) \lambda_i A_i(t) e_i(t).$$

Similarly, aggregate demand for the sector-specific factor is

$$H_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} m_{hi}(t) z \lambda_i a g_i(a, z; t)\, da\, dz = m_{hi}(t) \lambda_i A_i(t) e_i(t).$$

In equilibrium, the sector-specific factor is in fixed supply, so

$$H_i(t) = \bar{H}_i.$$

This condition pins down the rental price $q_i(t)$. As sector i expands, demand for the fixed factor rises; because $\bar{H}_i$ is fixed, $q_i(t)$ increases, reducing sectoral profitability.

H.3.2 Sectoral output and measured productivity

For active entrepreneurs,

$$y_i(a, z; t) = (z k_i(a, z; t))^{\alpha_i} \ell_i(a, z; t)^{\beta_i} h_i(a, z; t)^{\gamma_i}.$$

Using the policy rules above and $\alpha_i + \beta_i + \gamma_i = 1$,

$$\begin{aligned} y_i(a, z; t) &= (z\lambda_i a)^{\alpha_i} \left[m_{\ell i}(t) z \lambda_i a \right]^{\beta_i} \left[m_{hi}(t) z \lambda_i a \right]^{\gamma_i} \\ &= m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} z \lambda_i a. \end{aligned}$$

Aggregating over active entrepreneurs,

$$Y_i(t) = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t).$$

Now use the aggregate input demand expressions

$$K_i(t) = \lambda_i A_i(t) n_i(t),$$

$$L_i(t) = m_{\ell i}(t) \lambda_i A_i(t) e_i(t),$$

and

$$H_i(t) = m_{hi}(t) \lambda_i A_i(t) e_i(t).$$

Then

$$\begin{aligned} K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} &= [\lambda_i A_i(t) n_i(t)]^{\alpha_i} \left[m_{\ell i}(t) \lambda_i A_i(t) e_i(t)\right]^{\beta_i} \left[m_{hi}(t) \lambda_i A_i(t) e_i(t)\right]^{\gamma_i} \\ &= m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) n_i(t)^{\alpha_i} e_i(t)^{\beta_i + \gamma_i}. \end{aligned}$$

Since $\beta_i + \gamma_i = 1 - \alpha_i$,

$$Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} = \left(\frac{e_i(t)}{n_i(t)}\right)^{\alpha_i} m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) n_i(t)^{\alpha_i} e_i(t)^{1-\alpha_i}.$$

Therefore,

$$Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t) = Y_i(t).$$

Thus, sectoral output can be written as

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i},$$

where

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

Because $H_i(t) = \bar{H}_i$ is fixed in equilibrium, this can also be written as

$$Y_i(t) = \widetilde{Z}_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i}, \qquad \widetilde{Z}_i(t) \equiv Z_i(t) \bar{H}_i^{\gamma_i}.$$

Since

$$\alpha_i + \beta_i = 1 - \gamma_i < 1,$$

sectoral production exhibits decreasing returns in the scalable inputs $K_i(t)$ and $L_i(t)$.

H.3.3 Sectoral wealth law of motion

The reduced-form profit coefficient after optimizing over labor and the sector-specific factor is

$$\pi_i(t) = \alpha_i p_i^P(t) m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i}.$$

Equivalently,

$$\pi_i(t) = \alpha_i \left[p_i^P(t)\right]^{1/\alpha_i} \left(\frac{\beta_i}{w(t)}\right)^{\beta_i/\alpha_i} \left(\frac{\gamma_i}{q_i(t)}\right)^{\gamma_i/\alpha_i}.$$

The productivity cutoff satisfies

$$\pi_i(t)\underline{z}_i(t) = r(t) + \delta.$$

Using $s_i(z,t) = \Gamma_i(z,t) - \rho$, sectoral wealth growth is

$$\dot{A}_i(t) = A_i(t) \int \left[\lambda_i \max\{\pi_i(t)z - (r(t)+\delta), 0\} + r(t) - \rho\right] o_i(z,t)\, dz.$$

Splitting the integral at the cutoff,

$$\dot{A}_i(t) = \lambda_i \pi_i(t) A_i(t) e_i(t) - (r(t)+\delta)K_i(t) + (r(t)-\rho)A_i(t).$$

Since

$$Y_i(t) = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t),$$

and

$$\pi_i(t) = \alpha_i p_i^P(t) m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i},$$

we have

$$\lambda_i \pi_i(t) A_i(t) e_i(t) = \alpha_i p_i^P(t) Y_i(t).$$

Therefore,

$$\dot{A}_i(t) = \alpha_i p_i^P(t) Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t).$$

The sectoral wealth law is thus unchanged in form relative to the baseline model, but $Y_i(t)$, $Z_i(t)$, and equilibrium profitability now depend on the fixed factor and its rental price.

H.3.4 Factor-price conditions

Labor price. From the Cobb-Douglas first-order condition for labor,

$$w(t)\ell_i(a,z;t) = \beta_i p_i^P(t) y_i(a,z;t).$$

Aggregating across entrepreneurs in sector i,

$$w(t)L_i(t) = \beta_i p_i^P(t) Y_i(t).$$

Using the aggregate production representation,

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i},$$

the real wage condition can be written as

$$\frac{w(t)}{p_i^P(t)} = \beta_i Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i - 1} H_i(t)^{\gamma_i}.$$

Specific-factor price. Similarly, the first-order condition for the sector-specific factor implies

$$q_i(t) h_i(a, z; t) = \gamma_i p_i^P(t) y_i(a, z; t).$$

Aggregating,

$$q_i(t) H_i(t) = \gamma_i p_i^P(t) Y_i(t).$$

Hence,

$$\frac{q_i(t)}{p_i^P(t)} = \gamma_i Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i - 1}.$$

Since $H_i(t) = \bar{H}_i$, this equation pins down the sector-specific rental rate $q_i(t)$ given aggregate sectoral allocations.

Interest-rate condition. The productivity cutoff satisfies

$$\pi_i(t) \underline{z}_i(t) = r(t) + \delta.$$

Using

$$\pi_i(t) = \alpha_i p_i^P(t) m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i},$$

and the aggregate production identity,

$$Y_i(t) = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t),$$

one can express the cutoff condition in terms of aggregate sectoral variables as

$$r(t) + \delta = p_i^P(t) \alpha_i \zeta_i(t) Z_i(t) K_i(t)^{\alpha_i - 1} L_i(t)^{\beta_i} H_i(t)^{\gamma_i},$$

where the selection wedge is

$$\zeta_i(t) \equiv \frac{\underline{z}_i(t)}{Z_i(t)^{1/\alpha_i}} = \frac{\underline{z}_i(t)}{\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]} \in [0, 1].$$

Therefore,

$$r(t) = p_i^P(t) \alpha_i \zeta_i(t) Z_i(t) K_i(t)^{\alpha_i - 1} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} - \delta.$$

Aggregate factor-price implications. Using labor-market clearing, $\sum_{i\in I} L_i(t) = \bar{L}$, and the labor payment condition, the aggregate wage satisfies

$$w(t) = \frac{1}{\bar{L}} \sum_{i\in I} \beta_i p_i^P(t) Y_i(t).$$

Using capital-market clearing, $K(t) = \sum_{i\in I} K_i(t)$, and the interest-rate condition, the aggregate interest rate satisfies

$$r(t) = \frac{1}{K(t)} \sum_{i\in I} \alpha_i p_i^P(t) \zeta_i(t) Y_i(t) - \delta.$$

Finally, the sector-specific factor price is pinned down sector by sector:

$$q_i(t) = \frac{\gamma_i p_i^P(t) Y_i(t)}{\bar{H}_i}.$$

Collecting the main aggregation results,

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} \bar{H}_i^{\gamma_i},$$

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i},$$

and

$$\dot{A}_i(t) = \alpha_i p_i^P(t) Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t).$$

H.3.5 Domestic-origin prices, imports, and exports

In the open-economy environment, sectoral domestic absorption is interpreted as domestic use of a sectoral composite. For each tradable sector $i \in I^T$, let $D_i(t)$ denote domestic use: the quantity of sector i's composite used by resident households, firms, and the government, excluding exports. The sectoral composite is produced from a domestic-origin variety, $D_i^H(t)$, and an imported variety, $M_i(t)$. Domestic and imported varieties are imperfect substitutes.

I assume that the domestic-use composite in each tradable sector is given by a CES Armington aggregator:

$$D_i(t) = \left[\varphi_i^{1/\eta_i^\star} \left(D_i^H(t)\right)^{(\eta_i^\star - 1)/\eta_i^\star} + (1-\varphi_i)^{1/\eta_i^\star} M_i(t)^{(\eta_i^\star - 1)/\eta_i^\star}\right]^{\eta_i^\star/(\eta_i^\star - 1)}, \qquad i \in I^T,$$

where $\eta_i^\star > 0$ is the elasticity of substitution between domestic-origin and imported varieties, and $\varphi_i \in [0,1]$ is a home-bias parameter. The Cobb-Douglas case $\eta_i^\star = 1$ is understood as the usual limiting case.

Tradable sectors are associated with two prices. The first is the world price $p_i^*(t)$, at which the imported variety can be purchased and domestic output can be exported. The second is the domestic-origin price $p_i^H(t)$, which is the price received by domestic producers for goods sold to

domestic users. Because domestic producers can always export at the world price, the domestic-origin price satisfies

$$p_i^H(t) \geq p_i^*(t).$$

Given $p_i^H(t)$ and $p_i^*(t)$, the unit cost of the tradable domestic-use composite is

$$p_i^D(t) = \left[\varphi_i \left(p_i^H(t)\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^*(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, \qquad i \in I^T,$$

again with the Cobb-Douglas limiting expression when $\eta_i^\star = 1$. Cost minimization within the sectoral Armington nest implies

$$D_i^H(t) = \varphi_i \left(\frac{p_i^H(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T,$$

and

$$M_i(t) = (1-\varphi_i)\left(\frac{p_i^*(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T.$$

These demand functions imply the expenditure identity

$$p_i^D(t) D_i(t) = p_i^H(t) D_i^H(t) + p_i^*(t) M_i(t).$$

Domestic output is allocated between domestic-origin use and exports:

$$Y_i(t) = D_i^H(t) + X_i(t), \qquad i \in I^T.$$

Equivalently,

$$X_i(t) = Y_i(t) - D_i^H(t).$$

Feasibility requires

$$X_i(t) \geq 0.$$

The domestic-origin price satisfies the complementarity condition

$$\left[p_i^H(t) - p_i^*(t)\right] X_i(t) = 0.$$

Thus, if $X_i(t) > 0$, the sector exports and $p_i^H(t) = p_i^*(t)$. If domestic-origin demand exhausts domestic output, then exports are zero and the domestic-origin price may rise above the world price:

$$p_i^H(t) > p_i^*(t) \quad \Longrightarrow \quad X_i(t) = 0.$$

This condition gives domestic scarcity a price effect while preserving the open-economy discipline imposed by the world price. A sector can sell abroad at the world price, but when domestic-origin

goods are scarce, domestic users may pay a premium for the domestic-origin variety.

This notation is consistent with the producer-price formulation used in the firm problem. For tradable sectors, the producer price is

$$p_i^P(t) = p_i^H(t), \qquad i \in I^T.$$

When a tradable sector exports, $p_i^H(t) = p_i^*(t)$, so all units are valued at the world price. When $p_i^H(t) > p_i^*(t)$, exports are zero and all domestic output is sold domestically at the domestic-origin price. Hence producer revenue can be written as

$$p_i^P(t)Y_i(t) = p_i^H(t)Y_i(t)$$

in either case.

Sectoral net exports are defined as gross exports minus gross imports:

$$NX_i(t) = X_i(t) - M_i(t), \qquad i \in I^T.$$

Because $D_i(t)$ is now a CES composite quantity rather than a physical sum of domestic and imported units, sectoral net exports are not generally equal to $Y_i(t) - D_i(t)$. In the aggregate external-balance condition, imports and exports are valued at the world price:

$$p_i^*(t)NX_i(t) = p_i^*(t)\left[X_i(t) - M_i(t)\right].$$

For nontradable sectors $i \in I^N$, there is no world import/export price and no Armington nest. I set

$$M_i(t) = 0, \qquad X_i(t) = 0.$$

The domestic-use condition reduces to the usual nontradable market-clearing condition:

$$Y_i(t) = D_i(t), \qquad i \in I^N.$$

For these sectors, the producer price and the domestic-use price coincide with the domestic market-clearing price:

$$p_i^P(t) = p_i^D(t) = p_i^N(t), \qquad i \in I^N.$$

In the quantitative implementation, the Armington home-bias parameter φ_i is chosen to match the domestic-origin expenditure share in input-output data. Let

$$s_i^{H,IO} \equiv 1 - \frac{M_i^{IO}}{D_i^{IO}},$$

where D_i^{IO} denotes total domestic use expenditure in sector i, and M_i^{IO} denotes import expen-

diture used domestically. In the Peruvian input-output data used and described in Appendix J, D_i^{IO} corresponds to total demand net of exports and inventory changes. Under the benchmark normalization $p_i^H = p_i^*$, this gives

$$\varphi_i = s_i^{H,IO}.$$

More generally, if benchmark domestic-origin and import prices differ, φ_i can be chosen so that the Armington expenditure share

$$\frac{\varphi_i \left(p_i^H\right)^{1-\eta_i^\star}}{\varphi_i \left(p_i^H\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^*\right)^{1-\eta_i^\star}}$$

matches $s_i^{H,IO}$.

The next subsection defines final domestic demand across sectoral composites. Final users take as given the sectoral domestic-use price $p_i^D(t)$, which incorporates both the world price of imported varieties and the domestic-origin price of domestically supplied varieties.

H.3.6 Final composite demand with domestic-origin prices

The domestic final composite is produced from sectoral domestic-use composites. Let $D_i(t)$ denote the quantity of sector i's composite used domestically, and let aggregate domestic use be

$$D(t) = \left[\sum_{i \in I} \upsilon_i^{1/\eta} D_i(t)^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)}.$$

The corresponding sectoral domestic-use price is

$$p_i^D(t) = \begin{cases} \left[\varphi_i \left(p_i^H(t)\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^*(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, & i \in I^T, \\ p_i^N(t), & i \in I^N. \end{cases}$$

Thus, final users do not directly face the producer price of tradable domestic output $p_i^H(t)$, nor the world price $p_i^*(t)$ alone. They face the unit cost of the sectoral domestic-use composite.

Total expenditure on sectoral domestic use is denoted by $E(t)$, where

$$E(t) = P^D(t)C(t) + \dot{A}(t) + \delta A(t).$$

Equivalently, using aggregate factor payments and resource income,

$$E(t) = w(t)\bar{L} + \sum_{i \in I} q_i(t)\bar{H}_i + \rho A(t) + NX_{\text{res}}(t) + \dot{A}(t) + \delta A(t),$$

where

$$A(t) = \sum_{i \in I} A_i(t), \qquad \dot{A}(t) = \sum_{i \in I} \dot{A}_i(t).$$

Given the sectoral domestic-use prices $\{p_i^D(t)\}_{i \in I}$, final demand across sectoral composites solves the standard CES expenditure problem. The domestic-use price index is

$$P^D(t)^{1-\eta} = \sum_{i \in I} \upsilon_i \left[p_i^D(t)\right]^{1-\eta},$$

with the usual Cobb-Douglas limiting expression when $\eta = 1$. Sectoral domestic use satisfies

$$D_i(t) = \upsilon_i \left(\frac{p_i^D(t)}{P^D(t)}\right)^{-\eta} D(t), \qquad i \in I,$$

where

$$D(t) = \frac{E(t)}{P^D(t)}.$$

Equivalently,

$$D_i(t) = \upsilon_i \left(\frac{p_i^D(t)}{P^D(t)}\right)^{-\eta} \frac{E(t)}{P^D(t)}.$$

For tradable sectors, the domestic-origin and imported components of $D_i(t)$ are then determined by the Armington demand system:

$$D_i^H(t) = \varphi_i \left(\frac{p_i^H(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t),$$

and

$$M_i(t) = (1 - \varphi_i) \left(\frac{p_i^*(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t).$$

For nontradable sectors, market clearing requires

$$D_i(t) = Y_i(t), \qquad i \in I^N.$$

The domestic-origin price $p_i^H(t)$ in tradable sectors and the domestic price $p_i^N(t)$ in nontradable sectors are equilibrium objects. For tradable sectors, $p_i^H(t)$ adjusts subject to the world-price floor and the export complementarity condition

$$p_i^H(t) \geq p_i^*(t), \qquad X_i(t) = Y_i(t) - D_i^H(t) \geq 0, \qquad \left[p_i^H(t) - p_i^*(t)\right] X_i(t) = 0.$$

For nontradable sectors, $p_i^N(t)$ adjusts so that domestic use equals domestic output:

$$D_i(t) = Y_i(t).$$

This demand system replaces the constrained-CES allocation used in the previous formulation. Domestic scarcity affects the domestic-origin price $p_i^H(t)$, which changes both the relative price of the sectoral domestic-use composite and the allocation between domestic-origin and imported varieties within that composite.

H.4 Equilibrium of the Quantitative Open Economy

Definition 12 (Small open economy equilibrium with nontradables, sector-specific factors, and domestic-origin prices)**.** Let sectors be partitioned into tradable and nontradable sectors,

$$I = I^T \cup I^N, \qquad I^T \cap I^N = \emptyset.$$

Given initial wealth distributions $\{\omega_i(z,0)\}_{i\in I}$, wealth levels $\{A_i(0)\}_{i\in I}$, labor endowment $\bar{L}$, sector-specific factor endowments $\{\bar{H}_i\}_{i\in I}$, world-price sequences for tradable sectors $\{p_i^*(t)\}_{i\in I^T, t\geq 0}$, Armington home-bias parameters $\{\varphi_i\}_{i\in I^T}$, within-sector Armington elasticities $\{\eta_i^\star\}_{i\in I^T}$, and an exogenous sequence of resource net exports $\{NX_{\text{res}}(t)\}_{t\geq 0}$, a competitive small open economy equilibrium with financial autarky, sector-specific factors, nontradable sectors, exogenous resource income, and domestic-origin prices is a path of factor prices

$$\{w(t), r(t), q_i(t)\}_{i\in I, t\geq 0},$$

domestic-origin prices for tradable sectors

$$\{p_i^H(t)\}_{i\in I^T, t\geq 0},$$

domestic goods prices for nontradable sectors

$$\{p_i^N(t)\}_{i\in I^N, t\geq 0},$$

domestic-use prices

$$\{p_i^D(t)\}_{i\in I, t\geq 0},$$

allocation rules

$$\{k_i(z,t), \ell_i(z,t), h_i(z,t), c_i(z,t)\}_{i\in I, t\geq 0},$$

sectoral outputs, factor allocations, and wealth accumulation

$$\{Y_i(t), K_i(t), L_i(t), H_i(t), \dot{A}_i(t)\}_{i\in I, t\geq 0},$$

sectoral domestic use

$$\{D_i(t)\}_{i\in I, t\geq 0},$$

domestic-origin use, imports, exports, and net exports for tradable sectors

$$\{D_i^H(t), M_i(t), X_i(t), NX_i(t)\}_{i\in I^T, t\geq 0},$$

and wealth distributions

$$\{\omega_i(z,t)\}_{i\in I, t\geq 0},$$

such that:

1. Entrepreneurs optimize given producer prices and factor prices. The producer price relevant for sector i is

$$p_i^P(t) = \begin{cases} p_i^H(t), & i \in I^T, \\ p_i^N(t), & i \in I^N. \end{cases}$$

   Thus, tradable sectors face a domestic-origin producer price that is linked to the world price through the export condition below, while nontradable prices are determined domestically.

2. Sectoral output is generated by the production technology with sector-specific factors:

$$Y_i(t) = A(t) \int y_i(z,t)\, \omega_i(z,t)\, dz,$$

   where

$$y_i(z,t) = \big(zk_i(z,t)\big)^{\alpha_i} \ell_i(z,t)^{\beta_i} h_i(z,t)^{\gamma_i}, \qquad \alpha_i + \beta_i + \gamma_i = 1.$$

3. Factor markets clear:

$$\begin{aligned} \bar{L} &= A(t) \sum_{i\in I} \int \ell_i(z,t)\, \omega_i(z,t)\, dz, \\ A(t) &= A(t) \sum_{i\in I} \int k_i(z,t)\, \omega_i(z,t)\, dz, \\ \bar{H}_i &= A(t) \int h_i(z,t)\, \omega_i(z,t)\, dz, \qquad \forall i \in I. \end{aligned}$$

4. Domestic-use prices are determined by the sectoral price structure. For tradable sectors, domestic use combines domestic-origin and imported varieties through the Armington nest, so the domestic-use price is

$$p_i^D(t) = \left[\varphi_i \left(p_i^H(t)\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^*(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, \qquad i \in I^T,$$

   with the usual Cobb-Douglas limiting expression when $\eta_i^\star = 1$. For nontradable sectors,

$$p_i^D(t) = p_i^N(t), \qquad i \in I^N.$$

5. Aggregate expenditure on sectoral domestic use is

$$E(t) = w(t)\bar{L} + \sum_{i \in I} q_i(t)\bar{H}_i + \rho A(t) + NX_{\text{res}}(t) + \dot{A}(t) + \delta A(t),$$

where

$$A(t) = \sum_{i \in I} A_i(t), \qquad \dot{A}(t) = \sum_{i \in I} \dot{A}_i(t).$$

Given domestic-use prices, sectoral domestic use is determined by the CES final-composite demand system:

$$D(t) = \left[\sum_{i \in I} \upsilon_i^{1/\eta} D_i(t)^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)},$$

with domestic-use price index

$$P^D(t)^{1-\eta} = \sum_{i \in I} \upsilon_i \left[p_i^D(t) \right]^{1-\eta}.$$

Thus

$$D_i(t) = \upsilon_i \left(\frac{p_i^D(t)}{P^D(t)} \right)^{-\eta} \frac{E(t)}{P^D(t)}, \qquad i \in I.$$

6. For tradable sectors, domestic-use composites are allocated between domestic-origin and imported varieties according to the Armington demand system:

$$D_i^H(t) = \varphi_i \left(\frac{p_i^H(t)}{p_i^D(t)} \right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T,$$

and

$$M_i(t) = (1 - \varphi_i) \left(\frac{p_i^*(t)}{p_i^D(t)} \right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T.$$

The corresponding expenditure identity is

$$p_i^D(t) D_i(t) = p_i^H(t) D_i^H(t) + p_i^*(t) M_i(t).$$

7. Domestic output in tradable sectors is allocated between domestic-origin use and exports:

$$Y_i(t) = D_i^H(t) + X_i(t), \qquad i \in I^T.$$

Equivalently,

$$X_i(t) = Y_i(t) - D_i^H(t) \geq 0.$$

The domestic-origin price satisfies the world-price floor and export complementarity condi-

tion:

$$p_i^H(t) \geq p_i^*(t), \qquad \left[p_i^H(t) - p_i^*(t)\right] X_i(t) = 0, \qquad i \in I^T.$$

Thus, if a tradable sector exports, its domestic-origin price equals the world price. If its domestic-origin price exceeds the world price, domestic output is entirely absorbed by domestic-origin use.

8. Sectoral net exports in tradable sectors are gross exports minus gross imports:

$$NX_i(t) = X_i(t) - M_i(t), \qquad i \in I^T.$$

Because $D_i(t)$ is a CES composite quantity, $NX_i(t)$ is not generally equal to $Y_i(t) - D_i(t)$. Instead, the value identity implied by the Armington nest and the export complementarity condition is

$$p_i^*(t)NX_i(t) = p_i^H(t)Y_i(t) - p_i^D(t)D_i(t), \qquad i \in I^T.$$

9. Nontradable sectors satisfy

$$M_i(t) = X_i(t) = NX_i(t) = 0, \qquad i \in I^N.$$

Their goods markets clear domestically:

$$Y_i(t) = D_i(t), \qquad i \in I^N.$$

10. External balance holds for the complete economy:

$$\sum_{i \in I^T} p_i^*(t) \left[X_i(t) - M_i(t)\right] + NX_{\text{res}}(t) = 0.$$

11. Sectoral wealth evolves according to individual optimization:

$$\dot{A}_i(t) = \alpha_i p_i^P(t) Y_i(t) + r(t) \left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t), \qquad i \in I.$$

12. Wealth distributions evolve consistently with individual wealth accumulation and the productivity process.

The domestic-origin price system nests the standard small-open-economy case. If a tradable sector exports positive amounts, then $X_i(t) > 0$, and the complementarity condition implies

$$p_i^H(t) = p_i^*(t).$$

In that case, the domestic-origin price equals the world price, and the sector behaves like a standard

tradable sector. If domestic-origin demand exhausts domestic output, then exports are zero and $p_i^H(t)$ may exceed $p_i^*(t)$. The resulting domestic-origin scarcity premium raises producer revenue in the constrained sector and increases the domestic-use price $p_i^D(t)$, thereby reducing demand for the sectoral composite and shifting demand toward imported varieties within the sector.

The closed economy and the fully tradable small open economy are special cases of this definition. The closed economy corresponds to $I^N = I$, $I^T = \emptyset$, and $NX_{\text{res}}(t) = 0$. In that case all goods prices are determined domestically and goods-market clearing reduces to $Y_i(t) = D_i(t)$ for all sectors. The fully tradable small open economy corresponds to $I^T = I$ and $I^N = \emptyset$. In that case world import/export prices are exogenous for all sectors, while domestic-origin prices equal world prices whenever sectors export and may exceed world prices only when domestic-origin demand exhausts domestic output. The quantitative application uses the intermediate case with both tradable and nontradable sectors.

The resource-income term does not represent foreign borrowing or an exogenous current-account transfer. It is the net external income generated by a domestic sector omitted from the entrepreneurial production block. Financial autarky therefore continues to hold for the economy as a whole. Because the income is rebated to a hand-to-mouth resource owner, it affects modeled sectors through aggregate demand and equilibrium prices, wages, and factor rents, but does not directly alter entrepreneurs' wealth accumulation equations.

I Computation of Equilibrium and Dynamics

Equilibrium objects are computed in two steps at each point in time. First, conditional on the current distribution of wealth and productivities, I solve for the within-period equilibrium allocation. This fixed-point problem determines wages, interest rates, sector-specific factor rents, and the relevant goods prices, with the set of endogenous prices depending on the price regime. In the closed economy, all goods prices adjust to clear domestic goods markets. In the small open economy, world prices for tradable goods are taken as exogenous. In the quantitative open economy, world import/export prices discipline international transactions, nontradable prices adjust to clear domestic markets, and tradable domestic-origin prices adjust subject to the world-price floor and export complementarity conditions. Sector-specific factor rents clear fixed-factor markets. Domestic absorption is allocated across sectoral composites using a CES final-demand system, while tradable-sector domestic use is further allocated between domestic-origin and imported varieties through Armington nests. This step delivers factor prices, producer prices, domestic-use prices, sectoral output, domestic use, domestic-origin use, gross imports and exports, net exports, and the implied sectoral wealth dynamics at time t. Second, given these equilibrium objects, I update the distribution of wealth shares by solving the associated partial differential equation using a finite-difference approximation. The following subsections describe each of these steps in detail.

I.1 Equilibrium at point t

I.1.1 Closed Economy

Given sectoral wealth $\{A_i(t)\}$ and within-sector wealth distributions $\{o_i(z,t)\}$, the task is to find prices $\{p_i(t)\}_{i\in I}$, wage $w(t)$, and interest rate $r(t)$ such that all markets clear.

Prices are normalized so that the CES price index satisfies $P(t) = 1$.

The equilibrium is computed as a fixed point using the following steps:

1. **Initial guesses.** Guess $(w^0, r^0, \{p_i^0\})$, and normalize prices so that $P^0 = 1$.
2. **Static firm block.** Given prices, compute:
   - Profits:
   $$\pi_i^0 = p_i^0 \alpha_i \left(\frac{1-\alpha_i}{w^0/p_i^0}\right)^{\frac{1-\alpha_i}{\alpha_i}}$$
   - Productivity cutoffs:
   $$\underline{z}_i^0 = \frac{r^0 + \delta}{\pi_i^0}$$
   - Active-productivity integral:
   $$X_i^0 = \int_{\underline{z}_i^0}^{\infty} z\, o_i(z,t)\, dz$$

- Active mass:

$$1 - \Omega_i(\underline{z}_i^0)$$

- Sectoral productivity:

$$Z_i^0 = \left(\frac{X_i^0}{1 - \Omega_i(\underline{z}_i^0)} \right)^{\alpha_i}$$

- Selection wedge:

$$\zeta_i^0 = \frac{\underline{z}_i^0}{(Z_i^0)^{1/\alpha_i}}$$

3. **Factor demands and output.**

- Labor demand:

$$L_i^0 = \lambda_i A_i(t)\, X_i^0 \left(\frac{\pi_i^0}{p_i^0 \alpha_i} \right)^{\frac{1}{1-\alpha_i}}$$

- Capital demand:

$$K_i^0 = \lambda_i A_i(t)\, [1 - \Omega_i(\underline{z}_i^0)]$$

- Output:

$$Y_i^0 = Z_i^0 (K_i^0)^{\alpha_i} (L_i^0)^{1-\alpha_i}$$

4. **Wealth accumulation and absorption.**

- Sectoral wealth dynamics:

$$\dot{A}_i^0 = \alpha_i p_i^0 Y_i^0 + (r^0 - \rho) A_i(t) - (r^0 + \delta) K_i^0$$

- Total absorption:

$$P^0 D^0 = w^0 \bar{L} + \rho A(t) + \dot{A}(t) + \delta A(t)$$

- Sectoral demand:

$$D_i^0 = \upsilon_i \left(\frac{p_i^0}{P^0} \right)^{-\eta} D^0$$

5. **Excess demands.**

- Goods:

$$G_i^0 = Y_i^0 - D_i^0$$

- Labor:

$$L_{\text{diff}}^0 = \frac{\sum_i L_i^0 - \bar{L}}{\bar{L}}$$

- Capital:

$$K^0_{\text{diff}} = \frac{\sum_i K^0_i - A(t)}{A(t)}$$

6. **Update prices.**

   - Wage (labor clearing):

$$w^1 = w^0 \left(1 + \gamma_w L^0_{\text{diff}}\right)$$

   - Interest rate (capital clearing):

$$r^1 = r^0 + \gamma_r K^0_{\text{diff}}$$

   - Goods prices (goods clearing):

$$p^1_i = p^0_i \left(1 - \gamma_p \frac{G^0_i}{Y^0_i}\right)$$

   - Normalize prices so that $P^1 = 1$

7. **Iteration.** Repeat until convergence:

$$\max \left\{ |L_{\text{diff}}|, |K_{\text{diff}}|, \max_i |G_i/Y_i| \right\} < \varepsilon$$

I.1.2 Small Open Economy with Financial Autarky

Given sectoral wealth $\{A_i(t)\}$, within-sector wealth distributions $\{o_i(z,t)\}$, and exogenous goods prices $\{p^*_i\}_{i \in I}$, the task is to find the wage $w(t)$ and interest rate $r(t)$ such that factor markets clear.

Since goods prices are fixed, goods market clearing does not pin down prices, and the equilibrium reduces to clearing labor and capital markets given sectoral profitability.

The equilibrium is computed as follows:

1. **Initial guesses.** Guess (w^0, r^0).

2. **Static firm block.** Given (w^0, r^0) and world prices $\{p^*_i\}$, compute:

   - Profits:

$$\pi^0_i = p^*_i \alpha_i \left(\frac{1-\alpha_i}{w^0/p^*_i}\right)^{\frac{1-\alpha_i}{\alpha_i}}$$

   - Productivity cutoffs:

$$\underline{z}^0_i = \frac{r^0 + \delta}{\pi^0_i}$$

- Active-productivity integral:

$$X_i^0 = \int_{\underline{z}_i^0}^{\infty} z\, o_i(z,t)\, dz$$

- Active mass:

$$1 - \Omega_i(\underline{z}_i^0)$$

- Sectoral productivity:

$$Z_i^0 = \left(\frac{X_i^0}{1 - \Omega_i(\underline{z}_i^0)} \right)^{\alpha_i}$$

- Selection wedge:

$$\zeta_i^0 = \frac{\underline{z}_i^0}{(Z_i^0)^{1/\alpha_i}}$$

3. **Factor demands and output.**

- Labor demand:

$$L_i^0 = \lambda_i A_i(t)\, X_i^0 \left(\frac{\pi_i^0}{p_i^* \alpha_i} \right)^{\frac{1}{1-\alpha_i}}$$

- Capital demand:

$$K_i^0 = \lambda_i A_i(t)\, [1 - \Omega_i(\underline{z}_i^0)]$$

- Output:

$$Y_i^0 = Z_i^0 (K_i^0)^{\alpha_i} (L_i^0)^{1-\alpha_i}$$

4. **Factor market residuals.**

- Labor:

$$L_{\text{diff}}^0 = \frac{\sum_i L_i^0 - \bar{L}}{\bar{L}}$$

- Capital:

$$K_{\text{diff}}^0 = \frac{\sum_i K_i^0 - A(t)}{A(t)}$$

5. **Update factor prices.**

- Wage:

$$w^1 = w^0 \left(1 + \gamma_w L_{\text{diff}}^0\right)$$

- Interest rate:

$$r^1 = r^0 + \gamma_r K_{\text{diff}}^0$$

6. **Iteration.** Repeat until convergence:

$$\max\{|L_{\text{diff}}|, |K_{\text{diff}}|\} < \varepsilon$$

7. **Post-solution objects.**

   Given equilibrium (w, r), compute:

   - Sectoral wealth dynamics:

$$\dot{A}_i(t) = \alpha_i p_i^* Y_i(t) + (r(t) - \rho)A_i(t) - (r(t) + \delta)K_i(t)$$

   - Total absorption:

$$P^*(t)D(t) = w(t)\bar{L} + \rho A(t) + \dot{A}(t) + \delta A(t)$$

   - Sectoral absorption:

$$D_i(t) = \upsilon_i \left(\frac{p_i^*}{P^*(t)}\right)^{-\eta} D(t)$$

   - Net exports (residual):

$$NX_i(t) = Y_i(t) - D_i(t)$$

The quantitative application uses a richer hybrid environment than the two benchmark cases above. The next subsection describes the fixed-point procedure used for this environment.

I.1.3 Quantitative Open Economy with Nontradables, Sector-Specific Factors, and Domestic-Origin Prices

The quantitative implementation combines small-open-economy price discipline for international transactions with endogenous domestic-origin prices for tradable goods used domestically. Sectors are partitioned into tradable and nontradable sectors,

$$I = I^T \cup I^N, \qquad I^T \cap I^N = \emptyset.$$

For tradable sectors, the world import/export price sequence $\{p_i^*(t)\}_{i \in I^T}$ is taken as exogenous. Domestic producers in tradable sectors receive a domestic-origin price $p_i^H(t)$, which is bounded below by the world price and may exceed it when domestic-origin goods are scarce. For nontradable sectors, prices $p_i^N(t)$ adjust endogenously to clear domestic markets. The economy also includes sector-specific factors in fixed supply $\{\bar{H}_i\}_{i \in I}$, Armington home-bias parameters $\{\varphi_i\}_{i \in I^T}$, within-sector Armington elasticities $\{\eta_i^\star\}_{i \in I^T}$, and omitted resource-sector net exports $NX_{\text{res}}(t)$.

Given sectoral wealth $\{A_i(t)\}$, within-sector wealth distributions $\{o_i(z,t)\}$, tradable world prices $\{p_i^*(t)\}_{i \in I^T}$, and current resource net exports, the task is to find the wage $w(t)$, the interest rate $r(t)$,

sector-specific factor rents $\{q_i(t)\}_{i\in I}$, domestic-origin prices for tradable sectors $\{p_i^H(t)\}_{i\in I^T}$, and nontradable prices $\{p_i^N(t)\}_{i\in I^N}$ such that factor markets clear, nontradable goods markets clear, domestic-origin price complementarity holds in tradable sectors, and external balance holds for the complete economy.

For notational clarity in this subsection, I denote by $n_i(t)$ the active wealth share and by $e_i(t)$ the productivity-weighted active wealth share:

$$n_i(t) = 1 - \Omega_i(\underline{z}_i(t)), \qquad e_i(t) = \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz.$$

This avoids using $X_i(t)$ for both an active-productivity integral and gross exports.

The equilibrium at each point in time is computed as a fixed point using the following steps:

1. **Initial guesses.** Guess

$$\left(w^0, r^0, \{q_i^0\}_{i\in I}, \{p_i^{H,0}\}_{i\in I^T}, \{p_i^{N,0}\}_{i\in I^N}\right),$$

with

$$p_i^{H,0} \geq p_i^*(t), \qquad i \in I^T.$$

The producer price in each sector is

$$p_i^{P,0} = \begin{cases} p_i^{H,0}, & i \in I^T, \\ p_i^{N,0}, & i \in I^N. \end{cases}$$

The domestic-use price is

$$p_i^{D,0} = \begin{cases} \left[\varphi_i \left(p_i^{H,0}\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^*(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, & i \in I^T, \\ p_i^{N,0}, & i \in I^N. \end{cases}$$

with the usual Cobb-Douglas limiting expression when $\eta_i^\star = 1$.

2. **Static firm block.** Given $p_i^{P,0}$, w^0, r^0, and q_i^0, compute the reduced-form profit coefficient

$$\pi_i^0 = \alpha_i \left(p_i^{P,0}\right)^{1/\alpha_i} \left(\frac{\beta_i}{w^0}\right)^{\beta_i/\alpha_i} \left(\frac{\gamma_i}{q_i^0}\right)^{\gamma_i/\alpha_i}.$$

The productivity cutoff is

$$\underline{z}_i^0 = \frac{r^0 + \delta}{\pi_i^0}.$$

Then compute

$$n_i^0 = 1 - \Omega_i(\underline{z}_i^0), \qquad e_i^0 = \int_{\underline{z}_i^0}^{\infty} z o_i(z,t)\, dz,$$

and sectoral productivity

$$Z_i^0 = \left(\frac{e_i^0}{n_i^0}\right)^{\alpha_i}.$$

3. **Input demands and output.**

The input demand coefficients are

$$m_{\ell i}^0 = \left[\frac{p_i^{P,0}\beta_i}{w^0}\left(\frac{\gamma_i w^0}{\beta_i q_i^0}\right)^{\gamma_i}\right]^{1/\alpha_i},$$

and

$$m_{hi}^0 = \left[\frac{p_i^{P,0}\gamma_i}{q_i^0}\left(\frac{\beta_i q_i^0}{\gamma_i w^0}\right)^{\beta_i}\right]^{1/\alpha_i}.$$

Sectoral factor demands are

$$K_i^0 = \lambda_i A_i(t) n_i^0,$$

$$L_i^0 = m_{\ell i}^0 \lambda_i A_i(t) e_i^0,$$

and

$$H_i^{d,0} = m_{hi}^0 \lambda_i A_i(t) e_i^0.$$

Sectoral output is

$$Y_i^0 = Z_i^0 (K_i^0)^{\alpha_i} (L_i^0)^{\beta_i} (H_i^{d,0})^{\gamma_i}.$$

4. **Wealth accumulation and aggregate absorption expenditure.**

Sectoral wealth accumulation is

$$\dot{A}_i^0 = \alpha_i p_i^{P,0} Y_i^0 + (r^0 - \rho) A_i(t) - (r^0 + \delta) K_i^0.$$

Let

$$A(t) = \sum_{i \in I} A_i(t), \qquad \dot{A}^0 = \sum_{i \in I} \dot{A}_i^0.$$

Omitted resource net exports are treated as exogenous income rebated to a hand-to-mouth resource owner. In calibrated exercises, I write

$$NX_{\text{res}}^0(t) = \chi_{\text{res}}(t) \sum_{i \in I} p_i^{P,0} Y_i^0,$$

where $\chi_{\text{res}}(t)$ is the resource net-export share. Aggregate expenditure on sectoral domestic

use is

$$E^0 = w^0 \bar{L} + \sum_{i \in I} q_i^0 \bar{H}_i + \rho A(t) + NX_{\text{res}}^0(t) + \dot{A}^0 + \delta A(t).$$

5. **Final demand across sectoral domestic-use composites.**

   Given domestic-use prices $\{p_i^{D,0}\}_{i \in I}$ and expenditure E^0, compute the domestic-use price index

$$(P^{D,0})^{1-\eta} = \sum_{i \in I} \upsilon_i \left(p_i^{D,0}\right)^{1-\eta},$$

   with the usual Cobb-Douglas limiting expression when $\eta = 1$. Aggregate domestic use is

$$D^0 = \frac{E^0}{P^{D,0}}.$$

   Sectoral domestic use is

$$D_i^0 = \upsilon_i \left(\frac{p_i^{D,0}}{P^{D,0}}\right)^{-\eta} D^0, \qquad i \in I.$$

6. **Armington allocation and gross trade flows.**

   For tradable sectors, allocate sectoral domestic use between domestic-origin and imported varieties using the Armington demand system:

$$D_i^{H,0} = \varphi_i \left(\frac{p_i^{H,0}}{p_i^{D,0}}\right)^{-\eta_i^\star} D_i^0, \qquad i \in I^T,$$

   and

$$M_i^0 = (1 - \varphi_i) \left(\frac{p_i^*(t)}{p_i^{D,0}}\right)^{-\eta_i^\star} D_i^0, \qquad i \in I^T.$$

   The corresponding expenditure identity is

$$p_i^{D,0} D_i^0 = p_i^{H,0} D_i^{H,0} + p_i^*(t) M_i^0.$$

   Domestic output is allocated between domestic-origin use and exports:

$$X_i^0 = Y_i^0 - D_i^{H,0}, \qquad i \in I^T.$$

   Sectoral net exports are

$$NX_i^0 = X_i^0 - M_i^0, \qquad i \in I^T.$$

   For nontradable sectors,

$$M_i^0 = X_i^0 = NX_i^0 = 0.$$

7. **Equilibrium residuals.**

   The factor-market residuals are

$$L^0_{\text{diff}} = \frac{\sum_{i \in I} L^0_i - \bar{L}}{\bar{L}},$$

$$K^0_{\text{diff}} = \frac{\sum_{i \in I} K^0_i - A(t)}{A(t)},$$

   and

$$H^0_{\text{diff},i} = \frac{H^{d,0}_i - \bar{H}_i}{\bar{H}_i}, \qquad i \in I.$$

   For nontradable sectors, goods-market residuals are

$$G^{N,0}_i = Y^0_i - D^0_i, \qquad i \in I^N.$$

   For tradable sectors, the domestic-origin residual is

$$G^{H,0}_i = D^{H,0}_i - Y^0_i, \qquad i \in I^T.$$

   A positive value of $G^{H,0}_i$ indicates excess demand for the domestic-origin variety. A negative value indicates excess domestic output, which can be exported if the domestic-origin price is at the world-price floor.

   The external-balance residual is

$$TB^0 = \sum_{i \in I^T} p^*_i(t) \left[X^0_i - M^0_i \right] + NX^0_{\text{res}}(t).$$

   In a feasible equilibrium, this residual is zero up to numerical error. Equivalently, using the value identity implied by the Armington nest and export complementarity,

$$TB^0 = \sum_{i \in I^T} \left[p^{H,0}_i Y^0_i - p^{D,0}_i D^0_i \right] + NX^0_{\text{res}}(t).$$

8. **Update factor prices, nontradable prices, and domestic-origin prices.**

   Factor prices are updated using damped fixed-point steps:

$$w^1 = w^0 \exp\left(\gamma_w L^0_{\text{diff}}\right),$$

$$r^1 = r^0 + \gamma_r K^0_{\text{diff}},$$

   and

$$q^1_i = q^0_i \exp\left(\gamma_q H^0_{\text{diff},i}\right).$$

Nontradable prices are updated using the nontradable goods-market residuals:

$$p_i^{N,1} = p_i^{N,0} \exp\left(-\gamma_N \frac{G_i^{N,0}}{\max\{Y_i^0, D_i^0\}}\right), \qquad i \in I^N.$$

If $Y_i^0 > D_i^0$, there is excess supply of the nontradable good and its price falls. If $Y_i^0 < D_i^0$, there is excess demand and its price rises.

For tradable sectors, domestic-origin prices are updated using the domestic-origin residual:

$$p_i^{H,1} = \max\left\{p_i^*(t), p_i^{H,0} \exp\left(\gamma_H \frac{D_i^{H,0} - Y_i^0}{\max\{Y_i^0, D_i^{H,0}\}}\right)\right\}, \qquad i \in I^T.$$

If domestic-origin demand exceeds domestic output, $D_i^{H,0} > Y_i^0$, the domestic-origin price rises. If domestic output exceeds domestic-origin demand, $D_i^{H,0} < Y_i^0$, the domestic-origin price falls, but never below the world price $p_i^*(t)$. When $p_i^{H,0} = p_i^*(t)$ and domestic output exceeds domestic-origin demand, the surplus is exported.

9. **Iteration.**

   Repeat the previous steps until factor-market residuals, nontradable goods-market residuals, and tradable domestic-origin complementarity residuals are sufficiently small. The tradable complementarity conditions are

$$p_i^H(t) \geq p_i^*(t), \qquad X_i(t) = Y_i(t) - D_i^H(t) \geq 0, \qquad \left[p_i^H(t) - p_i^*(t)\right] X_i(t) = 0.$$

   A convenient normalized residual for the tradable domestic-origin block is

$$R_i^H = \frac{\max\{-X_i(t), 0\}}{\max\{Y_i(t), D_i^H(t)\}} + \frac{\max\{p_i^*(t) - p_i^H(t), 0\}}{p_i^*(t)} + \frac{\left[p_i^H(t) - p_i^*(t)\right] X_i(t)}{p_i^*(t) \max\{Y_i(t), D_i^H(t)\}}, \qquad i \in I^T.$$

   The fixed point is declared converged when

$$\max\left\{|L_{\text{diff}}|, |K_{\text{diff}}|, \max_i |H_{\text{diff},i}|, \max_{i \in I^N} \left|\frac{Y_i - D_i}{\max\{Y_i, D_i\}}\right|, \max_{i \in I^T} R_i^H\right\} < \varepsilon.$$

   The external-balance residual is reported as a diagnostic. When factor markets clear, nontradable goods markets clear, and the domestic-origin complementarity conditions hold, aggregate external balance follows from the absorption budget, the resource net-export term, and the value identity linking producer revenue, domestic absorption expenditure, and net exports.

10. **Post-solution objects.**

    Given the equilibrium objects at time t, compute sectoral wealth growth $\{\dot{A}_i(t)\}$, domestic

use $\{D_i(t)\}$, domestic-origin use $\{D_i^H(t)\}$, gross imports $\{M_i(t)\}$, gross exports $\{X_i(t)\}$, net exports $\{NX_i(t)\}$, and the saving policy

$$s_i(z,t) = \lambda_i \max\{\pi_i(t)z - (r(t)+\delta), 0\} + r(t) - \rho.$$

These objects are then used to update the distribution of sectoral wealth shares through the finite-difference approximation described below.

I.2 Solving the PDE via Approximation

1. Write equation G.1 as a generic second order PDE:

$$\frac{\partial \omega_i(z,t)}{\partial t} = u_i(z,t)\omega_i(z,t) + v_i(z)\frac{\partial \omega_i(z,t)}{\partial z} + x_i(z)\frac{\partial^2 \omega_i(z,t)}{\partial z^2} \tag{I.1}$$

2. Apply forward-difference approximation in time dimension and central-difference approximations in the productivity domain dimension:

$$\begin{aligned}
\frac{\partial \omega_i(z_k,t^n)}{\partial t} &\approx \frac{\omega_i(z_k,t^{n+1}) - \omega_i(z_k,t^n)}{\Delta t} \\
\frac{\partial \omega_i(z_k,t^n)}{\partial z} &\approx \frac{\omega_i(z_{k+1},t^n) - \omega_i(z_{k-1},t^n)}{2\Delta z} \\
\frac{\partial^2 \omega_i(z_k,t^n)}{\partial z^2} &\approx \frac{\omega_i(z_{k+1},t^n) - 2\omega_i(z_k,t^n) + \omega_i(z_{k-1},t^n)}{(\Delta z)^2}
\end{aligned}$$

where $\{z_k\}_{k=1}^K$ are the elements of the discrete grid for z and (t^n) is the discretized time sequence such that $t^{n+1} = t^n + \Delta t$.

3. Using these to approximate (I.1) in an implicit scheme (right-hand side evaluated in $n+1$):

$$\begin{aligned}
\frac{\omega_i(z_k,t^{n+1}) - \omega_i(z_k,t^n)}{\Delta t} &= u_i(z_k,t)\omega_i(z,t^n) \\
&\quad + v_i(z_k)\frac{\omega_i(z_{k+1},t^{n+1}) - \omega_i(z_{k-1},t^{n+1})}{2\Delta z} \\
&\quad + x_i(z_k)\frac{\omega_i(z_{k+1},t^{n+1}) - 2\omega_i(z_k,t^{n+1}) + \omega_i(z_{k-1},t^{n+1})}{(\Delta z)^2}
\end{aligned} \tag{I.2}$$

Rearranging:

$$\omega_i(z_{k-1},t^{n+1})f_i(z_k) + \omega_i(z_k,t^{n+1})g_i(z_k) + \omega_i(z_{k+1},t^{n+1})h_i(z_k) = \omega_i(z_k,t^n) \tag{I.3}$$

Where $f_i(z_k)$, $g_i(z_k)$, and $h_i(z_k)$ are the following functions of $a_i(z_k,t^n)$, $b_i(z_k)$, $c_i(z_k)$, Δt, and

Δz:

$$\begin{aligned} f_i(z_k) &= v_i(z_k)\frac{\Delta t}{2\Delta z} - x_i(z_k)\frac{\Delta t}{(\Delta z)^2} \\ g_i(z_k) &= 1 - u_i(z_k)\Delta t + 2x_i(z_k)\frac{\Delta t}{(\Delta z)^2} \\ h_i(z_k) &= -v_i(z_k)\frac{\Delta t}{2\Delta z} - x_i(z_k)\frac{\Delta t}{(\Delta z)^2} \end{aligned}$$

4. With boundary conditions $\omega_i(z_1, t^{n+1}) = 0$ and $\Delta z \sum_{k=1}^{K} \omega_i(z_k, t^{n+1}) = O_i(t^{n+1})$. The problem now becomes:

$$\begin{bmatrix} g_i(z_2) & h_i(z_2) & 0 & 0 & \dots & 0 \\ f_i(z_3) & g_i(z_3) & h_i(z_3) & 0 & \dots & 0 \\ 0 & f_i(z_4) & g_i(z_4) & h_i(z_4) & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ \dots & \dots & f_i(z_{K-2}) & g_i(z_{K-2}) & h_i(z_{K-2}) & 0 \\ \dots & \dots & \dots & f_i(z_{K-1}) & g_i(z_{K-1}) & h_i(z_{K-1}) \\ \Delta z & \Delta z & \dots & \dots & \Delta z & \Delta z \end{bmatrix} \begin{bmatrix} \omega_i(z_2, t^{n+1}) \\ \omega_i(z_3, t^{n+1}) \\ \omega_i(z_4, t^{n+1}) \\ \vdots \\ \omega_i(z_{K-2}, t^{n+1}) \\ \omega_i(z_{K-1}, t^{n+1}) \\ \omega_i(z_K, t^{n+1}) \end{bmatrix} = \begin{bmatrix} \omega_i(z_2, t^n) \\ \omega_i(z_3, t^n) \\ \omega_i(z_4, t^n) \\ \vdots \\ \omega_i(z_{K-2}, t^n) \\ \omega_i(z_{K-1}, t^n) \\ O_i(t^{n+1}) \end{bmatrix}$$

Which can be solved for $\omega_i(t^{n+1})$ by inverting the matrix in the left.

I.2.1 Specific Solution

From the OU Process for the log of productivity described in Appendix J, subsection J.1:

$$dz_i = \frac{1}{\theta_i}\left(-\log z_i + \frac{\sigma_i^2}{2}\right) z_i\, dt + \sigma_i \sqrt{1/\theta_i} z_i\, dW$$

Plugging $\mu_i(z_i) = \frac{1}{\theta_i}\left(-\log z_i + \frac{\sigma_i^2}{2}\right) z_i$ and $\sigma_i(z_i) = \sigma_i\sqrt{1/\theta_i} z_i$, the wealth share dynamics equation becomes:

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} + \frac{1}{\theta_i}\left(1 + \log z + \frac{\sigma_i^2}{2}\right)\right]\omega_i(z,t) + \frac{1}{\theta_i}\left(\log z + \frac{3}{2}\sigma_i^2\right) z \frac{\partial \omega_i(z,t)}{\partial t} + \frac{\sigma_i^2}{2\theta_i}\frac{\partial^2 \omega_i(z,t)}{\partial z^2}$$

That is: $u_i(z,t) = s_i(z,t) - \frac{\dot{A}(t)}{A(t)} + \frac{1}{\theta_i}\left(1 + \log z + \frac{\sigma_i^2}{2}\right)$, $v_i(z,t) = \frac{1}{\theta_i}\left(\log z + \frac{3}{2}\sigma_i^2\right) z$, and $x_i(z,t) = \frac{\sigma_i^2}{2\theta_i}$

J Numerical Implementation and Parametrization

J.1 Productivity Process

The theoretical model allows productivity to follow a general sector-specific ergodic diffusion,

$$dz_i(t) = \mu_i(z_i(t))\,dt + \sigma_i(z_i(t))\,dW_t.$$

In the quantitative exercises, I specialize this process to an Ornstein–Uhlenbeck process in log productivity. This specification captures two features that are useful for the simulations: productivity is persistent, but shocks are mean-reverting, so the cross-sectional distribution of productivity remains stationary over time.

Let

$$x_i(t) \equiv \log z_i(t).$$

Specifically, I assume that log productivity follows:

$$dx_i(t) = -\frac{1}{\theta_i} x_i(t)\,dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\,dW_t,$$

where $\theta_i > 0$ governs persistence and $\sigma_i > 0$ governs the dispersion of productivity shocks. A larger θ_i implies slower mean reversion and therefore more persistent productivity. The autocorrelation of log productivity over horizon s is

$$\operatorname{corr}\left(x_i(t), x_i(t+s)\right) = e^{-s/\theta_i}.$$

The stationary distribution of log productivity is

$$x_i \sim \mathcal{N}\left(0, \frac{\sigma_i^2}{2}\right).$$

Equivalently, productivity $z_i = \exp(x_i)$ has stationary density

$$\psi_i(z) = \frac{1}{z\sigma_i\sqrt{\pi}} \exp\left(-\frac{(\log z)^2}{\sigma_i^2}\right),$$

Thus, the parameter σ_i controls the stationary dispersion of log productivity, while θ_i controls the speed at which firms revert toward the sectoral mean.

To express the process in terms of $z_i(t)$, apply Itô's lemma to

$$z_i(t) = \exp(x_i(t)).$$

Since $f(x) = \exp(x)$ satisfies $f'(x) = f''(x) = \exp(x) = z_i(t)$, Itô's lemma[5] gives

$$dz_i(t) = z_i(t)\, dx_i(t) + \frac{1}{2} z_i(t)\, (dx_i(t))^2.$$

Using Itô multiplication rules[6] in the term $(dx_i(t))^2$, I obtain

$$dz_i(t) = \frac{1}{\theta_i}\left[-\log z_i(t) + \frac{\sigma_i^2}{2}\right] z_i(t)\, dt + \sigma_i \sqrt{\frac{1}{\theta_i}} z_i(t)\, dW_t.$$

Therefore, in the notation of the main text,

$$\mu_i(z) = \frac{1}{\theta_i}\left[-\log z + \frac{\sigma_i^2}{2}\right] z,$$

and

$$\sigma_i(z) = \sigma_i \sqrt{\frac{1}{\theta_i}} z.$$

[5] For a scalar Itô process $dx = \mu(x,t)\, dt + \sigma(x,t)\, dW_t$, Itô's lemma gives

$$df(x) = f'(x)\, dx + \frac{1}{2} f''(x)(dx)^2,$$

with $(dx)^2 = \sigma(x,t)^2 dt$ under the Itô rules $(dt)^2 = 0$, $dt\, dW_t = 0$, and $(dW_t)^2 = dt$.

[6] Starting from

$$dx_i(t) = -\frac{1}{\theta_i} x_i(t)\, dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\, dW_t,$$

we have

$$(dx_i(t))^2 = \left[-\frac{1}{\theta_i} x_i(t)\, dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\, dW_t\right]^2.$$

Using the rules

$$(dt)^2 = 0, \qquad dt\, dW_t = 0, \qquad (dW_t)^2 = dt,$$

only the diffusion term remains:

$$(dx_i(t))^2 = \sigma_i^2 \frac{1}{\theta_i} (dW_t)^2 = \frac{\sigma_i^2}{\theta_i}\, dt.$$

J.2 Baseline Model Parametrization

Table J.1: Baseline Parameters for Closed-Economy Simulations

Parameter / Object	Description	Baseline value
Panel A: Environment		
I	Number of sectors	2
$\bar{L}$	Aggregate labor endowment	1.0
Panel B: Preferences, Technology, and Constraints		
η	Elasticity of substitution across sectoral goods	2.0
υ_i	CES weights in final-good aggregator	$1/I$
ρ	Discount rate	0.05
δ	Depreciation rate	0.05
α_i	Capital share in sector i	0.50 (each sector)
λ_i	Collateral parameter in sector i	1.20 (each sector)
Panel C: Productivity Process and Initial Conditions		
θ_i	Mean-reversion parameter of log productivity	6.25 (each sector)
σ_i	Volatility of log productivity	0.65 (each sector)
$\mathrm{Corr}_i = e^{-1/\theta_i}$	One-period persistence	Implied
$\mathrm{std}_i = \sigma_i\sqrt{\theta_i/2}$	Stationary std. dev. of log productivity	Implied
$\rho_{a,z,i}(0)$	Initial correlation between wealth and productivity	0.50 (each sector)
$\mu_{\log z,i,0}$	Initial mean of log productivity	0.0
$\sigma_{\log z,i,0}$	Initial std. dev. of log productivity	std_i
$\mu_{\log a,i,0}$	Initial mean of log wealth	0.0
$\sigma_{\log a,i,0}$	Initial std. dev. of log wealth	std_i
$A_{i,0}$	Initial wealth in sector i	Implied by initialization
Panel D: Computational Parameters		
$z_{\min}$	Lower bound of productivity grid	10^{-4}
$z_{\max}$	Upper bound of productivity grid	$Q_{0.99}(\log\mathcal{N}(0, \max_i \mathrm{std}_i))$
N_z	Number of productivity grid points	1000
T	Simulation horizon	125
N_t	Number of time steps	1250
Δt	Time step size	$T/(N_t-1)$
$p_i(0)$	Initial guess for sectoral prices	1.0
$w(0)$	Initial wage guess	1.0
$r(0)$	Initial interest rate guess	0.03

Notes: Parameters follow standard values, and the productivity process generates persistent and dispersed firm-level heterogeneity. The one-sector and two-sector economies share the same aggregate wealth and marginal distributions, with asymmetry arising only from differences in the correlation between wealth and productivity across sectors.

The baseline parameterization is chosen to provide a transparent and symmetric benchmark for the quantitative exercises. Preference and technology parameters follow standard values in the macroeconomic literature. The parameters governing the productivity process are selected to generate persistent and dispersed productivity across entrepreneurs, consistent with empirical evidence on firm-level dynamics.

Initial conditions are constructed using sector-specific joint distributions over wealth and pro-

ductivity. For each sector i, I assume a lognormal specification for (a, z) characterized by $(\mu_{\log a,i,0}, \sigma_{\log a,i,0}, \mu_{\log z,i,0}, \sigma_{\log z,i,0})$ and a correlation parameter $\rho_{a,z,i}(0)$. This implies sector-specific wealth distributions and initial allocations of wealth across productivity levels.

To ensure comparability across specifications, I normalize initial wealth so that total wealth is fixed at a common value $A(0)$ and sectoral wealth shares are identical across sectors in the baseline. In the one-sector benchmark, I aggregate sectoral distributions so that both total initial wealth and the wealth-weighted productivity distribution coincide with those in the multisector economy. In the two-sector economy, asymmetry is introduced through differences in the correlation between wealth and productivity across sectors, holding marginal distributions and aggregate scale fixed.

J.3 Quantitative Model Calibration

This section describes the objects (to be) used to discipline the quantitative version of the model. The goal is to construct a disciplined baseline economy that matches the sectoral allocation, within-sector heterogeneity, and external shock exposure relevant for the Peruvian application. The calibration combines aggregate sectoral targets, firm-level moments, and normalizations used to initialize the hybrid open-economy model. Using these moments, the model is solved using the finite-difference and forward-simulation procedure described in Appendix I.

The calibration plan will be implemented prioritizing some elements before others. First, I will use economy-wide data sources to discipline the initial sectoral allocation and demand system: input-output tables and national accounts will inform sectoral absorption and output shares, while household and administrative sources discipline employment shares. Second, I will use firm-level EEA moments to discipline within-sector objects, including labor shares, productivity dispersion, and the initial alignment between productivity and wealth. Third, aggregate financial frictions and normalizations will be disciplined separately using accounting relationships, such as the capital-output ratio and external-finance targets. Parameters that are difficult to identify in isolation (including fixed-factor intensity, productivity persistence, and sector-specific collateral values) will be treated as benchmark choices with sensitivity exercises in a first implementation, and as objects to be disciplined jointly with dynamic moments in later implementations.

J.3.1 Sector structure and price regimes

The quantitative economy contains seven sectors: agroindustry, commerce, construction, fishing, machinery, (non-machinery) manufacturing, and (market) services. These sectors differ in their exposure to international prices. I therefore implement a "hybrid" small open-economy environment in which tradable sectors take external prices as given, while nontradable sectors clear domestically through endogenous relative prices.

In the SOE with non-tradables implementation, agroindustry, fishing, machinery, and manufacturing are treated as tradable sectors. Commerce, construction, and services are treated as nontradable. Commerce is interpreted primarily as domestic wholesale and retail distribution activity rather than as the production of internationally traded goods. Thus, its sectoral price is not assumed to be pinned directly by a world price. I also consider implementing robustness exercises in which commerce is classified as tradable.

J.3.2 Preferences, demand, and aggregate normalizations

I use the CES demand system defined in Section 2. This subsection describes how the demand-side objects are implemented in the quantitative exercise.

Table J.2: Sector structure and price regimes

Object	Description	Baseline treatment
I	Number of sectors	7
I^T	Tradable sectors	Agroindustry, Fishing, Machinery, Manufacturing
I^N	Nontradable sectors	Commerce, Construction, and Services
$p_i(t)$, $i \in I^T$	Tradable-sector price	$p_i(t) = p_i^*(t)$
$p_i(t)$, $i \in I^N$	Nontradable-sector price	Endogenous, clears $Y_i(t) = D_i(t)$
$NX_i(t)$	Net exports in sector i	$NX_i(t) = Y_i(t) - D_i(t)$

Elasticity of Substitution The elasticity of substitution across sectoral goods is set to $\eta = 0.99$ in the first calibration. This value is inside the CES parameter space but remains close to the Cobb–Douglas limit. Estimates in the structural-transformation literature for broad sectoral categories are generally below one, with magnitudes depending on whether sectoral goods are measured as final expenditure or value added.[7] I therefore treat $\eta = 0.99$ as a near-Cobb–Douglas benchmark and report sensitivity exercises for $\eta \in \{0.5, 0.8, 1.2, 1.5, 2.0\}$.

A complementary way to discipline η is to estimate the demand-side relationship implied by the CES aggregator using sectoral prices and expenditure shares. Let

$$s_{i,t} \equiv \frac{p_{i,t} D_{i,t}}{\sum_{j \in I} p_{j,t} D_{j,t}}$$

denote the expenditure share of sector i at time t. The CES demand system implies that, for any reference sector j,

$$\log \frac{s_{i,t}}{s_{j,t}} = \log \frac{\upsilon_i}{\upsilon_j} + (1 - \eta) \log \frac{p_{i,t}}{p_{j,t}}.$$

This relationship suggests estimating

$$\log \frac{s_{i,t}}{s_{j,t}} = \alpha_i + b \log \frac{p_{i,t}}{p_{j,t}} + \varepsilon_{i,t}, \qquad \widehat{\eta} = 1 - \widehat{b},$$

where α_i absorbs time-invariant differences in CES weights relative to the reference sector. In principle, this regression would provide a direct empirical estimate of the elasticity of substitution across the broad sectoral goods used in the model.

CES weights. The CES weights υ_i will be chosen to reproduce initial-period sectoral absorption shares. I will construct these shares using the 2007 Peruvian input-output tables from INEI. Let $E^D_{i,0}$ denote initial-period domestic absorption expenditure in sector i, aggregated to the seven sectors

[7] Kongsamut, Rebelo, and Xie (2001) use Stone–Geary preferences with implied intratemporal substitution across broad goods equal to 1. Herrendorf, Rogerson, and Valentinyi (2013) estimate an elasticity of substitution of 0.85 using U.S. final-expenditure data. Comin, Lashkari, and Mestieri (2021) report estimates close to one using expenditure data and lower estimates using value-added data. Boppart (2014) suggests an asymptotic value of 0.596 and notes that several calibrated values in the structural-transformation literature are below one.

used in the quantitative model. The target absorption share is

$$s_{i,0}^{D} \equiv \frac{E_{i,0}^{D}}{\sum_{j \in I} E_{j,0}^{D}}.$$

Given the initial sectoral price vector $\{p_{i,0}\}_{i \in I}$, the CES expenditure share of sector i is

$$s_{i,0}^{D} = \frac{\upsilon_i p_{i,0}^{1-\eta}}{\sum_{j \in I} \upsilon_j p_{j,0}^{1-\eta}}.$$

Therefore, imposing the normalization $\sum_{i \in I} \upsilon_i = 1$, I set

$$\upsilon_i = \frac{s_{i,0}^{D} p_{i,0}^{\eta-1}}{\sum_{j \in I} s_{j,0}^{D} p_{j,0}^{\eta-1}}.$$

This expression allows initial relative prices to differ across sectors. If all initial sectoral prices are equal, it reduces to $\upsilon_i = s_{i,0}^{D}$.

I will construct $E_{i,0}^{D}$ from domestic final absorption rather than sectoral value added whenever possible. This distinction is important for tradable sectors, for which domestic production and domestic absorption can differ because of imports and exports. In the first implementation, I will use absorption shares from the input-output table at basic prices, aggregated to the model sectors. Purchaser-price shares, which include trade margins, transport margins, and taxes, can be used as a robustness check. This choice is particularly relevant for Commerce, which is going to be treated as a nontradable distribution sector in the model.

Price index and numeraire. In the hybrid numerical implementation, the level of tradable prices pins down the numeraire. I will normalize the initial external price of one tradable sector, agroindustry, to one:

$$p_{\text{agro},0}^{*} = 1.$$

All other initial sectoral prices are measured relative to this price. For tradable sectors, domestic prices are given by the external price vector,

$$p_{i,t} = p_{i,t}^{*}, \qquad i \in I^{T},$$

while nontradable prices are determined by domestic market clearing.

The aggregate CES price index is therefore not normalized to one inside the hybrid solver. Instead, it is computed from the equilibrium sectoral price vector,

$$P(t) = \left(\sum_{i \in I} \upsilon_i p_i(t)^{1-\eta} \right)^{1/(1-\eta)}.$$

Movements in $P(t)$ reflect changes in the real cost of the composite absorption good relative to the chosen tradable numeraire, not monetary inflation. Since the model is real, I will report economically meaningful prices in relative terms, such as

$$\frac{p_i(t)}{P(t)}, \qquad \frac{w(t)}{P(t)}, \qquad \frac{q_i(t)}{P(t)}.$$

Aggregate quantities and absorption. Aggregate labor is normalized to

$$\bar{L} = 1.$$

Wealth and capital are measured in units of the model numeraire. In the hybrid implementation, this numeraire is pinned by the external price vector for tradable goods. The CES price index $P(t)$ converts quantities of the composite good into expenditure measured in the same numeraire units. Thus, $C(t)$ and $D(t)$ denote quantities of the composite good, while $P(t)C(t)$ and $P(t)D(t)$ denote the corresponding expenditures.

Aggregate consumption expenditure includes labor income, specific-factor income, and entrepreneurial consumption. In the sector-specific-factor version of the model, this gives

$$P(t)C(t) = w(t)\bar{L} + \sum_{i \in I} q_i(t)\bar{H}_i + \rho A(t),$$

where $w(t)\bar{L}$ is labor income, $\sum_i q_i(t)\bar{H}_i$ is specific-factor income, and $\rho A(t)$ is entrepreneurial consumption expenditure. The last term uses the fact that entrepreneurs consume a fraction ρ of their wealth, with wealth measured in numeraire units.

Aggregate absorption expenditure includes current consumption, net wealth accumulation, and capital replacement:

$$P(t)D(t) = P(t)C(t) + \dot{A}(t) + \delta K(t).$$

The capital replacement term is measured in numeraire units because the price of the homogeneous mobile capital good is normalized to one. When the aggregate capital market clears with $K(t) = A(t)$, this term can equivalently be written as $\delta A(t)$.

Table J.3: Preferences, demand, and aggregate normalizations

Object	Description	Initial-period value / discipline
η	Elasticity of substitution across sectoral goods	Set to 0.99 in the main calibration; sensitivity exercises use $\eta \in \{0.5, 0.8, 1.5, 2.0\}$
$E^D_{i,0}$	Domestic absorption expenditure in sector i	Constructed from the 2007 Peruvian input-output tables from INEI, aggregated to the seven model sectors
$s^D_{i,0}$	Initial-period absorption share	Defined as $s^D_{i,0} = E^D_{i,0} / \sum_{j \in I} E^D_{j,0}$
υ_i	CES demand weights	Chosen to reproduce initial absorption shares given initial sectoral prices, imposing $\sum_i \upsilon_i = 1$
$p^*_{\text{agro},0}$	Numeraire price	Initial external price of agroindustry normalized to one
$p^*_{i,t}$	External price index for tradable sector i	Exogenous path for tradable prices, measured relative to the chosen tradable numeraire
$P(t)$	Aggregate CES price index	Computed from equilibrium sectoral prices; not normalized to one inside the hybrid solver
ρ	Discount rate and entrepreneurial consumption rate	0.05
δ	Depreciation rate	0.10
$\bar{L}$	Aggregate labor endowment	Normalized to 1
$A(t), K(t)$	Aggregate wealth and mobile capital	Measured in units of the model numeraire; real values in units of the composite good are $A(t)/P(t)$ and $K(t)/P(t)$
$\sum_i q_i(t)\bar{H}_i$	Specific-factor income	Included in aggregate consumption expenditure in the sector-specific-factor implementation

J.3.3 Production technology and sector-specific factors

The quantitative model uses a sector-specific-factor technology,

$$y_{fi} = (z_{fi} k_{fi})^{\alpha_i} \ell_{fi}^{\beta_i} h_{fi}^{\gamma_i}, \qquad \alpha_i + \beta_i + \gamma_i = 1.$$

The firm-level technology remains constant returns to scale, while the fixed supply of the sector-specific factor generates decreasing returns at the sector level. This structure preserves the linearity of firm policies in wealth while limiting corner specialization in the open-economy environment.

Production choices are also shaped by the collateral constraint. Entrepreneur f in sector i can operate capital only up to a multiple of wealth,

$$k_{fi} \leq \lambda_i a_{fi}.$$

The parameter λ_i governs the tightness of financial frictions in sector i: higher values of λ_i allow entrepreneurs to scale up more for a given level of wealth. In the main implementation, I keep λ_i common across sectors in order to isolate the role of the wealth-productivity distribution and price regime. Sector-specific values of λ_i can be introduced in robustness exercises or in a richer calibration that allows financial frictions to differ across sectors.

Production elasticities. The labor share is derived from firm-level data. For each firm f in sector i, I compute the firm-level labor share as labor compensation relative to value added,

$$\beta_{fi} = \frac{w_{fi} L_{fi}}{p_i Y_{fi}}.$$

I then set the sectoral labor elasticity β_i equal to a sector-level moment of this firm-level distribution,

$$\beta_i = \mathcal{M}_{f \in i} \left(\beta_{fi} \right),$$

where $\mathcal{M}_{f \in i}(\cdot)$ denotes either the mean or the median across firms in sector i. The median specification is less sensitive to outliers and measurement error in value added, while the mean gives the direct average firm-level labor share.

The remaining share, $1 - \beta_i$, is split between mobile capital and the sector-specific factor. To organize this split, I define

$$\chi_i \equiv \frac{\gamma_i}{1 - \beta_i},$$

where χ_i is the fraction of non-labor income attributed to the sector-specific factor. The production elasticities are then constructed as

$$\gamma_i = \chi_i (1 - \beta_i), \qquad \alpha_i = (1 - \chi_i)(1 - \beta_i).$$

This parameterization guarantees that $\alpha_i + \beta_i + \gamma_i = 1$ and allows the measured labor share to vary across sectors.

I will work two implementations of this parametrization.

- In the first implementation, I will treat χ_i as a calibrated fixed-factor intensity profile. The values are chosen to reflect the relative importance of non-reproducible or slowly adjustable factors across sectors. As a reference, for sectors ordered

  $$(\text{Agroindustry}, \text{Commerce}, \text{Construction}, \text{Fishing}, \text{Machinery}, \text{Manufacturing}, \text{Services}),$$

  I will use

  $$\chi = (0.35,\ 0.25,\ 0.30,\ 0.45,\ 0.15,\ 0.10,\ 0.15).$$

  as place-holder values. Fixed factors are more important in sectors such as fishing and agroindustry, where natural resources, land, water, licenses, and location-specific constraints are central. They are less important in manufacturing, where production is more naturally scalable with mobile capital and labor. Commerce and services are assigned intermediate fixed-factor intensities, reflecting the role of distribution networks, local market access, locations, and sector-specific intangible capacity. I will also report sensitivity exercises that vary the fixed-factor intensity profile, including the limiting case $\chi_i = 0$ for all sectors, which nests

the model without sector-specific factors.

- A more structural calibration would discipline χ_i jointly with other dynamic parameters of the model. This is important because fixed-factor intensity affects the same moments as financial frictions, productivity persistence, and the initial wealth-productivity distribution. For example, slow sectoral responses to external demand or price shocks could reflect strong sector-level congestion, tight collateral constraints, persistent productivity differences, or initially poor wealth-productivity alignment. In a later implementation, I therefore will choose χ_i jointly with the financial and productivity parameters to match dynamic moments such as sectoral responses of output, employment, capital, and wealth shares to external shocks:

$$\widehat{\Theta} = \arg\min_{\Theta} \left[m^{data} - m^{model}(\Theta)\right]' W \left[m^{data} - m^{model}(\Theta)\right],$$

  where Θ includes the fixed-factor intensity parameters together with the financial and productivity parameters. In the first implementation, I instead use external discipline and sensitivity analysis for χ_i, leaving the joint moment-based calibration as a refinement once the rest of the quantitative structure is fixed, and the ideas in J.3.6 are implemented.

Collateral parameter. I consider both common and sector-specific versions.

- In the main implementation, I keep the collateral parameter common across sectors, $\lambda_i = \lambda$, in order to avoid attributing persistent specialization directly to exogenous sectoral differences in credit access. I discipline λ using the aggregate external-finance logic in the JMP version of Moll (2014). In a model with collateral constraints of the form $k_{fi} \leq \lambda a_{fi}$, the fraction of capital financed externally is

$$\frac{D}{K} = 1 - \frac{1}{\lambda}.$$

  Given an aggregate capital-output ratio, this implies

$$\frac{D}{Y} = \left(1 - \frac{1}{\lambda}\right)\frac{K}{Y},$$

  so that an empirical target for external finance relative to output, together with the calibrated capital-output ratio, pins down the aggregate value of λ:

$$\lambda = \frac{1}{1 - \frac{D/Y}{K/Y}}.$$

  This provides a transparent way to discipline the overall tightness of financial frictions without introducing sector-specific borrowing wedges in the main calibration.

- In a later extension, I allow λ_i to vary across sectors, using sectoral external-finance-to-capital

ratios when they can be measured credibly:

$$\lambda_i = \frac{1}{1 - \frac{D_i}{K_i}}.$$

Equivalently, when the data are organized in output ratios, this can be written as

$$\lambda_i = \frac{1}{1 - \frac{D_i/Y_i}{K_i/Y_i}}.$$

I will treat this sector-specific version as a refinement because sectoral borrowing capacity can become an independent source of persistent specialization.

Table J.4: Technology and financial frictions

Object	Description	Main implementation / discipline
β_{fi}	Firm-level labor share	Computed as labor compensation relative to firm value added, $\beta_{fi} = w_{fi}L_{fi}/(p_iY_{fi})$
β_i	Sectoral labor elasticity	Set equal to a sector-level moment of firm-level labor shares, $\beta_i = \mathcal{M}_{f\in i}(\beta_{fi})$; main implementation uses the median, with robustness using the mean
χ_i	Fixed-factor intensity within the non-labor share	Calibrated as a sector-specific profile in the first implementation; reference profile $(0.35, 0.25, 0.30, 0.45, 0.15, 0.10, 0.15)$ for (Agroindustry, Commerce, Construction, Fishing, Machinery, Manufacturing, Services); later disciplined jointly with dynamic moments
γ_i	Sector-specific factor share	Constructed as $\gamma_i = \chi_i(1 - \beta_i)$
α_i	Mobile-capital share	Constructed as $\alpha_i = (1-\chi_i)(1-\beta_i)$, so that $\alpha_i+\beta_i+\gamma_i = 1$
$\bar{H}_i$	Sector-specific factor endowment	Fixed supply in each sector; generates decreasing returns at the sector level while preserving firm-level constant returns
λ_i	Collateral parameter	Common across sectors in the main implementation, $\lambda_i = \lambda$; disciplined using aggregate external finance and capital-output targets following Moll (2014)
λ_i by sector	Sector-specific collateral parameter	Later extension using sectoral external-finance-to-capital ratios, $\lambda_i = 1/(1 - D_i/K_i)$, when these moments can be measured credibly

J.3.4 Productivity process and initial distributions

Productivity follows a sector-specific mean-reverting process in log productivity. Let

$$x_{fit} \equiv \log z_{fit}.$$

I assume that log productivity follows

$$dx_{fit} = -\frac{1}{\theta_i} x_{fit}\, dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\, dW_{fit},$$

where θ_i governs persistence and σ_i governs the stationary dispersion of productivity up to a scale factor. Under this convention, the one-period persistence of log productivity is

$$\mathrm{Corr}_i = \exp(-1/\theta_i),$$

and the stationary standard deviation is

$$\mathrm{sd}_i = \frac{\sigma_i}{\sqrt{2}}.$$

This normalization separates persistence from stationary dispersion: changing θ_i changes the autocorrelation of productivity without changing its stationary variance for a fixed value of σ_i.

In the first implementation, I use common persistence and volatility parameters across sectors. In the calibrated version, I will allow these parameters to vary by sector and discipline them using firm-level productivity proxies constructed within sector-year cells.

A complication is that firm-level use of the sector-specific factor is not observed. Moreover, value added in the data is observed in nominal terms. The empirical object is therefore

$$va_{fit}^{nom} = p_{it} \left(z_{fit} k_{fit}\right)^{\alpha_i} \ell_{fit}^{\beta_i} h_{fit}^{\gamma_i},$$

where p_{it} is the sector-year price of value added. A residual based only on capital and labor,

$$\log va_{fit}^{nom} - \alpha_i \log k_{fit} - \beta_i \log \ell_{fit},$$

therefore does not separately identify productivity from unobserved specific-factor use or from sector-year price components.

However, the model implies a useful within-sector-year return-shifter proxy. Conditional on sector-year prices and factor prices, optimal input choices satisfy

$$\ell_{fit} = m_{it}^{L} z_{fit} k_{fit}, \qquad h_{fit} = m_{it}^{H} z_{fit} k_{fit},$$

where m_{it}^{L} and m_{it}^{H} are sector-year components common across firms within sector i and year t.

Using firm-level constant returns to scale,

$$va_{fit}^{nom} = p_{it} \left(z_{fit} k_{fit}\right)^{\alpha_i} \ell_{fit}^{\beta_i} h_{fit}^{\gamma_i},$$

then implies

$$\log va_{fit}^{nom} - \log k_{fit} = x_{fit} + \kappa_{it},$$

where

$$\kappa_{it} = \log p_{it} + \beta_i \log m_{it}^{L} + \gamma_i \log m_{it}^{H}$$

is a sector-year component. This motivates the main productivity proxy

$$\widehat{x}_{fit} = \log va_{fit}^{nom} - \log k_{fit} - \overline{\left(\log va_{fit}^{nom} - \log k_{fit}\right)}_{it}.$$

Because the proxy is demeaned within sector-year cells, it removes sector-year price variation and common factor-price components. This object is a within-sector-year return shifter instead of a fully structural measure of physical productivity, but allows disciplining the model's within-sector productivity distribution without pre-specified values of α_i, β_i, γ_i, and specific factor use.

The model also implies a labor-capital relationship:

$$\ell_{fit} = m_{it}^{L} z_{fit} k_{fit}.$$

This motivates an alternative within-sector-year proxy that can be used as a robustness check:

$$\widehat{x}_{fit}^{\ell} = \log \ell_{fit} - \log k_{fit} - \overline{\left(\log \ell_{fit} - \log k_{fit}\right)}_{it}.$$

I will use the distribution of $\widehat{x}_{fit}$ to discipline the initial productivity distribution within each sector. In particular, the within-sector dispersion of $\widehat{x}_{fit}$ informs the stationary dispersion of log productivity. When firm identifiers can be linked reliably over time, sector-specific persistence can be estimated from

$$\widehat{x}_{fi,t+1} = \rho_i^x \widehat{x}_{fit} + \varepsilon_{fi,t+1}, \qquad \theta_i = -\frac{1}{\log \rho_i^x}.$$

When panel links are not reliable, I treat productivity persistence as externally calibrated and use the repeated cross-section of $\widehat{x}_{fit}$ mainly to discipline within-sector dispersion.

The initial joint distribution of productivity and wealth is disciplined using firm-level balance-sheet moments. Let a_{fit} denote the empirical wealth or asset proxy. I use moments such as

$$\mathrm{sd}(\widehat{x}_{fit}), \qquad \mathrm{corr}(\widehat{x}_{fit}, \log a_{fit}), \qquad \mathrm{corr}(\widehat{x}_{fit}, \log k_{fit})$$

within sector-year cells. The correlation between productivity and wealth, $\mathrm{corr}(\widehat{x}_{fit}, \log a_{fit})$, disciplines the initial wealth-productivity alignment in each sector. This moment is central because

the model's dynamic mechanism depends on whether wealth is initially held by more or less productive entrepreneurs. The correlation between productivity and capital is used as an additional diagnostic, since capital choices are endogenous in the model and reflect both productivity and borrowing constraints.

Table J.5: Productivity process and initial distributions

Object	Description	Main implementation / discipline
x_{fit}	Log productivity	Evolves according to a sector-specific mean-reverting process
θ_i	Mean-reversion parameter of log productivity	Common across sectors in the first implementation; later disciplined by firm-level persistence when panel links are reliable
σ_i	Innovation volatility of log productivity	Common across sectors in the first implementation; later chosen to match within-sector dispersion of productivity proxies
$\mathrm{Corr}_i = \exp(-1/\theta_i)$	One-period persistence of log productivity	Implied by θ_i under annual time steps
$\mathrm{sd}_i = \sigma_i/\sqrt{2}$	Stationary dispersion of log productivity	Implied by σ_i under the stated OU normalization
$\widehat{x}_{fit}$	Firm-level return-shifter proxy	Demeaned $\log va_{fit} - \log k_{fit}$ within sector-year cells
$\mathrm{sd}(\widehat{x}_{fit})$	Within-sector productivity dispersion	Used to discipline sectoral dispersion of log productivity
$\rho_{a,z,i}(0)$	Initial wealth-productivity alignment	Disciplined by $\mathrm{corr}(\widehat{x}_{fit}, \log a_{fit})$ within sector-year cells
$\mathrm{corr}(\widehat{x}_{fit}, \log k_{fit})$	Capital-productivity alignment	Used as a diagnostic moment for the interaction between productivity and capital allocation
$O_i(0)$	Initial sectoral wealth share	Disciplined by sectoral assets, net worth, or balance-sheet proxies

J.3.5 Sectoral aggregate targets

The model is initialized to match sectoral aggregates in an initial period. These targets discipline the initial allocation of output, labor, capital, and wealth across sectors. I distinguish between economy-wide sectoral aggregates and firm-level moments. Economy-wide shares are to be disciplined primarily with national accounts, input-output tables, and household-survey employment data. Firm-level EEA data are used to discipline within-sector moments, such as labor shares, capital proxies, assets, and the distribution of firm-level productivity and wealth. This distinction is important because the EEA covers formal medium and large firms and therefore may not coincide exactly with economy-wide sectoral aggregates.

The main initial targets are sectoral shares of output, labor, capital, and wealth:

$$\frac{Y_i(0)}{Y(0)}, \qquad \frac{L_i(0)}{\bar{L}}, \qquad \frac{K_i(0)}{K(0)}, \qquad \frac{A_i(0)}{A(0)}.$$

Output and labor shares anchor the initial sectoral composition of production. Capital shares discipline the allocation of the mobile capital input across sectors. Wealth shares discipline the initial distribution of entrepreneurial assets across sectors, which is central for the model's dynamic mechanism. Since the model features a homogeneous mobile capital good, aggregate capital-market clearing requires

$$K(0) = A(0),$$

but the sectoral allocation of capital need not coincide mechanically with the sectoral allocation of wealth. Differences between $K_i(0)$ and $A_i(0)$ reflect the initial allocation of capital across sectors relative to the distribution of entrepreneurial wealth.

I will use national accounts or input-output data for economy-wide sectoral output shares, ENAHO or national accounts for employment shares, and EEA balance-sheet information for capital and wealth proxies. The EEA moments are interpreted as informative about the formal firm sector rather than as exact economy-wide aggregates. For this reason, I will use economy-wide sources when possible to anchor sector sizes and use the EEA primarily to discipline within-sector production and balance-sheet moments.

Table J.6: Sectoral aggregate targets

Object	Description	Discipline
$Y_i(0)$	Initial sectoral output or value added	National accounts and input-output tables; aggregated to the seven model sectors
$Y_i(0)/Y(0)$	Initial sectoral output share	Main target for the initial sectoral composition of production
$L_i(0)$	Initial sectoral labor allocation	ENAHO, national accounts employment, and EEA employment moments
$L_i(0)/\bar{L}$	Initial sectoral employment share	Main target for the allocation of the normalized aggregate labor endowment
$K_i(0)$	Initial sectoral mobile capital	EEA fixed assets or capital-stock proxies; cross-checked against national accounts when available
$K_i(0)/K(0)$	Initial sectoral capital share	Used to discipline the initial allocation of the homogeneous mobile capital input
$A_i(0)$	Initial sectoral entrepreneurial wealth	EEA assets, net worth, or balance-sheet proxies
$A_i(0)/A(0)$	Initial sectoral wealth share	Disciplines the initial distribution of entrepreneurial wealth across sectors
β_i	Sectoral labor elasticity	Constructed from firm-level labor shares in the EEA; used in the production block
$K(0)/Y(0)$	Aggregate capital-output ratio	Used to scale aggregate capital and to discipline the collateral parameter jointly with external-finance targets
$A(0)$	Aggregate entrepreneurial wealth	Chosen so that aggregate capital-market clearing holds, $K(0) = A(0)$, given the model normalization

The calibration will prioritize matching economy-wide sectoral shares for output and employment. Capital and wealth shares are measured with more error and are therefore used both as

targets and as diagnostic moments. In particular, the EEA provides useful balance-sheet proxies, but these are not necessarily representative of all firms in the economy. I will interpret discrepancies between national accounts targets and EEA-based capital or wealth measures as informative about coverage and measurement.

Aggregate labor is normalized to $\bar{L} = 1$. Empirical labor moments will be mapped into sectoral employment shares or changes relative to aggregate employment. Allowing for an exogenous path of aggregate labor supply is a straightforward extension, but I abstract from it in the first implementations.

J.3.6 External shocks and dynamic moments

For tradable sectors, external conditions are summarized by sector-level price or demand shifters constructed from international trade data. In the hybrid model, tradable-sector prices are exogenous:

$$p_{i,t} = p^*_{i,t}, \qquad i \in I^T.$$

The level of the external price vector pins down the numeraire. In the numerical implementation, I normalize the initial external price of agroindustry to one,

$$p^*_{\text{agro},0} = 1,$$

and measure other external price indices relative to this price. I do not normalize all external prices to one, since initial relative prices across tradable sectors are economically meaningful.

In the empirical implementation, external shocks are constructed using international trade data. For each tradable sector, I use foreign demand conditions in Peru's export destinations, excluding Peru itself from the construction of world demand. A fixed-weight demand shock takes the form

$$\text{DemandShock}_{i,t} = \sum_d \omega_{id,0} \Delta \log M_{dit}^{World \setminus Peru},$$

where $\omega_{id,0}$ is the initial-period exposure of sector i to destination d, and $M_{dit}^{World \setminus Peru}$ denotes destination d's imports of goods in sector i from the rest of the world excluding Peru. I also construct a dynamic-exposure version,

$$\text{DynamicDemandShock}_{i,t} = \sum_d \omega_{id,t-1} \Delta \log M_{dit}^{World \setminus Peru},$$

where exposure weights are updated using lagged trade shares.

These shocks are used in two ways. First, they provide empirical discipline for sectoral dynamic responses. I estimate how sectoral output, labor, capital, and wealth shares respond to favorable

external demand changes over different horizons:

$$\Delta_h \log S_{i,t+h} = \mu_i + \lambda_t + \beta_h \text{Shock}_{i,t} + \varepsilon_{i,t+h},$$

where $S_{i,t}$ denotes a sectoral outcome such as output, employment, capital, or wealth shares. These moments discipline the persistence and magnitude of sectoral reallocation in the model.

Second, the shocks discipline heterogeneous firm responses within sectors. The model predicts that favorable external conditions should lead to stronger expansion among more productive firms, especially when wealth and productivity are initially aligned. I therefore estimate specifications of the form

$$\Delta_h \log X_{fi,t+h} = \mu_i + \lambda_t + \beta_h \text{Shock}_{i,t} + \gamma_h \widehat{x}_{fit} + \delta_h \left(\text{Shock}_{i,t} \times \widehat{x}_{fit}\right) + \varepsilon_{fi,t+h},$$

where X_{fit} denotes firm-level capital, employment, assets, or output, and $\widehat{x}_{fit}$ is the firm-level return-shifter proxy defined above. The interaction coefficient δ_h captures whether more productive firms expand more strongly after favorable external shocks.

In the quantitative exercises, external shocks can be mapped into paths for tradable prices or treated as empirical moments used to evaluate the model's transition dynamics as discussed in J.3.3. The main role of these moments will be to discipline the joint implications of productivity persistence, financial frictions, fixed-factor intensity, and the initial wealth-productivity distribution.

Table J.7: External shocks and dynamic moments

Object	Description	Discipline
$p_i^*(t)$	External price path for tradable sector i	Exogenous path in the hybrid model; measured relative to the tradable numeraire, with $p^*_{\text{agro},0} = 1$
$\text{DemandShock}_{i,t}$	Fixed-weight external demand shock	Constructed from BACI destination-level import demand excluding Peru, using initial exposure weights
$\text{DynamicDemandShock}_{i,t}$	Lagged-exposure external demand shock	Constructed from BACI destination-level import demand excluding Peru, using lagged exposure weights
$\Delta_h \log S_{i,t+h}$	Sectoral response over horizon h	Local projections for sectoral output, employment, capital, or wealth shares after external shocks
$\Delta_h \log K_{fi,t+h}$	Firm capital expansion	Firm-level EEA regressions after external shocks
$\Delta_h \log L_{fi,t+h}$	Firm labor expansion	Firm-level EEA regressions after external shocks
$\Delta_h \log A_{fi,t+h}$	Firm asset or wealth expansion	Firm-level EEA regressions using assets, net worth, or balance-sheet proxies
$\text{Shock}_{i,t} \times \widehat{x}_{fit}$	Heterogeneous response by productivity	Tests whether more productive firms expand more after favorable external shocks

J.3.7 Parameter values used in the first implementation

The values reported in this subsection are those used in the preliminary quantitative exercise presented in subsection 5.1. They combine three types of inputs. First, several common parameters are normalizations or provisional benchmark choices, including the elasticity of substitution, discount and depreciation rates, aggregate labor supply, and the common collateral parameter. Second, sector-specific production and productivity parameters are constructed using firm-level moments together with a provisional profile of fixed-factor intensities. Third, initial sectoral wealth shares and aggregate sectoral moments are used as initial conditions, calibration targets, or untargeted validation moments.

These values should therefore not be interpreted as the outcome of a joint moment-matching exercise. The purpose of the first implementation is to verify that the seven-sector hybrid economy can reproduce the targeted absorption shares, generate stable transition dynamics, and identify the parameters that must subsequently be adjusted to discipline employment allocations and sectoral trade balances. Table J.8 reports the parameters that are common across sectors.

Table J.8: Common parameters in the first quantitative implementation

Parameter	Value	Source / discipline
η	0.99	Near-Cobb–Douglas CES benchmark
ρ	0.05	Annual discount/consumption rate
δ	0.10	Approx. annual depreciation rate
$\bar{L}$	1	Labor normalization
λ	1.1	World Bank private credit and PWT aggregate capital-output ratio
$p^*_{\text{agro},0}$	1	Numeraire normalization

Sector-specific parameters combine measured moments and provisional assumptions. The labor elasticity β_i is constructed from firm-level labor shares, while χ_i denotes the fraction of non-labor income assigned to the sector-specific factor. Given these two objects, the mobile-capital and specific-factor elasticities are

$$\gamma_i = \chi_i(1-\beta_i), \qquad \alpha_i = (1-\chi_i)(1-\beta_i).$$

Under the normalization for the productivity process described in subsubsection J.3.4, productivity dispersion and persistence are mapped into

$$\sigma_i = \sqrt{2}\ \mathrm{sd}(\widehat{x}_{fi}), \qquad \theta_i = -\frac{1}{\log[\mathrm{corr}(\widehat{x}_t, \widehat{x}_{t-1})]}.$$

Table J.9 reports the resulting sector-specific values.

Table J.10 reports the sectoral classification, initial wealth allocation, and aggregate data moments used in the exercise. Sectoral wealth shares $O_{i,0}$ enter as initial conditions, while absorption shares are directly targeted through the CES weights. Employment shares and sectoral trade bal-

Table J.9: Sector-specific parameters in the first quantitative implementation

Sector	β_i	χ_i	α_i	γ_i	$\mathrm{sd}(\widehat{x}_{fi})$	σ_i	$\mathrm{corr}(\widehat{x}_t, \widehat{x}_{t-1})$	θ_i
Agroindustry	0.41	0.35	0.384	0.207	1.234	1.745	0.720	3.044
Commerce	0.34	0.25	0.495	0.165	1.431	2.024	0.869	7.122
Construction	0.42	0.30	0.406	0.174	1.632	2.308	0.781	4.046
Fishing	0.34	0.45	0.363	0.297	1.565	2.213	0.653	2.346
Machinery	0.44	0.15	0.476	0.084	1.276	1.805	0.845	5.938
Manufacturing	0.43	0.10	0.513	0.057	1.294	1.830	0.856	6.431
Services	0.47	0.15	0.451	0.080	1.626	2.300	0.873	7.363

ances identify the main remaining calibration margins, and production shares are left untargeted as validation moments. The resource-export residual offsets the aggregate trade deficit of the modeled sectors so that the complete economy satisfies external balance.

Table J.10: Preliminary sectoral calibration targets, initial conditions, and validation moments

Sector	Tradable	$O_{i,0}$	Employment share	Share of Absorption	Share of Production	$\mathrm{TB}_{i,0}/\sum_i p_{i,0}Y_{i,0}$
Calibration role	*Classification*	*Initial condition*	*To be target*	*Target*	*Untargeted*	*To be target*
Agroindustry	Yes	38.79%	32.66%	6.143%	6.291%	−0.335%
Commerce	No	17.64%	18.44%	10.602%	11.240%	–
Construction	No	4.96%	4.91%	8.594%	8.865%	–
Fishing	Yes	1.11%	0.77%	0.932%	0.971%	0.001%
Machinery	Yes	0.34%	0.38%	6.083%	1.452%	−5.034%
Manufacturing	Yes	10.41%	12.00%	31.384%	33.343%	−0.206%
Services	No	26.76%	30.84%	36.263%	37.839%	–
Total	–	100.00%	100.00%	100.00%	100.00%	−5.57%
					$NX_{\mathrm{res},0}/\sum_j p_{j,0}Y_{j,0}$	5.57%

I discipline the collateral parameter using the balance-sheet implication of the model's borrowing constraint. Entrepreneur f in sector i operates capital subject to

$$k_{fi} \leq \lambda a_{fi}.$$

If operated capital is financed with own wealth and external funds,

$$k_{fi} = a_{fi} + d_{fi},$$

then the collateral constraint implies

$$\frac{d_{fi}}{k_{fi}} = 1 - \frac{a_{fi}}{k_{fi}} \leq 1 - \frac{1}{\lambda}.$$

When the constraint binds, the external-finance share of operated capital is

$$\frac{d_{fi}}{k_{fi}} = 1 - \frac{1}{\lambda}.$$

I therefore use the aggregate relationship

$$\frac{D}{K} = 1 - \frac{1}{\lambda}$$

as a binding-constraint benchmark for mapping financial-depth data into the model. This is the same logic used to discipline collateral constraints in related models, but I use an external capital-output ratio rather than the model-implied capital-output ratio. This avoids mechanically tying the collateral parameter to ρ, δ, and the calibrated production shares. Specifically, I measure D/Y using domestic credit to the private sector as a share of GDP and K/Y using the aggregate capital-output ratio from Penn World Table, and compute

$$\lambda = \frac{1}{1 - \frac{D/Y}{K/Y}}.$$

For the first implementation, I discipline λ using the broad macro counterpart of the external-finance relationship

$$\frac{D}{K} = 1 - \frac{1}{\lambda}.$$

I measure D/Y using World Bank domestic credit to the private sector as a share of GDP, and K/Y using the aggregate capital-output ratio from Penn World Table. This yields a value close to $\lambda = 1.1$ for Peru. I use this value as a conservative benchmark and leave alternative constructions based on narrower capital concepts or firm-level balance-sheet moments for future refinements.

Because the empirical credit measure is broad and the model's credit object is gross borrowing used to finance operated capital, I interpret this value as a conservative macro benchmark rather than as a direct estimate of firm-level borrowing limits.

The common collateral parameter is disciplined separately using an aggregate external-finance benchmark. Table J.11 reports the credit and capital-output series entering this calculation and the associated annual values of λ.

Although the simulated transition starts in 2000, I discipline the CES demand weights using the 2007 input-output table. Let $s^D_{i,2007}$ denote sector i's share of domestic final absorption in the 2007 IO table. From CES demand,

$$s^D_{i,\tau} = \upsilon_i \left(\frac{p_{i,\tau}}{P_\tau} \right)^{1-\eta}.$$

Therefore, imposing the normalization $\sum_i \upsilon_i = 1$, I set

$$\upsilon_i = \frac{s^D_{i,2007} p^{\eta-1}_{i,2007}}{\sum_{j \in I} s^D_{j,2007} p^{\eta-1}_{j,2007}}.$$

In the first implementation, I normalize initial relative prices in the IO-based demand calibration, so that $p_{i,2007} = 1$ and hence $\upsilon_i = s^D_{i,2007}$.

Table J.11: Macro series benchmark for the collateral parameter

Year	D/Y (%)	K	Y	K/Y (%)	λ
2000	26.342	585,895	153,016	382.899	1.074
2001	23.448	603,471	154,087	391.642	1.064
2002	22.202	622,353	164,151	379.135	1.062
2003	19.986	642,855	173,867	369.739	1.057
2004	17.910	666,144	187,197	355.852	1.053
2005	19.066	693,788	207,762	333.933	1.061
2006	20.813	724,281	234,838	308.418	1.072
2007	24.607	756,706	259,291	291.837	1.092
2008	29.764	802,104	283,819	282.611	1.118
2009	30.015	835,966	284,735	293.594	1.114
2010	30.262	885,582	323,195	274.008	1.124
2011	32.309	948,442	355,340	266.911	1.138
2012	34.044	1,038,669	374,776	277.144	1.140
2013	37.720	1,132,344	378,570	299.111	1.144
2014	40.900	1,231,078	384,454	320.215	1.146
2015	43.868	1,341,270	388,238	345.476	1.145
2016	42.809	1,442,108	398,096	362.251	1.134
2017	42.415	1,557,388	411,379	378.577	1.126
2018	43.904	1,679,814	428,483	392.038	1.126
2019	44.724	1,830,022	450,548	406.177	1.124
2020	55.312	1,975,340	412,739	478.593	1.131
2021	47.115	2,149,082	478,284	449.332	1.117
2022	45.649	2,259,697	489,805	461.346	1.110
2023	42.949	2,341,043	483,527	484.160	1.097

Notes: D/Y is domestic credit to the private sector as a share of GDP from the World Bank's World Development Indicators, series `FS.AST.PRVT.GD.ZS`. K is the Penn World Table capital stock at current PPPs, series `cn`. Y is output-side real GDP at current PPPs, series `cgdpo`. Both K and Y are measured in millions of 2021 U.S. dollars. The collateral parameter is computed as

$$\lambda = \frac{1}{1 - \frac{D/Y}{K/Y}},$$

where D/Y and K/Y are expressed in the same units.

I treat construction as nontradable. While construction uses tradable intermediate inputs, the final construction good is location-specific and must be absorbed domestically. Its price therefore adjusts to clear the domestic construction market, as in the other nontradable sectors.

K Persistent Specialization in a Small Open Economy: A Sector-Specific Capital Benchmark

The purpose of this appendix is to provide a stripped-down example of the mechanism studied in the paper. The model has two sectors, mobile labor, fixed world prices, and sector-specific capital accumulation. The goal is to derive a closed-form characterization of how the interaction between capital accumulation and fixed world prices can generate persistent specialization patterns.

The appendix considers two benchmark exercises. First, if productivity differences across sectors are permanent, the economy converges to full specialization in the more productive sector. Second, if productivity differences are only temporary, aggregate capital eventually converges to the standard Solow steady state once technologies become identical again. However, the sectoral composition of capital continues to reflect the history of those temporary productivity differences. As a result, temporary productivity advantages can generate persistent differences in sectoral capital and output even after productivity differences disappear.

Importantly, the appendix shows that persistent specialization patterns can arise even without heterogeneous entrepreneurs, financial frictions, or a CES demand system. The key "trick" is that labor is mobile across sectors, while capital is accumulated within sectors. Mobile labor links the two sectors in general equilibrium. However, because the economy is small and open, goods prices are fixed at world levels. As a result, sectoral expansion does not lower the relative price of the expanding sector. This removes the closed-economy force that would otherwise reduce initial asymmetries.

The appendix therefore should not be interpreted as a literal reduced-form version of the main model. In particular, capital in this environment is accumulated within sectors, rather than being allocated through an integrated economy-wide financial market as in the heterogeneous-agent framework studied in the paper. As a result, persistence here operates through sectoral accumulation dynamics rather than through the endogenous wealth reallocation mechanism emphasized in the main model.

The purpose of the exercise is instead to isolate, in a transparent environment, the role of fixed world prices in shaping long-run sectoral dynamics. By shutting down the closed-economy relative price adjustment mechanism, temporary asymmetries in productivity translate into persistent differences in sectoral capital stocks and production patterns. The appendix therefore provides an illustrative benchmark for the broader mechanism emphasized in the main text, where similar forces operate through endogenous wealth accumulation and financial frictions in an integrated multisector economy.

K.1 Setup

Environment. Time is continuous. There are two sectors indexed by $i \in \{1, 2\}$. Sector i produces using capital $K_i(t)$ and labor $L_i(t)$ according to

$$Y_i(t) = A_i(t)K_i(t)^{\alpha}L_i(t)^{1-\alpha}, \qquad 0 < \alpha < 1. \tag{K.1}$$

Total labor supply is fixed and normalized to one:

$$\sum_{i\in\{1,2\}} L_i(t) = 1. \tag{K.2}$$

Labor is perfectly mobile across sectors. Goods prices are fixed at world levels. Without loss of generality, prices can be normalized to one so that all quantities are expressed in real terms:

$$p_i^* = 1, \qquad i \in \{1, 2\}.$$

Capital is accumulated by sector-specific entrepreneurs. To keep the model as close as possible to the Solow model, assume that a constant fraction $s_i \in (0, 1)$ of sectoral output is invested in the same sector. Capital depreciates at rate $\delta > 0$. Thus,

$$\dot{K}_i(t) = s_i Y_i(t) - \delta K_i(t). \tag{K.3}$$

Labor Allocation. At each point in time, labor is allocated competitively across sectors. Hence, the wage equals the value marginal product of labor in every active sector:

$$w(t) = (1 - \alpha)A_i(t)K_i(t)^{\alpha}L_i(t)^{-\alpha}. \tag{K.4}$$

For any two active sectors, this implies

$$(1 - \alpha)A_1(t)K_1(t)^{\alpha}L_1(t)^{-\alpha} = (1 - \alpha)A_2(t)K_2(t)^{\alpha}L_2(t)^{-\alpha}. \tag{K.5}$$

Canceling common terms and rearranging gives

$$\frac{L_1(t)}{L_2(t)} = \left(\frac{A_1(t)}{A_2(t)}\right)^{1/\alpha} \frac{K_1(t)}{K_2(t)}. \tag{K.6}$$

Using $\sum_{i\in\{1,2\}} L_i(t) = 1$, sectoral labor allocations are

$$L_i(t) = \frac{A_i(t)^{1/\alpha}K_i(t)}{\sum_{j\in\{1,2\}} A_j(t)^{1/\alpha}K_j(t)}. \tag{K.7}$$

Labor flows toward sectors with higher productivity and larger capital stocks.

Reduced-Form Sectoral Output. Substituting the labor allocation into production gives

$$Y_i(t) = A_i(t)K_i(t)^{\alpha}\left[\frac{A_i(t)^{1/\alpha}K_i(t)}{\sum_{j\in\{1,2\}} A_j(t)^{1/\alpha}K_j(t)}\right]^{1-\alpha}. \tag{K.8}$$

This simplifies to

$$Y_i(t) = \frac{A_i(t)^{1/\alpha}K_i(t)}{\left[\sum_{j\in\{1,2\}} A_j(t)^{1/\alpha}K_j(t)\right]^{1-\alpha}}. \tag{K.9}$$

Define aggregate effective capital as

$$X(t) \equiv \sum_{i\in\{1,2\}} A_i(t)^{1/\alpha}K_i(t). \tag{K.10}$$

Then sectoral output can be written compactly as

$$Y_i(t) = A_i(t)^{1/\alpha}X(t)^{-(1-\alpha)}K_i(t). \tag{K.11}$$

Capital Accumulation. The law of motion for sectoral capital becomes

$$\dot{K}_i(t) = s_i A_i(t)^{1/\alpha}X(t)^{-(1-\alpha)}K_i(t) - \delta K_i(t). \tag{K.12}$$

Therefore, the growth rate of sectoral capital is

$$g_{K,i} = \frac{\dot{K}_i(t)}{K_i(t)} = s_i A_i(t)^{1/\alpha}X(t)^{-(1-\alpha)} - \delta. \tag{K.13}$$

Taking the difference across sectors gives the law of motion for relative capital:

$$\frac{d}{dt}\log\left(\frac{K_1(t)}{K_2(t)}\right) = X(t)^{-(1-\alpha)}\left[s_1A_1(t)^{1/\alpha} - s_2A_2(t)^{1/\alpha}\right]. \tag{K.14}$$

This equation summarizes the mechanism. Relative sectoral capital grows toward the sector with higher productivity or a higher reinvestment rate.

K.2 A Persistent Productivity Gap

First, this toy model shows that if a productivity gap persists across sectors, the economy eventually specializes in the more productive sector.

Suppose both sectors begin with positive capital stocks,

$$K_1(0) > 0, \qquad K_2(0) > 0,$$

and have the same investment rate,

$$s_1 = s_2 = s.$$

Let sector 1 be permanently more productive:

$$A_1(t) = A, \qquad A_2(t) = (1-\tau)A, \qquad 0 < \tau < 1.$$

Using the relative capital growth derived above (Equation K.14),

$$\frac{d}{dt}\log\left(\frac{K_1(t)}{K_2(t)}\right) = sX(t)^{-(1-\alpha)}A^{1/\alpha}\left[1-(1-\tau)^{1/\alpha}\right] > 0. \tag{K.15}$$

Hence, sector 1 accumulates capital faster than sector 2, and its relative size increases over time.

In this case, the economy converges to a corner allocation in which only the more productive sector remains active:

$$K_1^* > 0, \qquad K_2^* = 0. \tag{K.16}$$

At the corner, aggregate capital is equal to capital in sector 1:

$$K^* = K_1^*. \tag{K.17}$$

Since sector 1 has productivity A, aggregate capital satisfies the standard Solow steady-state condition:

$$0 = sA(K^*)^{\alpha} - \delta K^*. \tag{K.18}$$

Thus,

$$K^* = K_1^* = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}, \qquad K_2^* = 0. \tag{K.19}$$

This benchmark shows that with a permanent productivity gap, fixed world prices imply full specialization in the more productive sector. Aggregate capital converges to the standard Solow steady state, but all capital is eventually concentrated in the more productive sector.

K.3 A Temporary Productivity Gap

The main model shows that under identical fundamentals, the self-financing mechanism tends to reduce initial productivity differences across sectors over time. To capture this idea in the toy model, now suppose that the productivity gap between sectors disappears after a finite period of time.

Suppose the two sectors begin with the same capital stock:

$$K_1(0) = K_2(0) = K_0. \tag{K.20}$$

Assume equal investment rates:

$$s_1 = s_2 = s. \tag{K.21}$$

As before, let sector 1 have productivity

$$A_1(t) = A,$$

while sector 2 has a temporary productivity disadvantage:

$$A_2(t) = \begin{cases} (1-\tau)A, & t < T, \\ A, & t \geq T, \end{cases}, \qquad \text{with } 0 < \tau < 1 \text{ and finite } T. \tag{K.22}$$

Then, for $t < T$, the same logic from the previous subsection implies

$$\frac{d}{dt} \log \left(\frac{K_1(t)}{K_2(t)} \right) > 0. \tag{K.23}$$

Hence, while the productivity gap persists, sector 1 accumulates capital faster than sector 2.

Now suppose that at time T, the productivity gap disappears:

$$A_1(t) = A_2(t) = A, \qquad t \geq T. \tag{K.24}$$

From that moment onward, both sectors grow at the same rate as the factor $\left[1 - (1-\tau)^{1/\alpha}\right]$ goes to zero:

$$\frac{d}{dt} \log \left(\frac{K_1(t)}{K_2(t)} \right) = 0 \qquad \Leftrightarrow \qquad g_{K,1} = g_{K,2}. \tag{K.25}$$

Therefore, the relative capital allocation reached at time T persists:

$$R_T \equiv \frac{K_1(t)}{K_2(t)} = \frac{K_1(T)}{K_2(T)} > 1 \qquad \text{for all } t \geq T. \tag{K.26}$$

Even though productivity differences disappear in finite time, the asymmetry generated during the transition remains permanently embedded in the sectoral composition of capital.

Steady State. After time T, both sectors have the same productivity A and the same investment rate s. Therefore, aggregate capital evolves according to:

$$\dot{K}(t) = sAK(t)^{\alpha} - \delta K(t), \tag{K.27}$$

which is exactly the same law of motion as in the standard Solow model.[8]

Hence, aggregate capital converges again to the standard Solow steady state:

$$K^* = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}} \qquad \Leftrightarrow \qquad 0 = sA(K^*)^{\alpha} - \delta K^*. \tag{K.28}$$

However, the sectoral composition of capital depends on the history of temporary productivity differences. Since relative capital remains constant after time T,

$$\frac{K_1(t)}{K_2(t)} = R_T \qquad \text{for all } t \geq T, \tag{K.29}$$

the steady-state sectoral capital stocks must satisfy

$$K_1^* = \frac{R_T}{1+R_T}K^*, \qquad K_2^* = \frac{1}{1+R_T}K^*, \tag{K.30}$$

with

$$\frac{K_1^*}{K_2^*} = R_T > 1. \tag{K.31}$$

This shows that even though both sectors eventually share identical technologies, the long-run allocation of capital depends on the history of temporary productivity differences. Aggregate capital converges to the standard Solow steady state, but the sectoral composition of that steady

[8] After time T, both sectors share the same productivity level A, so

$$X(t) = A^{1/\alpha} \sum_{i\in\{1,2\}} K_i(t).$$

The sectoral capital accumulation equation becomes

$$g_{K,i} = sA^{1/\alpha}\left[A^{1/\alpha}\sum_{j\in\{1,2\}} K_j(t)\right]^{-(1-\alpha)} - \delta.$$

Using

$$\left[A^{1/\alpha}\right]^{-(1-\alpha)} = A^{-(1-\alpha)/\alpha},$$

and defining aggregate capital as

$$K(t) \equiv \sum_{i\in\{1,2\}} K_i(t),$$

this implies

$$g_{K,i} = sAK(t)^{-(1-\alpha)} - \delta.$$

Since both sectors grow at the same rate after time T, aggregate capital evolves according to

$$\dot{K}(t) = \sum_{i\in\{1,2\}} g_{K,i}K_i(t) = \left[sAK(t)^{-(1-\alpha)} - \delta\right]K(t),$$

which simplifies to

$$\dot{K}(t) = sAK(t)^{\alpha} - \delta K(t).$$

state inherits the asymmetries generated during the transition.

The Persistence of Specialization. This toy model delivers a simplified version of the history dependence mechanism in the paper. With a permanent productivity gap, the economy converges to full specialization in the more productive sector. However, even when productivity differences are only temporary, the sectoral composition of capital continues to reflect the history of those differences in the long run. Once technologies become identical again, aggregate capital converges to the standard Solow steady state, but sectoral capital shares remain pinned down by the relative accumulation that occurred during the transition.

The key mechanism is that world prices are fixed. As a result, temporary productivity differences are not offset by relative-price adjustments, allowing them to translate into persistent differences in capital accumulation. In the full model, the relevant state variable is not only sectoral capital, but the wealth-weighted distribution of entrepreneurs within each sector. The full model therefore endogenizes the accumulation mechanism through wealth heterogeneity and financial frictions, allowing the magnitude and persistence of sectoral differences to depend on the evolving distribution of wealth and productivity within sectors.

L Empirical Appendix

This appendix reports additional empirical results supporting the mechanisms emphasized in the paper, but using fixed initial exposure shares instead of dynamically changing ones. It presents local-projection estimates of the dynamic response of sectoral employment shares to external demand shocks, showing persistent sectoral reallocation patterns following favorable external conditions. The appendix also reports firm-level regressions linking capital and labor growth to lagged external demand shocks, firm productivity, and their interaction. Together, these results provide preliminary evidence consistent with the paper's central mechanism, in which external demand conditions interact with heterogeneous productivity and financial frictions to shape accumulation and specialization dynamics over time.

Table L.1: External demand shocks and firm expansion

	Panel A. Capital growth				
	ℓ=1	ℓ=2	ℓ=3	ℓ=4	ℓ=5
Demand shock	0.209	-0.249*	0.084	-0.073	-0.105
	(0.147)	(0.136)	(0.157)	(0.074)	(0.118)
Lagged productivity	0.062***	0.063***	0.062***	0.062***	0.063***
	(0.012)	(0.011)	(0.011)	(0.011)	(0.011)
Shock × productivity	-0.026*	0.036**	-0.024	0.013	0.014
	(0.015)	(0.015)	(0.023)	(0.012)	(0.020)
Observations	4,521	4,521	4,521	4,521	4,521
	Panel B. Labor growth				
	ℓ=1	ℓ=2	ℓ=3	ℓ=4	ℓ=5
Demand shock	-1.305***	0.223	-0.346	0.144	-0.105
	(0.316)	(0.590)	(0.422)	(0.557)	(0.226)
Lagged productivity	0.067***	0.063***	0.066***	0.064***	0.065***
	(0.019)	(0.019)	(0.018)	(0.017)	(0.018)
Shock × productivity	0.133***	-0.023	0.075	-0.009	0.005
	(0.037)	(0.078)	(0.055)	(0.085)	(0.032)
Observations	4,532	4,532	4,532	4,532	4,532

Notes: The table reports firm-level regressions of capital and labor growth on lagged sector-level dynamic demand shocks, lagged firm productivity, and their interaction. Panel A uses capital growth as the dependent variable, and Panel B uses labor growth. Columns correspond to shock lags from one to five years. Dynamic demand shocks are constructed from changes in external import demand using initial Peruvian export shares as exposure weights. Lagged productivity is measured using the firm-level productivity residual described in the text. All regressions include sector and year fixed effects. Standard errors are clustered at the sector level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

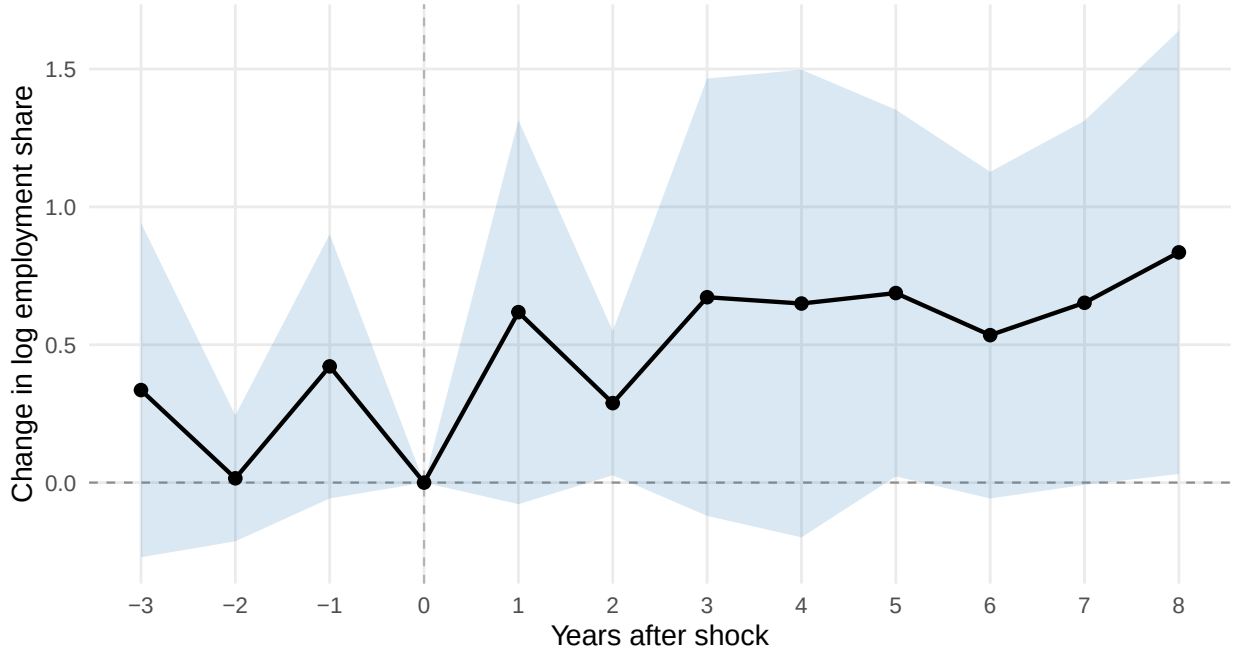
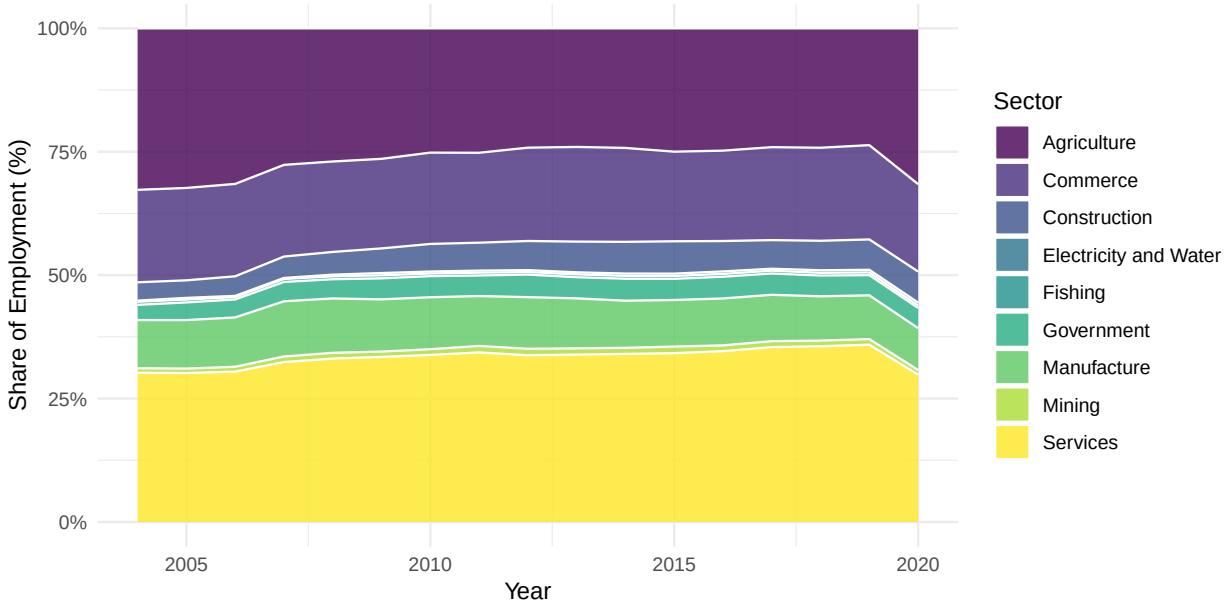
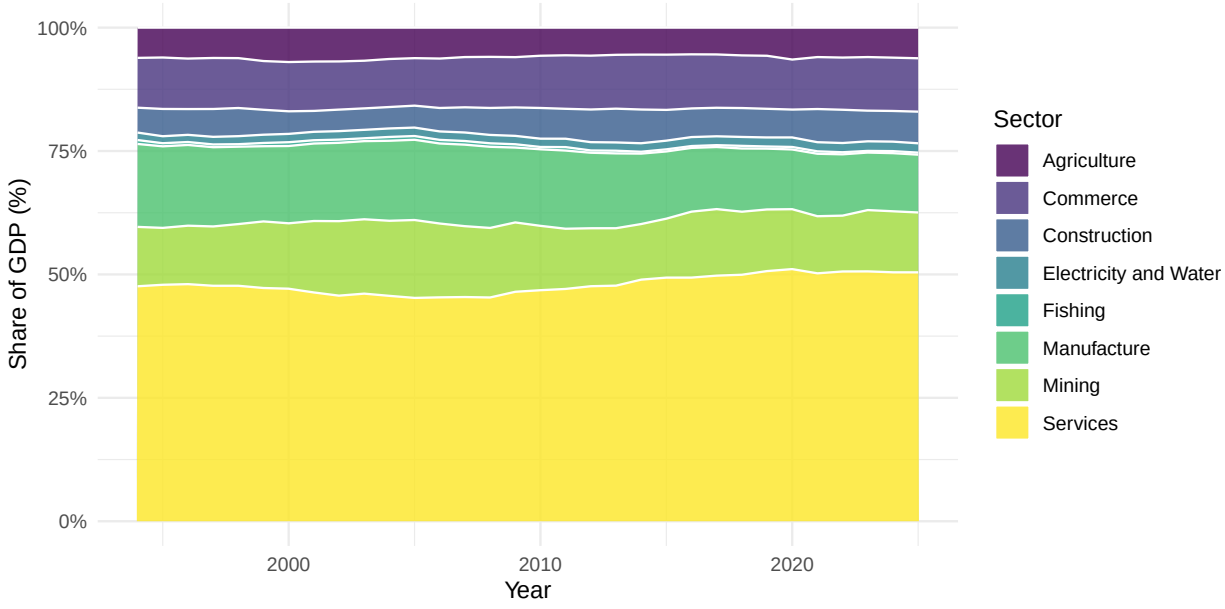


Figure L.1: Dynamic Response of Sectoral Employment Shares to External Demand Shocks

Notes: The figure plots local-projection estimates of the dynamic response of sectoral employment shares to external demand shocks in Peru. The horizontal axis reports years relative to the shock, where negative values correspond to pre-trends and positive values to post-shock responses. The vertical axis reports cumulative changes in log sectoral employment shares. The shock measure is constructed using changes in external import demand weighted by the initial Peruvian export exposure across destination markets and sectors. Shaded bands denote 95 percent confidence intervals based on standard errors clustered at the sector level. Sectoral employment shares are computed using household survey data from ENAHO aggregated to ISIC Rev. 3 two-digit sectors. The dashed vertical line marks the shock period ($h = 0$), and the dashed horizontal line marks zero change in sectoral employment shares.



(a) Employment



(b) Output

Figure L.2: Sector Composition of Peruvian Economy

Notes: Panel a): Shares represent each sector's percentage of total employment in Peru, ENAHO 2004–2020. Panel b) Shares represent each sector's percentage of GDP, BCRP Series.